

# RADIANT Theory Manual

Physics Reference for the RADIANT EO Sensor Performance Model

RADIANT Project

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# Notation and Symbols

This front-matter chapter is the manual’s single notation home. Every symbol, canonical unit, and sign convention used in the chapters that follow is defined here once; the chapters use the symbols without redefining them. Where a chapter needs a symbol that is local to one derivation, it introduces it at the point of use and the local meaning wins for that section only.

Two rules govern everything below:

- **One canonical unit per quantity.** RADIANT computes internally in the units of the canonical-units table below — wavelength in  $\mu\text{m}$ , angles in radians, length in meters, time in seconds, radiance in  $\text{W}/\text{m}^2/\text{sr}/\mu\text{m}$ , noise in e- RMS. No module works in any other unit.
- **Conversion happens exactly once, at a boundary.** User input converts on entry; external files convert in their reader. A unit conversion inside a physics equation is a defect, not a convenience.

## 0.1 Canonical units

| Quantity                 | Symbol                       | Canonical unit                                            | Notes                                                                                                          |
|--------------------------|------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Wavelength               | $\lambda$                    | $\mu\text{m}$                                             | The primary spectral variable; arrays ascend in $\lambda$                                                      |
| Wavelength in SI         | $\lambda_m$                  | m                                                         | $\lambda_m = \lambda \cdot 10^{-6}$ ; used wherever $h$ , $c$ , or $k_B$ appears                               |
| Wavenumber               | $\nu_{cm}$                   | $\text{cm}^{-1}$                                          | Derived only ( $\nu_{cm} = 10^4/\lambda$ ), never stored as the primary axis                                   |
| Spectral radiance        | $L(\lambda)$                 | $\text{W}/\text{m}^2/\text{sr}/\mu\text{m}$               | MODTRAN’s $\text{W}/\text{cm}^2/\text{sr}/\mu\text{m}$ converts by $\times 10^4$ in the reader                 |
| In-band radiance         | $L$                          | $\text{W}/\text{m}^2/\text{sr}$                           | $\int L(\lambda) d\lambda$ over the bandpass                                                                   |
| Spectral irradiance      | $E(\lambda)$                 | $\text{W}/\text{m}^2/\mu\text{m}$                         |                                                                                                                |
| In-band irradiance       | $E$                          | $\text{W}/\text{m}^2$                                     |                                                                                                                |
| Spectral intensity       | $I(\lambda)$                 | $\text{W}/\text{sr}/\mu\text{m}$                          | Point sources                                                                                                  |
| Spectral photon radiance | $L_q(\lambda)$               | $\text{photon}/\text{s}/\text{m}^2/\text{sr}/\mu\text{m}$ | Derived: $L_q = L\lambda_m/(hc)$                                                                               |
| Temperature              | $T$                          | K                                                         | Users may enter K, $^{\circ}\text{C}$ , or $^{\circ}\text{F}$ ; the affine conversion is applied once at entry |
| Angle (all internal)     | $\theta, \eta, \zeta, \dots$ | rad                                                       | Users enter large angles in degrees, small angles in $\mu\text{rad}$                                           |
| Length                   | $D, f, p, R_s$               | m                                                         |                                                                                                                |
| Time                     | $t_{int}, T_{frame}$         | s                                                         | Displays may use ms or $\mu\text{s}$ ; storage is seconds                                                      |
| Signal                   | $S$                          | e-                                                        | Photoelectrons at the sense node, after integration                                                            |
| Noise                    | $\sigma$                     | e- RMS                                                    | Every noise term, before any DN conversion                                                                     |

| Quantity                        | Symbol                        | Canonical unit | Notes                                                                           |
|---------------------------------|-------------------------------|----------------|---------------------------------------------------------------------------------|
| Digital number                  | DN                            | DN             | $DN = S/g$ with $g$ in e-/DN                                                    |
| Spatial frequency (focal plane) | $\nu$                         | cy/m           | Quoted as cy/mm in worked numbers; always labeled                               |
| Spatial frequency (angular)     | $\nu_{ang}$                   | cy/mrad        | $\nu f$ with $f$ in m is cy/rad, so $\nu_{ang} = \nu f / 1000$ ; always labeled |
| Transfer function               | MTF( $\nu$ )                  | dimensionless  | $0 \leq MTF \leq 1$                                                             |
| Dimensionless fractions         | $\varepsilon, \rho, \tau, EE$ | dimensionless  | Stated on the interval $[0, 1]$ where physical                                  |

**Reading the tables that follow.** Units in square brackets are the canonical units above. A quantity listed as dimensionless still carries a physical range, given where it matters.

## 0.2 Coordinate, attitude, and indexing conventions

| Property       | Convention                                                                                            |
|----------------|-------------------------------------------------------------------------------------------------------|
| Handedness     | Right-handed                                                                                          |
| +Z             | Toward the target, along the boresight / optical axis                                                 |
| +X             | Cross-track (perpendicular to the flight direction)                                                   |
| +Y             | Along-track (flight direction projected into the image plane)                                         |
| Euler sequence | 3-2-1 (ZYX): yaw about +Z, then pitch about the once-rotated +Y, then roll about the twice-rotated +X |
| Pixel indexing | $[\text{row}, \text{col}] = [y, x] = [\text{along-track}, \text{cross-track}]$ , 0-indexed            |

For a nadir-pointing space sensor +Z is nadir; for a ground observer +Z is toward the target in the sky. The convention holds for every viewing geometry the model supports without redefinition.

Viewing geometry is referenced to the **target**, not the sensor: the primary angle is the line-of-sight zenith angle at the target,  $\theta_o$ , from which the look angle  $\eta$  at the sensor, the Earth central angle  $\Lambda$ , and the slant range  $R_s$  follow.

## 0.3 Radiometric symbols

| Symbol                    | Meaning                                                     | Units                          |
|---------------------------|-------------------------------------------------------------|--------------------------------|
| $B(\lambda, T)$           | Planck blackbody spectral radiance                          | W/m <sup>2</sup> /sr/μm        |
| $B_q(\lambda, T)$         | Planck photon spectral radiance, $B/E_{ph}$                 | photon/s/m <sup>2</sup> /sr/μm |
| $\partial B / \partial T$ | temperature derivative of Planck radiance (the NEDT kernel) | W/m <sup>2</sup> /sr/μm/K      |
| $\bar{L}$                 | band-averaged radiance                                      | W/m <sup>2</sup> /sr/μm        |
| $L_{\text{band}}$         | band-integrated radiance                                    | W/m <sup>2</sup> /sr           |

| Symbol                             | Meaning                                                                   | Units                   |
|------------------------------------|---------------------------------------------------------------------------|-------------------------|
| $L_{\text{target}}, L_{\text{bg}}$ | target and background spectral radiance at the source                     | W/m <sup>2</sup> /sr/μm |
| $L_{\text{ap}}$                    | at-aperture spectral radiance                                             | W/m <sup>2</sup> /sr/μm |
| $L_{\text{path}}$                  | atmospheric path radiance (scattered plus emitted into the line of sight) | W/m <sup>2</sup> /sr/μm |
| $L_{\text{sky}}$                   | hemispheric downwelling sky radiance                                      | W/m <sup>2</sup> /sr/μm |
| $E_{\text{sun}}(\lambda)$          | top-of-atmosphere solar spectral irradiance                               | W/m <sup>2</sup> /μm    |
| $E_{ph}$                           | photon energy, $hc/\lambda_m$                                             | J                       |
| $\varepsilon(\lambda)$             | emissivity                                                                | dimensionless, 0–1      |
| $\rho(\lambda)$                    | hemispherical reflectance (albedo)                                        | dimensionless, 0–1      |
| $f_r$                              | bidirectional reflectance distribution function (BRDF)                    | sr <sup>-1</sup>        |
| $\tau_{\text{atm}}(\lambda)$       | atmospheric transmittance along the slant path                            | dimensionless, 0–1      |
| $\tau_{\text{opt}}(\lambda)$       | net optical-train transmittance                                           | dimensionless, 0–1      |
| $R(\lambda), T(\lambda)$           | surface reflectance and transmittance of an optical element               | dimensionless, 0–1      |
| $\theta_{\text{sun}}$              | solar zenith angle at the surface                                         | rad                     |
| $x$                                | dimensionless Planck argument, $x = hc/(\lambda_m k_B T)$                 | dimensionless           |
| $ff$                               | sub-pixel fill fraction, target area over pixel footprint                 | dimensionless, 0–1      |
| $QE(\lambda)$                      | quantum efficiency                                                        | e-/photon               |

## 0.4 Geometry and sampling symbols

| Symbol                | Meaning                                                                                | Units          |
|-----------------------|----------------------------------------------------------------------------------------|----------------|
| $R$                   | mean Earth radius, 6371.0 km                                                           | m              |
| $h$                   | sensor altitude above the surface                                                      | m              |
| $R_s$                 | slant range, sensor to target                                                          | m              |
| $\eta$                | look (off-nadir) angle at the sensor                                                   | rad            |
| $\theta_o$            | line-of-sight zenith angle at the target (equals incidence angle for a surface target) | rad            |
| $\varepsilon_{el}$    | grazing/elevation angle at the target, $\pi/2 - \theta_o$                              | rad            |
| $\Lambda$             | Earth central angle between sub-sensor point and target                                | rad            |
| $\zeta$               | path zenith angle used by the atmosphere models                                        | rad            |
| $A_t$                 | target projected area                                                                  | m <sup>2</sup> |
| $A_{\text{ap}}$       | collecting aperture area                                                               | m <sup>2</sup> |
| $\Omega_{\text{pix}}$ | pixel solid angle, $p_x p_y / f^2$                                                     | sr             |
| $\Omega_t$            | target solid angle, $A_t / R_s^2$                                                      | sr             |
| $D$                   | aperture diameter                                                                      | m              |
| $f$                   | focal length                                                                           | m              |
| $F_{\#}$              | f-number, $f / D$                                                                      | dimensionless  |
| $p, p_x, p_y$         | pixel pitch (cross-track, along-track)                                                 | m              |
| IFOV                  | instantaneous field of view of one pixel, $p / f$                                      | rad            |
| GSD                   | ground sample distance                                                                 | m              |
| $v_{\text{img}}$      | image-plane motion rate during integration                                             | m/s            |
| $\theta$              | target angular extent, $\sqrt{A_t} / R_s$                                              | rad            |

## 0.5 Spatial symbols — PSF and MTF

| Symbol                       | Meaning                                                        | Units                                      |
|------------------------------|----------------------------------------------------------------|--------------------------------------------|
| $P$                          | complex pupil function, $P = A e^{i2\pi W/\lambda}$            | dimensionless                              |
| $W$                          | wavefront error across the pupil                               | m (quoted in waves at a stated $\lambda$ ) |
| $\sigma_{WFE}$               | RMS wavefront error                                            | waves at the stated reference wavelength   |
| $Z_n^m$                      | Zernike coefficient (Noll-ordered)                             | waves RMS                                  |
| PSF                          | point spread function (the degraded, chain-assembled one)      | dimensionless, normalized to unit volume   |
| OTF, MTF( $\nu$ )            | optical transfer function and its modulus                      | dimensionless                              |
| MTF <sub>sys</sub> ( $\nu$ ) | system MTF, the product of all contributor MTFs                | dimensionless                              |
| $\epsilon$                   | central-obscuration ratio, $D_{sec}/D$                         | dimensionless, 0–1                         |
| $\nu_c$                      | optical cutoff frequency, $1/(\lambda F_\#)$                   | cy/m                                       |
| $\nu_{Nyq}$                  | detector Nyquist frequency, $1/(2p)$                           | cy/m                                       |
| $Q$                          | sampling parameter, $\lambda F_\#/p$                           | dimensionless                              |
| $S_{tr}$                     | Strehl ratio (degraded PSF peak over diffraction-limited peak) | dimensionless, 0–1                         |
| EE <sub>box</sub>            | ensquared energy inside one pixel, from the degraded PSF       | dimensionless, 0–1                         |
| RER                          | relative edge response                                         | dimensionless                              |
| $\sigma_j$                   | RMS angular jitter                                             | rad (entered in $\mu$ rad)                 |
| $L_{sm}$                     | smear length during integration                                | m at the focal plane                       |
| $r_0$                        | Fried parameter                                                | m                                          |
| $C_n^2(z)$                   | refractive-index structure constant profile                    | $m^{-2/3}$                                 |
| $L_d$                        | minority-carrier diffusion length                              | m                                          |
| $\alpha_{IPC}$               | interpixel-capacitance coupling coefficient                    | dimensionless                              |

## 0.6 Detector, readout, and noise symbols

| Symbol          | Meaning                                                 | Units  |
|-----------------|---------------------------------------------------------|--------|
| $S, S_{bg}$     | target and background signal                            | e-     |
| $N_{fw}$        | full-well capacity                                      | e-     |
| $g$             | conversion gain, $N_{fw}/2^n$ for an $n$ -bit converter | e-/DN  |
| $t_{int}$       | integration time                                        | s      |
| $T_{frame}$     | frame period                                            | s      |
| $J_{dark}$      | dark-current rate                                       | e-/s   |
| $E_a$           | dark-current activation energy                          | eV     |
| $\sigma_i$      | one noise term, input-referred to electrons             | e- RMS |
| $\sigma_{tot}$  | total noise, the RSS of all contributing terms          | e- RMS |
| $\sigma_{read}$ | read noise                                              | e- RMS |
| $\sigma_{kTC}$  | reset (kTC) noise, $\sqrt{k_B T C_{node}}/q$            | e- RMS |

| Symbol                        | Meaning                                                                                                                                                                                                                                                             | Units                |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| $\sigma_{ADC}$                | quantization noise, $g/\sqrt{12}$                                                                                                                                                                                                                                   | e- RMS               |
| $k$                           | PRNU fraction (multiplicative gain dispersion) — the <i>residual</i> after NUC in the noise chapter's $\sigma_{PRNU} = kS$ , the <i>pre-correction</i> value in the calibration chapter's one-point residual, which is what an offset-only correction leaves intact | dimensionless        |
| $\sigma_{DSNU}$               | dark-signal non-uniformity                                                                                                                                                                                                                                          | e- RMS               |
| $\sigma_{clutter}$            | scene-induced spatial variance term                                                                                                                                                                                                                                 | e- RMS               |
| $N_{TDI}, N_{coadd}, M_{bin}$ | TDI stages, co-added frames, binned pixels                                                                                                                                                                                                                          | dimensionless counts |
| $dS/dT$                       | signal derivative with respect to scene temperature                                                                                                                                                                                                                 | e-/K                 |
| $\oplus$                      | root-sum-square combination, $a \oplus b = \sqrt{a^2 + b^2}$                                                                                                                                                                                                        | —                    |

## 0.7 Calibration symbols

| Symbol            | Meaning                                                                                        | Units              |
|-------------------|------------------------------------------------------------------------------------------------|--------------------|
| $S_1, S_2, S_3$   | calibration-point signals (low, mid/high, high)                                                | e-                 |
| $T_{cal,j}$       | calibration-source temperature at point $j$                                                    | K                  |
| $f_j$             | calibration-point signal declared as a fraction of the scene signal                            | dimensionless      |
| $S_{ref}$         | full-scale reference for the nonlinearity coefficient                                          | e-                 |
| $\beta$           | 1 $\sigma$ per-pixel response-nonlinearity fraction at full scale                              | dimensionless      |
| $D_j$             | calibration-point signal derivative, $dS/dT$ at $T_{cal,j}$                                    | e-/K               |
| $\Delta T_{unif}$ | 1 $\sigma$ spatial non-uniformity of the calibration source                                    | K                  |
| $r_g$             | gain-drift rate since calibration                                                              | s <sup>-1</sup>    |
| $r_o$             | offset-drift rate since calibration                                                            | e-/s               |
| $t_{cal}$         | elapsed time since the calibration event                                                       | s                  |
| $\sigma_{cal}$    | RSS of the calibration residual noise terms                                                    | e- RMS             |
| $b_i, b_{tot}$    | per-source and total fractional radiometric bias (1 $\sigma$ )                                 | dimensionless      |
| $g(T)$            | band-shift log-derivative of the photon-weighted band integral                                 | $\mu\text{m}^{-1}$ |
| $\delta\lambda$   | band-center (spectral-calibration) uncertainty                                                 | $\mu\text{m}$      |
| $f_{fore}$        | photon-weighted fraction of near-field emission from elements ahead of an internal cal shutter | dimensionless, 0–1 |

## 0.8 Physical constants

Every constant is defined once in the code and never re-entered in an equation. The first six rows are the CODATA 2018 values — the first five exact by the SI definitions, the second radiation constant derived from them. The last row is not a CODATA quantity: it is the mean Earth radius, a geodetic convention (see the geometry chapter).

| Constant                  | Symbol         | Value                         | Units                            |
|---------------------------|----------------|-------------------------------|----------------------------------|
| Speed of light            | $c$            | $2.99792458 \times 10^8$      | m/s                              |
| Planck constant           | $h$            | $6.62607015 \times 10^{-34}$  | J·s                              |
| Boltzmann constant        | $k_B$          | $1.380649 \times 10^{-23}$    | J/K                              |
| Stefan-Boltzmann constant | $\sigma_{SB}$  | $5.670374419 \times 10^{-8}$  | W/m <sup>2</sup> /K <sup>4</sup> |
| Elementary charge         | $q$            | $1.602176634 \times 10^{-19}$ | C                                |
| Second radiation constant | $c_2 = hc/k_B$ | $1.4387769 \times 10^{-2}$    | m·K                              |
| Mean Earth radius         | $R$            | $6.3710 \times 10^6$          | m                                |

## 0.9 Operators, accents, and reused symbols

| Form                   | Meaning                                                 |
|------------------------|---------------------------------------------------------|
| $\bar{X}$              | band average of $X$ over the bandpass                   |
| $X_m$                  | $X$ expressed in SI meters (used only for $\lambda_m$ ) |
| $a \oplus b$           | root-sum-square, $\sqrt{a^2 + b^2}$                     |
| $\star$                | correlation (used for the pupil autocorrelation)        |
| $\mathcal{F}\{\cdot\}$ | Fourier transform                                       |
| $\prod_i, \sum_i$      | product and sum over contributor index $i$              |

Four symbols carry more than one meaning across the classical literature this manual follows. They are disambiguated by chapter, and never used in two senses in one equation:

- $\eta$  is the **look (off-nadir) angle** everywhere except the performance chapter's NEI/NEP/D\* section, where the detector literature's  $\eta$  (quantum efficiency) and  $\eta_{sys}$  (end-to-end photon-to-electron efficiency) are kept; that section defines both at the point of use.
- $\varepsilon$  is **emissivity** in the radiometric, atmosphere, and calibration chapters. The geometry chapter's grazing/elevation angle is written  $\varepsilon_{el}$  here to keep the two apart.
- $\epsilon$  is the **central-obscuratio ratio** in the spatial chapter only.
- $\nu$  is **spatial frequency** [cy/m] throughout the spatial and performance chapters; spectral wavenumber, which the atmospheric literature also writes  $\nu$ , appears only as  $\nu_{cm}$  [cm<sup>-1</sup>] at MODTRAN interfaces. Angular spatial frequency is written  $f_a$  [cy/rad] in the spatial chapter's turbulence section, where the classical results are stated that way.

Two chapters keep a legacy spelling that predates this table and is retained rather than churned: the radiometric-chain chapter writes the slant range as  $R$  (here and in the geometry chapter it is  $R_s$ , with  $R$  reserved for the mean Earth radius), and it writes spectral power as  $\Phi(\lambda)$  [W/μm].

## 0.10 Acronyms

| Acronym | Expansion                                            |
|---------|------------------------------------------------------|
| ADC     | analog-to-digital converter                          |
| BLIP    | background-limited infrared performance              |
| BRDF    | bidirectional reflectance distribution function      |
| CDS     | correlated double sampling                           |
| CSNR    | contrast signal-to-noise ratio                       |
| DN      | digital number                                       |
| DROIC   | digital readout integrated circuit (photon-counting) |
| DSNU    | dark-signal non-uniformity                           |
| EE      | ensquared (or encircled) energy                      |
| ERF     | edge response function                               |
| FPA     | focal plane array                                    |
| FPN     | fixed-pattern noise                                  |
| FWHM    | full width at half maximum                           |
| GIQE    | General Image Quality Equation                       |
| GSD     | ground sample distance                               |
| IFOV    | instantaneous field of view                          |
| IPC     | interpixel capacitance                               |
| LOS     | line of sight                                        |
| LSF     | line spread function                                 |
| MRT     | minimum resolvable temperature difference            |
| MTF     | modulation transfer function                         |
| NEDL    | noise-equivalent differential radiance               |
| NEDT    | noise-equivalent differential temperature            |
| NEI     | noise-equivalent irradiance                          |
| NEP     | noise-equivalent power                               |
| NIIRS   | National Imagery Interpretability Rating Scale       |
| NUC     | non-uniformity correction                            |
| OTF     | optical transfer function                            |
| PRNU    | photo-response non-uniformity                        |
| PSF     | point spread function                                |
| QE      | quantum efficiency                                   |
| RER     | relative edge response                               |
| ROIC    | readout integrated circuit                           |
| RSS     | root-sum-square                                      |
| SCNR    | signal-to-clutter-plus-noise ratio                   |
| SNR     | signal-to-noise ratio                                |
| TDI     | time delay and integration                           |
| TOA     | top of atmosphere                                    |
| WFE     | wavefront error                                      |

# Introduction

RADIANT is a first-principles performance model for electro-optical sensors, space-based and airborne, from the ultraviolet through the long-wave infrared. It answers questions of the form *what will this sensor, looking at that scene through this atmosphere, actually measure* — and it answers them by carrying a physical quantity, in canonical units, from the emitting surface to a photoelectron count and then to a performance metric, with no unexplained factor anywhere in between.

This chapter states what the chain is, what selects the radiometric regime a target falls into, and where the model’s boundaries lie. Every equation named here is developed in a later chapter; the purpose of this one is that a reader always knows which chapter to open.

## 1.1 The chain at a glance

RADIANT models the end-to-end signal chain as ten sequential stages. Each stage is a pure function of an immutable chain state: it consumes what earlier stages published, adds its own radiometric frames, noise terms, bias terms, and transfer-function contributions, and returns a new state. No stage reads a file, calls another stage, or edits what an earlier stage published.

| # | Stage                | What it establishes                                                                                                                                                                                                                                  | Detailed in  |
|---|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 0 | Geometry             | Slant range $R_s$ [m], target-referenced zenith angle $\theta_o$ [rad], solar geometry, ground sampling, motion rates                                                                                                                                | Ch. 2        |
| 1 | Source               | Target and background spectral radiance $L_{\text{target}}(\lambda)$ , $L_{\text{bg}}(\lambda)$ [W/m <sup>2</sup> /sr/μm]; tentative regime                                                                                                          | Ch. 3        |
| 2 | Atmosphere           | Transmittance $\tau_{\text{atm}}(\lambda)$ , path radiance $L_{\text{path}}(\lambda)$ , downwelling $L_{\text{sky}}(\lambda)$ , and the at-aperture radiance they produce                                                                            | Ch. 3, Ch. 4 |
| 3 | Optics               | Aperture area $A_{\text{ap}}$ [m <sup>2</sup> ], pixel solid angle $\Omega_{\text{pix}}$ [sr], throughput $\tau_{\text{opt}}(\lambda)$ , warm-optics self-emission, the diffraction-plus-aberration PSF and optical MTF, and the <b>final</b> regime | Ch. 3, Ch. 5 |
| 4 | Platform             | Jitter and smear degradation of the PSF and the MTF budget; $EE_{\text{box}}$ from the fully degraded PSF                                                                                                                                            | Ch. 5        |
| 5 | Spectral integration | The single spectral-to-scalar collapse: spectra in, photoelectrons out                                                                                                                                                                               | Ch. 3        |
| 6 | Detector             | Quantum efficiency, dark current, the noise budget’s detector terms, detector-plane MTF kernels                                                                                                                                                      | Ch. 5, Ch. 6 |
| 7 | Readout              | TDI, binning, co-adding, gain, ADC — signal scaling and the readout noise terms                                                                                                                                                                      | Ch. 6        |
| 8 | Calibration          | Post-NUC residual fixed-pattern noise, drift, and the radiometric bias budget                                                                                                                                                                        | Ch. 7        |
| 9 | Performance          | SNR, NEDT, NIIRS, system MTF, detection range, and the rest of the metric layer                                                                                                                                                                      | Ch. 8        |

### 1.1.1 Geometry first

Geometry runs before radiometry, not alongside it. The reason is that almost every radiometric quantity downstream is conditioned on a geometric one: the atmospheric slant column depends on the path zenith angle, the solar reflected term depends on the solar zenith angle, the sub-pixel fill fraction depends on the target's angular extent at the slant range, and the smear kernel depends on the image-plane motion rate. Deriving those inside the stages that consume them would mean deriving several of them more than once, which is how two parts of a model come to disagree about the same angle. The geometry stage solves the viewing triangle once, publishes  $R_s$ ,  $\theta_o$ ,  $\eta$ , and the sampling and motion quantities, and every later stage reads them.

### 1.1.2 The radiometric spine

Stages 1 through 5 form the radiometric spine. The source stage builds spectral radiance from Planck emission, graybody emissivity, reflected solar irradiance through a BRDF, or a user-supplied spectrum. The atmosphere stage attenuates it and adds what the atmosphere itself contributes along the line of sight:

$$L_{\text{ap}}(\lambda) = L_{\text{target}}(\lambda) \tau_{\text{atm}}(\lambda) + L_{\text{path}}(\lambda)$$

The optics stage supplies the collecting area, the solid angle the pixel subtends, the train's transmittance, and — in the thermal infrared — the train's own emission, whose emissivity is never an independent input but is always derived from the element's reflectance and transmittance by Kirchhoff's law. The spectral-integration stage then performs the one collapse the whole model is allowed:

$$S = \int_{\lambda_1}^{\lambda_2} L_{\text{ap}}(\lambda) \tau_{\text{opt}}(\lambda) A_{\text{ap}} \Omega_{\text{pix}} \text{QE}(\lambda) \frac{\lambda_m}{hc} t_{\text{int}} d\lambda \quad [\text{e-}]$$

Before that integral every radiometric quantity is a spectrum; after it every quantity is a per-pixel scalar in electrons. There is no second collapse and no stage that partially collapses one.

### 1.1.3 The spatial spine

In parallel with the radiometric spine, the model carries a spatial description of the same system along two paths that must agree. The **PSF path** starts from the complex pupil, forms the point spread function, and convolves onto it every spatial degradation in turn — detector aperture, charge diffusion, interpixel capacitance, jitter, smear, turbulence. Ensquared energy, relative edge response, full width at half maximum, Strehl ratio, and the line and edge responses are computed from that one object. The **MTF product path** computes the optical transfer function directly from the autocorrelation of the same pupil and multiplies it by an analytic or kernel-derived MTF for each downstream contributor:

$$\text{MTF}_{\text{sys}}(\nu) = \prod_i \text{MTF}_i(\nu)$$

MTF budgets, MTF at Nyquist, folded MTF, and the image-quality equation consume this path. Because both paths originate in the same pupil, the Fourier transform of the convolved PSF must reproduce the MTF

product, and the model checks that on every run that computes the spatial path. A failure means a degradation was added to one path and not the other. The two paths and the check are the subject of Chapter 5.

### 1.1.4 Terms, not transformations

Two stages — platform and calibration — deliberately transform nothing. They contribute *terms*: a kernel and an MTF factor in the platform’s case, noise terms and bias terms in the calibration’s. The calibration stage’s position is itself load-bearing physics. Readout applies the  $\sqrt{N}$  averaging of TDI and co-adding to the temporal noise terms; calibration runs after, so its residuals are structurally exempt from that averaging, which is exactly right because correlated errors do not average down.

## 1.2 The three radiometric regimes

How much of a target’s flux reaches one pixel depends on whether the target fills the pixel, underfills it, or is small enough that the point spread function — not the target — sets the energy distribution. RADIANT classifies every scene into one of three regimes, and the classification changes the signal equation, not merely a label.

| Regime       | Physical meaning                                                   | $EE_{\text{box}}$ applied?                   |
|--------------|--------------------------------------------------------------------|----------------------------------------------|
| Extended     | Target fills the pixel footprint completely                        | No — the pixel sees a uniform radiance field |
| Sub-pixel    | Target smaller than the pixel but larger than the diffraction blur | Yes, to the target term only                 |
| Point source | Target smaller than the diffraction limit                          | Yes, to the target term                      |

**What selects the regime.** The source stage computes the target’s angular extent  $\theta = \sqrt{A_t}/R_s$  [rad] and compares it with the pixel instantaneous field of view  $\text{IFOV} = p/f$  [rad]. A declared fill fraction below 1.0 forces the sub-pixel regime; otherwise an explicit override is honored if present; otherwise  $\theta \geq 2 \text{ IFOV}$  gives extended,  $\theta \leq 0.25 \text{ IFOV}$  gives point source, and anything between is sub-pixel. That classification is *tentative*. The optics stage, which is the first stage that knows the actual PSF size, finalizes it. Every downstream stage reads the final classification and no stage re-derives it.

**Where the ensquared energy enters.**  $EE_{\text{box}}$  — the fraction of the degraded PSF’s energy falling inside one pixel — is computed in the platform stage from the *fully* degraded PSF, so jitter, smear, and turbulence are already in it, and it is applied in the spectral-integration stage and nowhere else. It multiplies the target term in the sub-pixel and point-source regimes, never the background term, and never anything in the extended regime, where the pixel is filled by construction and the energy that leaves it is replaced by energy from the neighboring scene.

Getting the regime wrong is the most common cause of an SNR that looks inexplicable: an extended scene treated as a point source loses the factor  $EE_{\text{box}} < 1$  for no physical reason, and a point source treated as extended assumes a uniformly filled pixel that does not exist.

### 1.3 What the model predicts

The metric layer turns the final chain state into the quantities an analyst reports. They fall into five families, each of which can be enabled or disabled per run:

- **Radiometric** — SNR, contrast SNR, signal-to-clutter-plus-noise ratio, NEDT [K], radiometric accuracy [% and K], detection range [m], minimum resolvable temperature at Nyquist [K]. The classical sensitivity relatives — noise-equivalent irradiance, power, specific detectivity, noise-equivalent radiance and reflectance, and the temperature retrieval with its Jacobians — are developed in Chapter 8 as computed relatives of the same chain state.
- **Spatial and MTF** — system MTF at Nyquist, folded MTF, aliased fraction, Strehl ratio, relative edge response, ensquared energy in  $1\times 1$  and  $3\times 3$  pixel boxes, PSF full width at half maximum [m], per-contributor MTF budgets.
- **Sampling** — ground sample distance [m] cross-track, along-track, and geometric mean; target-plane sample distance [m]; the sampling parameter  $Q$ ; swath width [m]; ground range [m]; access rate [ $\text{m}^2/\text{s}$ ]; diffraction limit in  $\mu\text{rad}$  and in meters.
- **Saturation and dynamic range** — well margin [dB], ADC margin [dB], dynamic range [dB], maximum integration time before saturation [s].
- **Interpretability** — NIIRS via the General Image Quality Equation, with an explicit flag when an input falls outside the equation's calibration envelope.

Every metric is reported with its units, and a metric whose inputs do not support it is reported as a named failure rather than as a silent NaN.

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### 1.4 Model scope — what is deliberately absent

A performance model earns trust by being explicit about its edges. The following are not modeled, and a result that depends on one of them is outside the model's validity rather than merely imprecise.

**Scene and source.** No molecular band spectroscopy or plume chemistry; no fluorescence; no scene-level interreflection or ray tracing; no temporal scene variability within or between frames; no lunar illumination and no sun-glint geometry. Target emission is Lambertian, and the reflectance structure available is the Lambertian and normalized-Phong BRDF pair rather than a microfacet model or measured BRDF tables.

**Atmosphere.** No refraction, and therefore no bending of the geometric path; no scintillation; no adjacency effect; no polarization; no cloud or precipitation attenuation. The simple parametric model and the library-backed models are two families with one contract, and their limits are the subject of Chapter 4.

**Optics.** No ghost images, no bidirectional scatter distribution function beyond the scalar-roughness halo, no chromatic aberration, no field-dependent vignetting map. Aberrations enter as wavefront error in the pupil — either a bulk RMS value or a Zernike set — and their interaction with diffraction is handled in the pupil, never as a separate multiplicative MTF factor.

**Detector and readout.** No optical crosstalk between pixels, no charge-trapping model, no ADC integral or differential nonlinearity, no electronics-level coupling or supply noise, no cosmic-ray transient or bad-pixel statistics. Blooming, charge-transfer efficiency in CCDs, and radiation-damage accumulation are outside the current model.

**Spatial.** No geometric distortion, no frame-to-frame registration error, no whiskbroom scan-mechanism transfer function, and no short-exposure turbulence branch — turbulence enters as the long-exposure Kolmogorov result, from a Fried parameter entered directly or integrated from a  $C_n^2$  profile (Chapter 4), and is omitted entirely when that parameter resolves to zero or negligible.

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## 1.5 How to read this manual

Chapter 2 establishes the viewing geometry and sampling on which everything else is conditioned. Chapter 3 is the radiometric core: the governing equations from Planck’s law to the photoelectron count, each with its validity limits, its classic implementation pitfalls, and a pinned numeric anchor. Chapter 4 covers the two atmosphere model families. Chapter 5 is the spatial model, PSF and MTF together. Chapter 6 is the noise taxonomy and its acquisition scaling. Chapter 7 is the calibration error model — what survives a non-uniformity correction, and how it sets a floor that integration time cannot lower. Chapter 8 is the metric layer. The appendix treats the mixed refractive-reflective optical train in full, and the front matter holds the notation on which all of it depends.

Throughout, an equation is stated in the form the model actually evaluates. Where the implementation makes an approximation, the approximation is named where the equation is given, not deferred to a limitations list.

Every quantitative claim in the source documents behind this manual is anchored to the implementation function that evaluates it and to a committed test that pins its value; the repository sources carry those anchors as the project’s traceability record. The typeset manual omits the pointers and keeps the physics.

# Geometry & Sampling

Spherical-Earth viewing geometry, orbit kinematics, ground sampling, smear kinematics, solar geometry, and attitude conventions as implemented in RADIANT, which resolves the scene geometry before any radiometry. Numeric anchors were re-derived from the literature independently of the code and then checked against it. RADIANT uses the IUGG **mean Earth radius**  $R = 6371.0$  **km**, so the anchors below are quoted for that radius: an  $\sim 8$  km radius difference — the gap to the WGS-84 equatorial value — moves a 500 km/30° slant range by  $\sim 9$  m and the orbit period by  $\sim 9$  s, so always match radii before comparing numbers.

**Symbols and the code’s naming.** Classical texts parameterize the viewing triangle by the look angle at the satellite; RADIANT is  $\theta_o$ -referenced:

| Classical symbol                                | Meaning                               | RADIANT name                                        |
|-------------------------------------------------|---------------------------------------|-----------------------------------------------------|
| $\eta$                                          | look/off-nadir angle at the sensor    | <code>eta(viewing_triangle.eta_from_theta_o)</code> |
| $\theta_o$ ( $= \theta_i$ for a surface target) | LOS zenith angle at the <b>target</b> | <code>theta_o, path_zenith_rad</code>               |
| $\varepsilon = \pi/2 - \theta_o$                | grazing/elevation angle               | <code>elevation_angle_rad</code>                    |
| $\Lambda$                                       | Earth central angle                   | via ground range $R\Lambda$                         |
| $R_s$                                           | slant range                           | <code>slant_range_m</code>                          |
| $h$                                             | sensor altitude                       | <code>geometry.sensor_altitude_m</code>             |

## 2.1 The viewing triangle

**Equations.** Triangle Earth-center–sensor–target with sides  $R$ ,  $R + h$ ,  $R_s$ . Angle closure  $\Lambda = \theta_o - \eta$  and the law of sines give the load-bearing identity

$$\sin \theta_o = \frac{R + h}{R} \sin \eta$$

— note the **amplifying** factor  $(R + h)/R$ : incidence at the target always *exceeds* the look angle on a sphere ( $R/(R + h)$  belongs only in the  $\varepsilon \rightarrow \eta$  direction). Closed-form solution sets exist from any one of  $\{\eta, \theta_o, \Lambda\}$  plus  $h$ ; the  $\Lambda$ -parameterized form uses the branch-safe  $\eta = \text{atan2}(R \sin \Lambda, (R + h) - R \cos \Lambda)$ .

**Assumptions & validity.** Spherical Earth, surface (or specified-altitude) target, no refraction (refraction lifts apparent elevation up to  $\sim 0.5^\circ$  at  $\varepsilon \approx 0$ ).

**Pitfalls.** The flat-Earth intuition  $\theta_o = \eta$  (wrong by  $2.6^\circ$  at  $h = 500$  km,  $\eta = 30^\circ$  and unbounded toward the limb);  $\Lambda = \eta - \theta_o$  sign flips; degrees fed to radian trig.

**Numeric anchor.**  $h = 500$  km,  $\eta = 30^\circ$ :  $\theta_o = 32.6319^\circ$ ,  $\Lambda = 2.6319^\circ$  (mean-R).

## 2.2 Slant range

**Equation.** Direct form in the look angle:

$$R_s = (R + h) \cos \eta - \sqrt{R^2 - (R + h)^2 \sin^2 \eta}.$$

The **minus** root is the near (visible-surface) intersection; the plus root exits the far side of the Earth. A negative discriminant means the LOS misses the Earth —  $\sin \eta > R/(R+h)$ , beyond the limb; the boundary is  $\eta_{max} = \arcsin \frac{R}{R+h}$  with tangent-ray range  $\sqrt{2Rh + h^2}$ .

**Pitfalls.** Plus-root selection (returns ~11 Mm at nadir instead of  $h$ ); silent NaN on a beyond-horizon geometry instead of an actionable error; near  $\eta_{max}$ ,  $dR_s/d\eta \rightarrow \infty$  — solvers should re-parameterize in  $\Lambda$  or  $\varepsilon$  there.

**Numeric anchors.**  $h = 500$  km,  $\eta = 30^\circ$ :  $R_s = 585,110.5$  m (mean-R). Nadir:  $R_s = h$  exactly.  $\eta_{max}(500 \text{ km}) = 68.01^\circ$  (mean-R).

## 2.3 Circular-orbit kinematics

**Equations.**

$$v = \sqrt{\frac{\mu}{R+h}}, \quad T = 2\pi \sqrt{\frac{(R+h)^3}{\mu}}, \quad v_g = v \frac{R}{R+h}.$$

The  $R/(R+h)$  factor: satellite and nadir point share one angular rate  $\omega$ ; linear speed is  $\omega$  times each circle's radius. It is a projection of angular motion, not a velocity-vector projection.

**Assumptions & validity.** Two-body, circular, no  $J_2$  (~0.1 % LEO period effect); non-rotating Earth for  $v_g$  — Earth rotation adds up to  $\pm 465 \cos(\text{lat})$  m/s (~6.6 % equatorial) to the true relative ground speed, unmodeled.

**Pitfalls.**  $\sqrt{\mu/R}$  instead of  $\sqrt{\mu/(R+h)}$  (3.8 % at 500 km); **orbital**  $v$  **where ground**  $v_g$  **belongs** in smear/line-rate math — a +7.8 % error at 500 km that looks plausible;  $\mu$  in km<sup>3</sup>/s<sup>2</sup> mixed with meters.

**Numeric anchors** (mean-R):  $v = 7616.6$  m/s,  $T = 94.469$  min at  $h = 500$  km.

## 2.4 GSD — nadir and off-nadir

**Equations.** IFOV =  $p/f$ ; nadir GSD =  $(p/f) h$ . Off-nadir, the beam width IFOV  $\cdot R_s$  (measured  $\perp$  to the LOS) projects onto the local tangent plane; the LOS makes angle  $\theta_o$  (**incidence at the target, not the look angle**  $\eta$ ) with the surface normal:

$$\text{GSD}_{\text{in-plane}} = \frac{\text{IFOV} \cdot R_s}{\cos \theta_o}, \quad \text{GSD}_{\perp} = \text{IFOV} \cdot R_s.$$

The  $1/\cos\theta_o$  applies **only** in the tilt plane (the plane containing the LOS and the local vertical); the orthogonal horizontal direction is itself  $\perp$  to the LOS and projects length-preserving. In RADIANT's convention the LOS tilt is carried by `geometry.path_zenith_rad` in the **along-track** plane, so `along_track_m` receives the  $1/\cos\theta_o$  elongation and `cross_track_m` the range-only factor. The formula is exact on the sphere: differentiating the viewing triangle gives  $R d\Lambda/d\eta = R_s/\cos\theta_o$  identically.

**Pitfalls.**  $\cos\eta$  for  $\cos\theta_o$  (2.75 % at the anchor geometry, unbounded toward the limb);  $\cos$  vs  $1/\cos$  (GSD must *grow* off-nadir); applying the obliquity factor to both directions (a further  $\sim 19$  % area error at the anchor); quoting one off-nadir “GSD” without naming the direction.

**Numeric anchors** (mean-R,  $p = 10\ \mu\text{m}$ ,  $f = 2\ \text{m}$ ,  $h = 500\ \text{km}$ ): nadir 2.500000 m;  $\eta = 30^\circ$ : in-plane 3.473901 m — the wrong-angle value 3.378 m is explicitly rejected by the anchor test.

## 2.5 Swath and access

**Equations.** Swath from the cross-track GSD and detector format:  $W = \text{GSD}_{\text{cross}} \cdot N_{\text{pix}}$ ; area access rate  $\dot{A} = W v_g$ ; ground range from the central angle,  $R\Lambda$ . Wide-FOV swath uses the full spherical mapping  $\Lambda(\eta)$  — the flat-Earth  $2h \tan \eta_{\text{half}}$  is 0.3 % low at  $\pm 15^\circ$  and diverges beyond  $\sim 30^\circ$ .

## 2.6 Smear kinematics and TDI line-rate matching

**Equations.** Uncompensated nadir pushbroom: ground motion  $v_g t_{\text{int}}$  maps to the focal plane through the magnification  $f/R_s$ :

$$d_{\text{img}} = v_g t_{\text{int}} \frac{f}{R_s} \quad (= v_g t_{\text{int}} f/h \text{ at nadir}).$$

TDI requires the line clock to match the image velocity:  $p f_{\text{line}} = v_{\text{img}} \iff t_{\text{line}} = \text{GSD}/v_g$ ; residual per-stage mismatch smear multiplies by  $N$  stages. The MTF consequence of  $d_{\text{img}}$  is the smear sinc of the spatial chapter's *Platform kernels: jitter and smear*.

**Pitfalls.** Orbital  $v$  for  $v_g$  (+7.8 %);  $f/R$  or  $f/(R+h)$  for the magnification; ground meters compared to focal-plane microns without  $f/R_s$ ; conflating  $t_{\text{int}}$  with  $t_{\text{line}}$  in TDI ( $t_{\text{int}} = N t_{\text{line}}$ ; the matching condition constrains  $t_{\text{line}}$ ).

**Dwell-time feasibility guard.** The full scan subsystem (line-rate derivation, cross-track and target-motion smear) is not implemented; what is implemented is the one guard whose absence would let an unphysical TDI timing report an authoritative SNR. The ground advances one along-track sample in the dwell time, and each TDI stage must integrate within it:

$$t_{\text{dwell}} = \frac{\text{GSD}_{\text{along}}}{v_g}, \quad t_{\text{int}} \leq t_{\text{dwell}}, \quad \text{smear} = \frac{t_{\text{int}}}{t_{\text{dwell}}} [\text{pixels}].$$

The chain publishes `max_integration_time_s` ( $= t_{dwell}$ ), `smear_pixels` and a `feasible` flag; a longer integration smears the target across more than one pixel per stage and breaks TDI registration.

**Numeric anchor.**  $h = 500$  km,  $p = 10$   $\mu$ m,  $f = 2$  m,  $t_{int} = 1$  ms:  $d_{img} \approx 2.8$  pixels — uncompensated millisecond integration is not viable; the matched line time is  $\sim 354$   $\mu$ s, which is also the guard's `max_integration_time_s` ( $2.5$  m /  $7.06$  km/s), so the same configuration reports `smear_pixels`  $\approx 2.8$  and `feasible` = False.

## 2.7 Solar geometry

**Equation.** Spherical law of cosines on the astronomical triangle:

$$\cos \theta_z = \sin \phi \sin \delta + \cos \phi \cos \delta \cos H$$

( $\phi$  latitude,  $\delta$  declination,  $H$  hour angle from solar noon). Declination comes from the day of year, and the hour angle from either the local solar time or the local time of the ascending node.

**Pitfalls.** Clock time vs apparent solar time (equation of time,  $\pm 4^\circ$  in  $H$ ); elevation returned where zenith is expected (downstream  $\cos \theta_z$  irradiance factors get sin/cos swapped); sign conventions on  $\delta$  across hemispheres.

**Numeric anchor.**  $\phi = 35^\circ$ N,  $\delta = 23.44^\circ$ ,  $H = 15^\circ$ :  $\theta_z = 17.4256^\circ$ . Noon collapses to  $\theta_z = \phi - \delta$  exactly.

## 2.8 Attitude: Euler ZYX

**Equation.** Intrinsic  $z$ - $y'$ - $x''$  (yaw  $\psi \rightarrow$  pitch  $\theta \rightarrow$  roll  $\varphi$ ), active, right-handed, column vectors:

$$\mathbf{R} = \mathbf{R}_z(\psi) \mathbf{R}_y(\theta) \mathbf{R}_x(\varphi),$$

gimbal lock at  $\theta = \pm 90^\circ$ . This matches `scipy Rotation.from_euler('ZYX', ...)` exactly; extraction via  $\theta = -\arcsin R_{31}$ ,  $\psi = \text{atan2}(R_{21}, R_{11})$ ,  $\varphi = \text{atan2}(R_{32}, R_{33})$ .

**Pitfalls.** Intrinsic  $z$ - $y'$ - $x''$  equals *extrinsic*  $x$ - $y$ - $z$  (same product) but not intrinsic  $x$ - $y'$ - $z''$ ; active/passive transposes; the  $+\sin \theta$  corner of  $\mathbf{R}_y$  sits opposite to  $\mathbf{R}_z/\mathbf{R}_x$ .

## 2.9 Sampling on the ground

Focal-plane Nyquist  $1/(2p)$  projects to ground Nyquist  $1/(2 \text{ GSD})$  — direction-dependent off-nadir (use the direction's GSD from *GSD — nadir and off-nadir* above). The optics-vs-sampling budget ( $Q = \lambda F_\# / p$ , aliasing, folded MTF) lives in the spatial chapter's *Sampling: Nyquist, Q, and folded MTF*; the geometry chapter's contribution is the GSD that scales it to the ground.

**Target-plane sample distance.** An air or space target has no ground plane to project onto. The reference there is the plane through the target *normal to the line of sight*, and the sample distance in it is the pixel’s angular subtense at the slant range:

$$d_x = \frac{p_x R_s}{f}, \quad d_y = \frac{p_y R_s}{f}, \quad d = \sqrt{d_x d_y}$$

— GSD without the  $1/\cos$  projection (the chord/arc distinction is below any modelling fidelity:  $\tan(\text{IFOV})/\text{IFOV} - 1 < 3 \times 10^{-7}$  for  $\text{IFOV} \lesssim 1$  mrad). There is deliberately no target-orientation term: the metric answers how far apart two adjacent samples are *where the target is*, an optics-and-range question with one answer, not how much of the target’s skin a pixel covers, which needs an attitude the framework does not carry. The geometric mean is defined the way GIQE-5 averages GSD, so the two families are directly comparable and meet at zero incidence when a target crosses the air/ground scene-class boundary. **Numeric anchor.**  $p = 10 \mu\text{m}$ ,  $f = 2 \text{ m}$ ,  $R_s = 500 \text{ km}$ : 2.5 m per axis. **In RADIANT.** `performance/target_plane_sample_distance.py::target_plane_sample_distance` · anchored by `performance/tests/test_target_plane_sample_distance.py`.

## 2.10 Input modes and scene classes

**Input modes.** The scene is expressed in exactly one mode per family, and the mode is detected by *provenance*: a parameter left at its default was not provided, so there is no mode switch to set. The **viewing** family resolves to  $\theta_o$ : V0 direct range (`geometry.target_range_m`; on a level path with no angle entered the range is the chord that builds the triangle,  $\varphi = 2 \arcsin(d/2r)$ ,  $\theta_o = \pi/2 + \varphi/2$ ), V1 path zenith at the lower endpoint (`path_zenith_rad`), V2 off-boresight angle at the sensor (`sensor_off_boresight_rad`, its reference axis nadir when the sensor is the upper endpoint and zenith when it is the lower one), V3 ground range (`ground_range_m`, direction-free: the surface arc fixes  $\Lambda = \arccos(R/r)$  whichever endpoint is higher), V4 elevation above the horizontal at the lower endpoint (`elevation_angle_rad`, signed:  $\zeta_{low} = \pi/2 - \varepsilon$ ), and V6 circular orbit (ground speed and period from the sensor altitude). Every entered viewing angle is referenced to the path’s **lower** endpoint — exactly what it always meant when the sensor was above the target; when the sensor is the lower endpoint,  $\theta_o = \pi - \zeta_{up}$  is derived and the published mode label says so. The **solar** family resolves to  $\theta_s, \Delta\phi$ : S0 night (thermal-only), S1 direct zenith, S2 elevation, S3 site and time (latitude, day of year, local solar time or LTAN  $\rightarrow$  declination and hour angle). The **LOS-rate** family resolves to  $\omega$ : K0 platform-only  $\omega = v_g/R_s$  (the default), K1 direct rate, K2 target velocity  $\omega = |\mathbf{v}_{rel,\perp}|/R_s$  with  $\mathbf{v}_{rel} = \mathbf{v}_{target} - \mathbf{v}_{sensor}$ .

**Resolution rules.** Redundant entries for one canonical quantity must agree within 1 % (1  $\mu\text{rad}$  absolute floor for angles) or the stage raises, naming every entry and the value it implies; every derived quantity is published with its mode label; no entries at all gives the documented defaults (nadir view, 0.5 rad solar zenith by day) — never a silent NaN.

**LOS direction and scene class.** Up, down or level follows from the two altitudes and  $\theta_o$  — never a user switch. The canonical  $\theta_o$  lives on  $[0, \pi)$ : below  $\pi/2$  the sensor is above the target’s horizon plane (every classic scene), above it the sensor is below — the same Earth-centre / target / sensor triangle read from the other vertex. The scene class is the observer  $\times$  target band: **ground** ( $h < 1 \text{ km}$ ), **air** ( $1 \leq h \leq 100 \text{ km}$ ) and **space** ( $h > 100 \text{ km}$ , the atmosphere-top convention), both boundaries closed from below. Physics never branches on the class — a scene at 999 m and one at 1001 m compute identically — it drives defaults,

metric relevance, validation and the GUI composition only. `geometry.scene_class` is an optional assertion cross-checked against the derivation: what it catches is a wrong-magnitude altitude (600 m typed for 600 km), which pure derivation would render as a self-consistent scene of the wrong class. The **horizon guard** refuses  $|\theta_o - \pi/2| < 0.5^\circ$  (refraction, unmodelled, dominates there) and computes between  $0.5^\circ$  and  $\sim 2^\circ$  with a warning. `geometry.site_elevation_m` is a *third* altitude — the terrain under the line of sight — not an input mode; the Hufnagel-Valley surface term (atmosphere chapter) is its consumer, and the terrain-bearing endpoint must sit at or above it.

**Pitfalls.** Entering the sensor’s own zenith as V1 for an up-looking scene and expecting it to be  $\theta_o$  (V1 is the lower endpoint’s zenith, so it *is* the sensor’s there and  $\theta_o = \pi - \zeta$  follows); ground range typed where slant range was meant; two doors set, one of them stale, which the 1 % rule reports rather than resolves.

**Numeric anchors.** V4 with  $\varepsilon = -10^\circ$  at the lower endpoint:  $\zeta_{low} = 100^\circ$ . A level chord  $d = 200$  km at  $r = R + 5$  km:  $\varphi = 2 \arcsin(100/6376) = 1.7973^\circ$ ,  $\theta_o = 90.8987^\circ$ .

# Radiometric Signal Chain

The radiometry foundations of RADIANT: the governing equations that carry a photon from source emission to a photoelectron count, each stated with its derivation, validity limits, classic implementation pitfalls, and a pinned numeric anchor. This chapter covers the radiometric quantities themselves; the spatial (PSF/MTF) treatment is the Spatial Model chapter and the noise taxonomy is the Noise Model chapter.

Every numeric anchor below was independently re-derived from the physics literature, without reference to the implementation, and then verified against the implementation.

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## 3.1 Notation

Symbols and canonical units are defined once for the whole manual in the front matter, Notation and Symbols; this chapter uses them without redefinition. RADIANT canonical units throughout: wavelength in  $\mu\text{m}$ , angles in radians, lengths in meters, spectral radiance in  $\text{W}/\text{m}^2/\text{sr}/\mu\text{m}$ .

Where a formula needs  $\lambda$  in meters (anything with  $h$ ,  $c$ ,  $k_B$  in it), we write  $\lambda_m$  explicitly. The  $\mu\text{m}$ -to-m bookkeeping is the single most common bug source in this entire subject; every entry below calls out where the  $10^{-6}$  lives.

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## 3.2 The chain at a glance

RADIANT models the end-to-end signal chain as ten sequential stages, the first of which resolves the scene geometry. Each stage is a pure function that transforms an immutable `ChainState`, adding radiometric frames, noise terms, and MTF contributions:

```
GeometryStage → SourceStage → AtmosphereStage → OpticsStage → PlatformStage  
→ SpectralIntegrationStage → DetectorStage → ReadoutStage → CalibrationStage  
→ PerformanceStage
```

The final `ChainState` contains everything needed to compute performance metrics (SNR, NEDT, NIIRS, MTF). The foundations in this chapter are the radiometric content of that chain: Geometry establishes ranges and angles; Source builds  $L(\lambda)$  from the equations below; Atmosphere applies  $\tau_{\text{atm}}$  and adds path radiance; Optics applies throughput and finalizes the radiometric regime; SpectralIntegration collapses spectra to photoelectrons using the photon-conversion entry. A stage-by-stage map is given in [How the Foundations Feed the Chain](#) at the end.

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### 3.3 Foundations

#### 3.3.1 Planck spectral radiance

**Equation.**

$$B(\lambda, T) = \frac{2hc^2}{\lambda_m^5} \cdot \frac{1}{e^x - 1} \cdot 10^{-6}, \quad x = \frac{hc}{\lambda_m k_B T}$$

The first factor evaluates in W/m<sup>2</sup>/sr/m; the trailing 10<sup>-6</sup> is the Jacobian  $d\lambda_m/d\lambda_{\mu m}$  that converts to the canonical W/m<sup>2</sup>/sr/μm.

**Symbols.**  $\lambda_m$  [m] wavelength in meters;  $T$  [K];  $h$  [J·s];  $c$  [m/s];  $k_B$  [J/K];  $x$  dimensionless; output  $B$  [W/m<sup>2</sup>/sr/μm].

**Derivation.** Planck's law follows from Bose–Einstein statistics of photon modes in a cavity: the mode density per unit volume per unit frequency is  $8\pi\nu^2/c^3$ , each mode carries mean energy  $h\nu/(e^{h\nu/k_B T} - 1)$ , and converting energy density to radiance (multiply by  $c/4\pi$ ) and changing variables from  $\nu$  to  $\lambda$  (Jacobian  $c/\lambda^2$ ) gives the per-meter form  $2hc^2/\lambda_m^5 \cdot (e^x - 1)^{-1}$ . A per-μm spectral interval is 10<sup>-6</sup> of a per-m interval, hence the final factor. The dimensionless argument can equivalently be written  $x = c_2/(\lambda_m T)$  with the second radiation constant  $c_2 = hc/k_B$ .

**Assumptions & validity.** Thermal-equilibrium emission into vacuum (refractive index  $n = 1$ ; in a medium  $B$  scales as  $n^2$ ). Unpolarized, Lambertian (direction-independent) radiance. Valid for all  $\lambda, T > 0$ . Numerically,  $x \gtrsim 700$  overflows  $\exp()$  in float64; RADIANT truncates the Wien tail to exactly zero there (the true value is below float64 underflow, so no physics is lost).  $T = 0$  returns exactly zero radiance.

**Pitfalls.**

- The per-μm 10<sup>-6</sup> applied zero times or twice — results off by 10<sup>6</sup> or 10<sup>12</sup>.
- Using  $\lambda$  in μm inside  $\lambda^5$  without conversion — a 10<sup>30</sup> scale error partially masked by the 10<sup>-6</sup> Jacobian.
- $\exp(x) - 1$  instead of  $\expm1(x)$ : catastrophic cancellation for  $x \ll 1$  (Rayleigh–Jeans regime). RADIANT uses `numpy.expm1`.
- Confusing the radiance constant  $c_{1L} = 2hc^2$  with the exitance constant  $c_1 = 2\pi hc^2$  — a stray factor of  $\pi$ . Radiance  $B$  and exitance  $M = \pi B$  are different quantities.

**Numeric anchor.**  $B(10 \mu\text{m}, 300 \text{ K}) = 9.92403333 \text{ W/m}^2/\text{sr}/\mu\text{m}$  ( $x = 4.796$ , near the LWIR peak of a 300 K blackbody at  $\lambda_{\text{max}} = 2898 \mu\text{m}\cdot\text{K}/300 \text{ K} \approx 9.66 \mu\text{m}$ ). Also  $B(4 \mu\text{m}, 300 \text{ K}) = 0.721976423 \text{ W/m}^2/\text{sr}/\mu\text{m}$ .

**References.** [Planck 1901], [Siegel & Howell §1.6], [CODATA 2018].

#### 3.3.2 Temperature derivative of Planck radiance — the NEDT kernel

**Equation.**

$$\frac{\partial B}{\partial T} = B(\lambda, T) \cdot \frac{x}{T} \cdot \frac{e^x}{e^x - 1} \quad \left[ \text{W/m}^2/\text{sr}/\mu\text{m/K} \right]$$

**Symbols.** As in the Planck entry; output in  $\text{W/m}^2/\text{sr}/\mu\text{m}/\text{K}$ . The fully expanded form is  $\partial B/\partial T = (2h^2c^3 \cdot 10^{-6})/(\lambda_m^6 k_B T^2) \cdot e^x/(e^x - 1)^2$ .

**Derivation.** Write  $B = A/(e^x - 1)$  with  $A = 2hc^2/\lambda_m^5$  fixed in  $T$ . Since  $x = hc/(\lambda_m k_B T)$ ,  $dx/dT = -x/T$ . Chain rule:  $dB/dT = A e^x (e^x - 1)^{-2} \cdot (x/T) = B \cdot (x/T) \cdot e^x/(e^x - 1)$ . This is the kernel of every NEDT computation: a scene temperature change  $\Delta T$  produces a radiance change  $\Delta L \approx (\partial B/\partial T) \Delta T$ .

**Assumptions & validity.** The derivative itself is exact — no linearization. Using it as the NEDT kernel assumes  $\Delta T$  small enough that  $B$  is locally linear: excellent for  $\Delta T \lesssim$  a few K at terrestrial temperatures, degrading in the Wien regime where the fractional error of the linearization grows like  $(x/2)(\Delta T/T)$ . Undefined at  $T = 0$  (RADIANT raises).

#### Pitfalls.

- Dropping the  $e^x/(e^x - 1)$  factor (writing  $\partial B/\partial T = B \cdot x/T$ ) — a Wien-limit shortcut that is  $\sim 0.8\%$  low at  $10\ \mu\text{m}/300\ \text{K}$  and silently regime-dependent.
- Carrying  $(e^x - 1)^1$  where the expanded form needs  $(e^x - 1)^2$ .
- Sign check:  $\partial B/\partial T > 0$  for every  $\lambda, T$ . A negative value anywhere is a bug.
- All the per- $\mu\text{m}$   $10^{-6}$  hazards of the Planck entry, now propagating straight into NEDT.

A physical sanity check worth internalizing: at  $300\ \text{K}$  the MWIR derivative is  $\sim 5.5\times$  smaller than LWIR in absolute per- $\mu\text{m}$  terms, but its *fractional* contrast  $(\partial B/\partial T)/B$  is larger ( $4.00\ \%/K$  at  $4\ \mu\text{m}$  vs  $1.61\ \%/K$  at  $10\ \mu\text{m}$ ) — the classic MWIR-vs-LWIR contrast trade.

**Numeric anchor.**  $\partial B/\partial T(10\ \mu\text{m}, 300\ \text{K}) = 0.159971567\ \text{W/m}^2/\text{sr}/\mu\text{m}/\text{K}$  (audit value, verified against central finite differences to  $\leq 3 \times 10^{-10}$  relative).

**References.** [Planck 1901], [Holst 2008], [CODATA 2018].

### 3.3.3 Band-integrated and band-averaged radiance

#### Equation.

$$L_{\text{band}} = \int_{\lambda_1}^{\lambda_2} L(\lambda) d\lambda \approx \sum_{i=1}^{N-1} \frac{1}{2} (L_i + L_{i+1}) (\lambda_{i+1} - \lambda_i) \quad [\text{W/m}^2/\text{sr}]$$

$$\bar{L} = \frac{L_{\text{band}}}{\lambda_2 - \lambda_1} \quad [\text{W/m}^2/\text{sr}/\mu\text{m}]$$

**Symbols.**  $\lambda_1, \lambda_2$  [ $\mu\text{m}$ ] band edges;  $L_i$  [ $\text{W/m}^2/\text{sr}/\mu\text{m}$ ] samples on the spectral grid; grid spacing in  $\mu\text{m}$ .

**Derivation.** Integrating a per- $\mu\text{m}$  spectral density over a wavelength interval expressed in  $\mu\text{m}$  yields  $\text{W/m}^2/\text{sr}$  with *no* extra conversion factor — the units cancel by construction. The trapezoid rule is the RADIANT quadrature (via `numpy.trapezoid`); its general form handles non-uniform grids correctly because the spacing enters per interval. The band average divides by the full span  $\lambda_2 - \lambda_1$ ; if a relative spectral response  $R(\lambda)$  is involved, the correct band average is instead  $\int LR d\lambda / \int R d\lambda$ .

**Assumptions & validity.** Trapezoid accuracy requires the grid to resolve the integrand: trivially satisfied for a smooth blackbody (201 points across a band gives  $< 10^{-6}$  relative error in LWIR/MWIR), but for

atmospherically filtered radiance the grid must resolve the  $\tau(\lambda)$  line structure or the result is quadrature-limited, not physics-limited.

#### Pitfalls.

- $\Delta\lambda$  in the wrong units (m or nm instead of  $\mu\text{m}$ ) —  $10^{-6}$  or  $10^3$  scale errors on the integral.
- Dividing by the sample count  $N$  instead of the wavelength span when band-averaging.
- Applying the band average *and* multiplying by  $\Delta\lambda$  somewhere downstream — integrating twice.
- Treating the band as a top-hat when a real spectral response weighting exists.
- Coarse-grid bias: even trapezoid on a 5-point grid across 8–12  $\mu\text{m}$  is 0.4 % low; a left-Riemann sum is worse.

**Numeric anchor.**  $\int_8^{12} B(\lambda, 300 \text{ K}) d\lambda = 38.5004239 \text{ W/m}^2/\text{sr}$  (adaptive-quadrature truth; band average 9.62510598 W/m<sup>2</sup>/sr/ $\mu\text{m}$ ). Sanity: that is 26.3 % of the total  $\sigma T^4/\pi = 146.15 \text{ W/m}^2/\text{sr}$ , consistent with blackbody band-fraction tables.

**References.** [Siegel & Howell §1.6], [Holst 2008].

### 3.3.4 Brightness temperature — spectral (monochromatic) inversion

**Equation.** Given spectral radiance  $L$  [W/m<sup>2</sup>/sr/ $\mu\text{m}$ ] at wavelength  $\lambda$ , the brightness temperature is the closed-form inverse of Planck:

$$T_B = \frac{hc}{\lambda_m k_B} \cdot \frac{1}{\ln\left(1 + \frac{2hc^2 \cdot 10^{-6}}{\lambda_m^5 L}\right)} \quad [\text{K}]$$

**Symbols.**  $\lambda_m$  [m];  $L$  [W/m<sup>2</sup>/sr/ $\mu\text{m}$ ]; the prefactor  $hc/(\lambda_m k_B) = c_2/\lambda_m$  [K]; the log argument is dimensionless (both terms in W/m<sup>2</sup>/sr/ $\mu\text{m}$  at that  $\lambda$ ).

**Derivation.** Solve  $L = (2hc^2 \cdot 10^{-6}/\lambda_m^5)/(e^x - 1)$  for  $x$ :  $e^x - 1 = 2hc^2 \cdot 10^{-6}/(\lambda_m^5 L)$ , so  $x = \ln(1 + r)$  with  $r$  the radiance ratio, and  $T_B = hc/(\lambda_m k_B x)$ . Because  $B(\lambda, T)$  is strictly monotonic in  $T$  at fixed  $\lambda$ , the inverse is unique. For a blackbody,  $T_B$  at every wavelength equals the physical temperature;  $\lambda$ -dependence of  $T_B$  diagnoses a graybody or selective emitter.

**Assumptions & validity.** Blackbody spectral shape at that wavelength ( $\varepsilon = 1$ ); a graybody's brightness temperature sits *below* its thermodynamic temperature. Monochromatic:  $L$  must be the spectral radiance at  $\lambda$ , or a band narrow enough that  $L$  is flat across it. Defined for any  $L > 0$ .

#### Pitfalls.

- The  $10^{-6}$  inside the log:  $L$  is per- $\mu\text{m}$ , so the  $2hc^2/\lambda_m^5$  term must be converted to per- $\mu\text{m}$  before forming the ratio — omitting it is a  $10^6$  error in the log argument.
- Dropping the “1+” (Wien approximation) — fine for  $x \gg 1$ , but a hidden ~1 K bias near the band peak (at 10  $\mu\text{m}$ /300 K,  $x \approx 4.8$ ).
- $\log(1 + r)$  evaluated naively for tiny  $r$  (very hot / Rayleigh–Jeans cases) — use  $\log1p$ .
- $\lambda$  in  $\mu\text{m}$  inside  $\lambda^5$  (same as the Planck entry).

**Numeric anchor.**  $T_B(10\ \mu\text{m}, L = 9.0\ \text{W/m}^2/\text{sr}/\mu\text{m}) = 294.054730\ \text{K}$ ; re-forwarding gives  $B(10\ \mu\text{m}, 294.054730\ \text{K}) = 9.00000000\ \text{W/m}^2/\text{sr}/\mu\text{m}$  (round-trip exact to 10 digits).

**References.** [Planck 1901], [NIST ITS-90], [Holst 2008].

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### 3.3.5 Band radiance temperature — band-integrated inversion

**Equation.** Given a measured band radiance  $L_{\text{meas}}$  [ $\text{W/m}^2/\text{sr}$ ] over  $[\lambda_1, \lambda_2]$ , the band radiance temperature  $T_R$  solves

$$F(T_R) \equiv \int_{\lambda_1}^{\lambda_2} B(\lambda, T_R) d\lambda - L_{\text{meas}} = 0$$

No closed form exists; solve numerically (RADIANT uses Brent’s method on a bracketing interval).

**Symbols.**  $L_{\text{meas}}$  [ $\text{W/m}^2/\text{sr}$ ];  $T_R$  [K]; band edges in  $\mu\text{m}$ .

**Derivation.** Existence and uniqueness follow from monotonicity:  $\partial B/\partial T > 0$  for every  $\lambda, T$  (previous entry, strictly positive), so  $F'(T) = \int \partial B/\partial T d\lambda > 0$  —  $F$  is strictly increasing.  $F(T \rightarrow 0^+) = -L_{\text{meas}} < 0$  and  $F(T \rightarrow \infty) \rightarrow +\infty$ , so a strictly increasing continuous function with a sign change has exactly one root. This licenses bracketing solvers with a guaranteed single solution — which is exactly why RADIANT validates the bracket before calling Brent.

**Assumptions & validity.** Blackbody spectral *shape* within the band; a graybody or atmospherically filtered scene yields an *effective*  $T_R$ , not the physical temperature. With a spectral response  $R(\lambda) \geq 0$  the same monotonicity argument holds. The quadrature grid used inside the solver must match the one used to produce  $L_{\text{meas}}$ , or a quadrature mismatch masquerades as a temperature bias (a 0.15 % radiance error is  $\sim 0.1\ \text{K}$  at LWIR).

#### Pitfalls.

- Inverting the *band-averaged* radiance through the *monochromatic* Planck at band center — close for narrow bands, a systematic bias for wide bands (Planck is nonlinear across the band).
- Solver bracket too narrow (e.g. [200, 400] K fails on hot targets). RADIANT brackets [1, 10,000] K and raises an actionable error if  $L_{\text{meas}}$  falls outside the image of the bracket.
- Comparing per- $\mu\text{m}$  to band-integrated units ( $\text{W/m}^2/\text{sr}/\mu\text{m}$  vs  $\text{W/m}^2/\text{sr}$ ) — a  $(\lambda_2 - \lambda_1)$  factor error.
- Confusing tolerance on radiance with tolerance on temperature: at 3–5  $\mu\text{m}$ ,  $dL_{\text{band}}/dT \approx 4\ \%/K$  of  $L_{\text{band}}$ , so a 1 % radiance tolerance is  $\sim 0.25\ \text{K}$ .

**Numeric anchor.** For 8–12  $\mu\text{m}$  and  $L_{\text{meas}} = 30.0\ \text{W/m}^2/\text{sr}$ :  $T_R = 285.450727\ \text{K}$  (re-forward reproduces  $30.0000000\ \text{W/m}^2/\text{sr}$ ). The self-consistency case  $L_{\text{meas}} = 38.5004239\ \text{W/m}^2/\text{sr}$  recovers exactly  $300.000000\ \text{K}$ .

**References.** [Siegel & Howell §1.6], [Press et al. §9.3] (Brent’s method).

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### 3.3.6 Graybody emission (scene targets)

**Equation.**

$$L_{\text{emit}}(\lambda, T) = \varepsilon(\lambda) \cdot B(\lambda, T) \quad \left[ \text{W/m}^2/\text{sr}/\mu\text{m} \right], \quad 0 \leq \varepsilon \leq 1$$

The complete at-surface leaving radiance in the thermal IR adds the reflected downwelling term:

$$L_{\text{leave}}(\lambda) = \varepsilon(\lambda) B(\lambda, T_{\text{surf}}) + [1 - \varepsilon(\lambda)] L_{\text{down}}(\lambda)$$

**Symbols.**  $\varepsilon(\lambda)$  dimensionless emissivity (scalar graybody or spectral);  $T_{\text{surf}}$  [K];  $L_{\text{down}}$  [W/m<sup>2</sup>/sr/μm] downwelling sky radiance.

**Derivation.** Kirchhoff's law in local thermodynamic equilibrium equates spectral directional absorptance and emissivity,  $\alpha(\lambda) = \varepsilon(\lambda)$ . Energy balance for an opaque surface ( $\tau = 0$ ) gives  $\rho + \alpha = 1$ , hence  $\rho = 1 - \varepsilon$ : the emission and sky-reflection terms are complementary. High- $\varepsilon$  surfaces emit; low- $\varepsilon$  surfaces mirror the sky.

For **scene targets and backgrounds**,  $\varepsilon$  is a legitimate *independent* material input — it describes what the material is, and RADIANT accepts it directly (contrast with optical elements, next entry).

**Assumptions & validity.** Local thermodynamic equilibrium (fails only for luminescent/lasing media — not passive-EO scenes). Strictly, Kirchhoff holds per direction:  $\varepsilon(\lambda, \theta, \phi) = \alpha(\lambda, \theta, \phi)$ ; using a hemispherical  $\varepsilon$  for directional radiance assumes a diffuse emitter (water at grazing angles is the classic violator).  $\varepsilon$  is assumed independent of  $T$  over the range of interest. RADIANT validates  $\varepsilon \in [0, 1]$  and refuses to extrapolate a spectral emissivity table outside its wavelength range.

**Pitfalls.**

- Applying  $\rho + \varepsilon = 1$  across bands — pairing a *solar-band*  $\rho$  with a *thermal-band*  $\varepsilon$ . Kirchhoff is spectral; visible albedo says nothing about 10 μm emissivity.
- Forgetting the reflected-downwelling term: for  $\varepsilon = 0.95$  at LWIR it is a ~5 % correction, and omitting it biases retrieved brightness temperatures.
- Allowing  $\varepsilon > 1$  or  $\varepsilon + \rho > 1$  through unvalidated inputs.

**Numeric anchor.**  $\varepsilon = 0.95$  at  $\lambda = 10 \mu\text{m}$ ,  $T = 300 \text{ K}$ :  $L = 0.95 \times 9.92403333 = 9.42783166 \text{ W/m}^2/\text{sr}/\mu\text{m}$ . Reflected-sky term for  $\rho = 0.05$ ,  $L_{\text{down}} = 5.0 \text{ W/m}^2/\text{sr}/\mu\text{m}$ :  $0.250000000 \text{ W/m}^2/\text{sr}/\mu\text{m}$  (2.6 % of the emitted term).

**References.** [Kirchhoff 1860], [Siegel & Howell §3], [Holst 2008].

### 3.3.7 Kirchhoff-derived emissivity of optical elements

**Equation.** For any optical element *inside the sensor*, emissivity is never an input — it is derived:

$$\text{mirrors: } \varepsilon(\lambda) = 1 - R(\lambda) \quad \text{transmissive elements: } \varepsilon(\lambda) = 1 - T(\lambda) - R(\lambda)$$

**Symbols.**  $R(\lambda)$  reflectance,  $T(\lambda)$  transmittance,  $\varepsilon(\lambda)$  emissivity — all dimensionless, all per wavelength, constrained by  $T + R \leq 1$ .

**Derivation.** Same physics as the graybody entry — Kirchhoff plus energy balance — but the *architectural* consequence differs. A scene target's  $\varepsilon$  is a material property the user legitimately knows. An optical element's  $\varepsilon$ ,  $R$ , and  $T$  are locked together by energy conservation: a user who specifies both  $R$  and  $\varepsilon$  for a mirror has over-specified the energy balance, and the two claims can contradict. RADIANT therefore accepts  $R$  (and  $T$  where applicable) and *derives*  $\varepsilon$ ; the derived value then drives the warm-optics self-emission term  $L_{\text{nf}}(\lambda) = \varepsilon_{\text{opt}}(\lambda) B(\lambda, T_{\text{opt}})$  that the optics stage adds to the background.

For a refractive element modeled with the cavity model, the generalized form  $\varepsilon_{\text{eff}} = T_2 n^2 (1 - \beta) / D$  applies ( $n^2$  enhancement for emission inside a dielectric medium); for a simple refractive element with no bulk absorption model,  $\varepsilon = 0$  — the residual  $1 - T$  is predominantly reflection, and assigning it to emission would fabricate thermal background.

**Assumptions & validity.** Same LTE assumptions as Kirchhoff generally. Per-wavelength: the constraint is enforced on the full spectral grid, not on band averages.

#### Pitfalls.

- Accepting  $\varepsilon$  as an independent parameter for an optical surface — an over-specification of the energy balance. RADIANT raises KirchhoffViolationError instead.
- Applying  $\varepsilon = 1 - R$  to a *transparent* element (missing the  $T$  term).
- Letting  $T + R > 1$  slip through unvalidated — RADIANT enforces  $T + R \leq 1$  within tolerance and raises otherwise.
- Assigning the whole  $1 - T$  of a lens to emission when most of it is reflection — over-predicts warm-optics background.

**Numeric anchor.** Gold mirror with  $R = 0.98$  (flat):  $\varepsilon = 1 - 0.98 = 0.02$  (dimensionless), and its self-emission at  $T_{\text{opt}} = 300$  K,  $\lambda = 10$   $\mu\text{m}$  is  $0.02 \times 9.92403333 = 0.198480667$  W/m<sup>2</sup>/sr/ $\mu\text{m}$ .

**References.** [Kirchhoff 1860], [Wolfe & Zissis §5].

### 3.3.8 Top-of-atmosphere solar irradiance

#### Equation.

$$E_{\text{sun}}(\lambda) = \pi B(\lambda, T_{\text{sun}}) \left( \frac{R_{\text{sun}}}{d} \right)^2 k_{\text{scale}} \quad \left[ \text{W/m}^2/\mu\text{m} \right]$$

with distance scaling  $E(d) = E(1 \text{ AU})/d_{\text{AU}}^2$ .

**Symbols.**  $T_{\text{sun}} = 5778$  K effective photospheric temperature;  $R_{\text{sun}} = 6.957 \times 10^8$  m;  $d$  Sun–target distance [m] ( $d_{\text{AU}}$  in AU);  $(R_{\text{sun}}/d)^2 \approx 2.18 \times 10^{-5}$  dimensionless geometric dilution;  $k_{\text{scale}}$  the flat calibration factor that makes  $\int E_{\text{sun}} d\lambda = S_0 = 1361$  W/m<sup>2</sup> exactly.

**Derivation.** The Sun subtends  $\Omega_{\text{sun}} = \pi(R_{\text{sun}}/d)^2$  at the observer (small-angle; the  $\pi$  here is the projected-solid-angle integral of a uniform disk). Irradiance on a normal plane is radiance  $\times$  solid angle:  $E = \Omega_{\text{sun}} B(\lambda, T_{\text{sun}})$ , which regroups into the form above. Because a 5778 K blackbody with IAU nominal

$R_{\text{sun}}/\text{AU}$  integrates to  $\sim 1365 \text{ W/m}^2$  rather than the observed  $S_0 = 1361 \text{ W/m}^2$ , RADIANT applies the flat  $k_{\text{scale}}$  correction: the spectral *shape* stays exactly Planck while the integral recovers  $S_0$  to machine precision. Distance scaling is conservation of intensity:  $E \propto 1/d^2$ ; Earth's seasonal range  $d = 0.9833\text{--}1.0167 \text{ AU}$  gives a  $\pm 3.4 \%$  annual swing.

**Assumptions & validity.** Blackbody spectral shape — real solar spectra deviate at the few-percent level (Fraunhofer lines, UV/VIS deficit). A tabulated `astm_e490` model is planned to replace the shape without breaking the API. Small-angle solid angle is excellent ( $\Omega_{\text{sun}} \approx 6.8 \times 10^{-5} \text{ sr}$ ).

### Pitfalls.

- Confusing irradiance with radiance:  $E_{\text{sun}}$  is  $\text{W/m}^2/\mu\text{m}$ ; the equivalent-Lambertian radiance used in single-scatter path-radiance formulas is  $E_{\text{sun}}/\pi$  [ $\text{W/m}^2/\text{sr}/\mu\text{m}$ ].
- Applying  $1/d^2$  twice (once in the irradiance table, once in code) or not at all.
- Using the solar radius/distance ratio unsquared.

**Numeric anchor.** An uncalibrated 5778 K blackbody Sun gives  $E(0.55 \mu\text{m}) = 1756.84 \text{ W/m}^2/\mu\text{m}$  at 1 AU ( $\Omega_{\text{sun}} = 6.79427 \times 10^{-5} \text{ sr}$ ). The flat calibration multiplies that by  $k_{\text{scale}} = 0.99574$ , so RADIANT's shipped model returns  $1749.35 \text{ W/m}^2/\mu\text{m}$  — within 7 % of the measured  $\approx 1870 \text{ W/m}^2/\mu\text{m}$  [ASTM E490], the real Sun exceeding the blackbody near  $0.55 \mu\text{m}$ . The calibrated model integrates to  $S_0 = 1361.0 \text{ W/m}^2$  over  $0.05\text{--}50 \mu\text{m}$  to  $\sim 10^{-6}$  relative.

A note on the temperature, because two values circulate: the IAU nominal solar effective temperature is 5772 K, while 5778 K is the older value that pairs with the nominal radius and 1 AU to reproduce  $S_0$  most closely. RADIANT uses **5778 K** (`blackbody_5778`). The choice moves  $E(0.55 \mu\text{m})$  by 0.5 % before calibration and *nothing at all* after it in integrated terms, since  $k_{\text{scale}}$  absorbs the difference — it is a shape parameter, not an energy one.

**References.** [Kopp & Lean 2011] ( $S_0 = 1361 \text{ W/m}^2$ ), [ASTM E490], [Wehrli 1985].

### 3.3.9 Reflected solar radiance

#### Equation.

$$L_{\text{refl}}(\lambda) = f_r(\lambda; \theta_{\text{sun}}, \theta_{\text{obs}}) \cdot E_{\text{sun}}(\lambda, d) \cdot \cos \theta_{\text{sun}} \quad \left[ \text{W/m}^2/\text{sr}/\mu\text{m} \right]$$

For the Lambertian special case  $f_r = \rho/\pi$ :

$$L_{\text{refl}}(\lambda) = \frac{\rho(\lambda) E_{\text{sun}}(\lambda) \cos \theta_{\text{sun}}}{\pi}$$

**Symbols.**  $f_r$  [ $\text{sr}^{-1}$ ] BRDF;  $E_{\text{sun}}$  [ $\text{W/m}^2/\mu\text{m}$ ] solar spectral irradiance on a plane normal to the rays at the target (exo-atmospheric if  $\tau_{\text{atm}}$  is applied separately downstream — which is RADIANT's convention);  $\theta_{\text{sun}}$  [rad] solar zenith angle;  $\cos \theta_{\text{sun}}$  projects normal-incidence irradiance onto the tilted surface;  $d$  [AU] Sun–target distance.

**Derivation.** By definition of BRDF,  $dL_o = f_r dE_i$ : outgoing radiance is BRDF times incident irradiance *on the surface*, which is the normal-plane irradiance times  $\cos \theta_{\text{sun}}$ . For a Lambertian surface, reflected exitance

is  $M = \rho E \cos \theta_{\text{sun}}$  and a perfectly diffuse surface has  $M = \pi L$ , giving the  $1/\pi$  [sr<sup>-1</sup>]. The Sun is treated as a single collimated directional source (no hemisphere integration) — v1 convention.

**Assumptions & validity.** BRDF model fidelity is the limit — Lambertian is the baseline, Phong adds a glint lobe (next entry). Collimated-sun is fine except in near-terminator penumbra.  $\cos \theta_{\text{sun}} \geq 0$  required: RADIANT clamps sun-below-horizon to zero radiance *with intent* (returns zeros, never silent negative radiance) and rejects zenith angles outside  $[0, \pi/2]$  at construction.

#### Pitfalls.

- Missing  $1/\pi$  (returns an exitance-like value, high by  $\pi \approx 3.14$ ) or dividing by  $2\pi$  (confusing hemisphere solid angle  $2\pi$  with the projected-solid-angle integral  $\int \cos \theta d\Omega = \pi$ ).
- Omitting  $\cos \theta_{\text{sun}}$ , or using *elevation* where *zenith* is expected (a  $\cos \leftrightarrow \sin$  swap).
- Degrees passed to  $\cos()$  expecting radians.
- Applying a  $\cos \theta_{\text{view}}$  as well — for a Lambertian surface the radiance has no view-angle cosine (the  $\cos \theta_v$  in received power exactly cancels the  $1/\cos \theta_v$  in projected source area).
- Double-counting the atmosphere:  $E_{\text{sun}}$  must be exo-atmospheric if  $\tau_{\text{atm}}$  is applied separately (RADIANT's case), at-surface if not.

**Numeric anchor.** With the audit's adopted  $E_0(0.55 \mu\text{m}) = 1870 \text{ W/m}^2/\mu\text{m}$  (exo-atmospheric,  $\tau = 1$ ),  $\rho = 0.3$ ,  $\theta_{\text{sun}} = 30^\circ$ :  $L = 154.647755 \text{ W/m}^2/\text{sr}/\mu\text{m}$ . This anchor scales linearly with the adopted solar table — RADIANT's `blackbody_5778` model gives a few-percent-lower  $E_0$  at  $0.55 \mu\text{m}$  (previous entry), so rescale by  $E_{0,\text{impl}}/1870$  before comparing implementation output against it.

**References.** [Nicodemus 1977], [Holst 2008], [ASTM E490].

### 3.3.10 BRDF normalization — Lambertian and Phong

**Equation.** BRDF definition:  $f_r(\omega_i, \omega_o) = dL_o/dE_i$  [sr<sup>-1</sup>]. Energy conservation requires  $\int_{\text{hemi}} f_r \cos \theta_o d\Omega_o \leq 1$  for every incidence direction. RADIANT implements both of:

$$\text{Lambertian: } f_r = \frac{\rho}{\pi} \quad \text{Phong: } f_r = \frac{\rho_d}{\pi} + \rho_s \cdot \frac{n+2}{2\pi} \cdot \cos^n \alpha$$

**Symbols.**  $\rho$  [—] total hemispherical reflectance;  $\rho_d, \rho_s$  [—] diffuse and specular shares with  $\rho_d + \rho_s = \rho$ ;  $n \geq 0$  [—] Phong exponent (higher = narrower lobe);  $\alpha$  [rad] angle between the observer direction and the mirror-reflection direction.

**Derivation.** *Lambertian:* the projected-solid-angle integral over the hemisphere is  $\int \cos \theta d\Omega = 2\pi \int_0^{\pi/2} \cos \theta \sin \theta d\theta = \pi$  — so  $f_r = \rho/\pi$  integrates to exactly  $\rho$ . That is why the divisor is  $\pi$ , not  $2\pi$ . *Phong:* with the lobe centered on the normal ( $\alpha = \theta$ ),  $\int_{\text{hemi}} \cos^n \alpha \cos \theta d\Omega = 2\pi \int_0^{\pi/2} \cos^{n+1} \theta \sin \theta d\theta = 2\pi/(n+2)$ ; the factor  $(n+2)/(2\pi)$  therefore makes the specular lobe integrate to exactly  $\rho_s$  at normal incidence, so  $\rho_d + \rho_s \leq 1$  suffices for energy conservation there. The weaker  $(n+1)/(2\pi)$  variant seen in graphics literature normalizes  $\int \cos^n \alpha d\Omega$  *without* the  $\cos \theta$  throughput factor — it conserves lobe solid-angle weight, not reflected energy, and can over-reflect by up to  $(n+2)/(n+1)$ . RADIANT uses the energy-normalizing  $(n+2)/(2\pi)$ .

**Assumptions & validity.** Lambertian: exact energy conservation for any  $\rho \leq 1$ ; view-independent radiance. Phong: phenomenological (no reciprocity in the classic form, no Fresnel dependence); the normalization is exact only when the mirror direction coincides with the normal — at grazing incidence part of the lobe falls below the horizon and the actual reflected fraction is  $< \rho_s$  (energy lost, never gained: conservative but biased). Adequate for glint order-of-magnitude budgets; use microfacet models for rendering-grade physics. RADIANT evaluates  $\alpha = |\theta_{\text{obs}} - \theta_{\text{sun}}|$  in the plane of incidence (azimuthally symmetric v1 geometry).

#### Pitfalls.

- $\rho$  instead of  $\rho/\pi$  — off by  $\pi$  in every reflected radiance; arguably the single most common radiometry bug in existence. Or  $\rho/(2\pi)$  from the hemisphere-vs-projected-solid-angle confusion.
- $(n+1)/(2\pi)$  where energy normalization  $(n+2)/(2\pi)$  is intended, or vice versa without documenting the convention.
- Measuring  $\alpha$  from the surface normal instead of from the mirror-reflection direction.
- $\cos^n \alpha$  not clamped at  $\alpha > \pi/2$  — odd  $n$  on a negative cosine produces negative radiance. RADIANT clamps  $\cos \alpha$  at zero.
- Adding a Lambertian *radiance* term to a Phong *BRDF* term (unit mismatch: one already has the  $1/\pi$  folded into a radiance, the other is  $\text{sr}^{-1}$ ).

**Numeric anchor.** Lambertian,  $\rho = 0.3$ :  $\int f_r \cos \theta d\Omega = 0.300000000$  (dimensionless,  $= \rho$  exactly). Phong with  $(n+2)/(2\pi)$  at normal incidence: the hemispherical integral over  $\rho_s$  equals 1.000000000 (dimensionless) for  $n = 1, 10, 100$  (audit adaptive quadrature).

**References.** [Nicodemus 1977], [Phong 1975], [Lewis 1994].

#### 3.3.11 Point-source irradiance and pixel solid angle

**Equation.** A point source of spectral intensity  $I(\lambda)$  [W/sr/μm] at range  $R$  [m] through path transmittance  $\tau_{\text{atm}}(\lambda)$  delivers at-aperture spectral irradiance

$$E_{\text{ap}}(\lambda) = \frac{I(\lambda) \tau_{\text{atm}}(\lambda)}{R^2} \quad \left[ \text{W/m}^2/\mu\text{m} \right]$$

The pixel-subtended solid angle (small-angle) is

$$\Omega_{\text{pix}} = \frac{p_x p_y}{f^2} = \text{IFOV}_x \cdot \text{IFOV}_y \quad [\text{sr}]$$

and the target solid angle is  $\Omega_t = A_t/R^2$  [sr].

**Symbols.**  $I(\lambda)$  [W/sr/μm] — for a small Lambertian-emitting facet  $I = L \cdot A_t$ ;  $R$  [m] slant range;  $p_x, p_y$  [m] pixel pitches;  $f$  [m] focal length;  $A_t$  [m<sup>2</sup>] projected target area.

**Derivation.** Inverse square is conservation of energy through expanding spheres: intensity is fixed, irradiance  $= I/R^2$ . Aperture spectral power is then  $\Phi(\lambda) = E_{\text{ap}} A_{\text{ap}}$  [W/μm]. RADIANT's spectral-integration stage implements the equivalent radiance route:  $\Phi = L_{\text{target}} \cdot A_{\text{ap}} \cdot \Omega_t$  with  $\Omega_t = A_t/R^2$  — identical to  $I/R^2 \cdot A_{\text{ap}}$  because  $I = L A_t$ . The blur spreads this energy per the PSF; the ensquared-energy fraction  $\text{EE}_{\text{box}}$  multiplies the in-pixel signal, applied exactly once, downstream in spectral integration.

**Assumptions & validity.** Point-source regime valid when the source's angular extent  $\ll$  IFOV and  $\ll$  PSF width.  $\Omega = A/R^2$  is the small-angle approximation — exact solid-angle formulas matter only for  $\text{IFOV} \gtrsim 0.1$  rad, never for imaging sensors. Flat normal-incidence footprint assumed; off-nadir the ground footprint grows by  $1/\cos(\text{incidence})$ .  $\tau_{\text{atm}}$  is the *slant-path* transmittance, not the vertical column.

**Pitfalls.**

- Applying both  $1/R^2$  and  $\Omega_{\text{pix}}$  to the same term — double-counting the geometry. Radiance-based extended-scene math ( $L \cdot \Omega_{\text{pix}} \cdot A_{\text{ap}}$ ) and irradiance-based point-source math ( $I/R^2 \cdot A_{\text{ap}}$ ) are *alternative routes*, never multiplied together.
- $\tau_{\text{atm}}^2$  (two-way path) sneaking in from radar heritage — passive EO is one-way.
- Range in km fed to a formula expecting m — a  $10^6$  error in  $E_{\text{ap}}$ .
- Applying  $EE_{\text{box}}$  anywhere but once, in spectral integration.

**Numeric anchor.**  $I = 100 \text{ W/sr/}\mu\text{m}$ ,  $\tau = 0.7$ ,  $R = 500 \text{ km}$ :  $E_{\text{ap}} = 2.80000000 \times 10^{-10} \text{ W/m}^2/\mu\text{m}$ . Pixel solid angle for  $\text{GSD} = 3 \text{ m}$  at  $R = 500 \text{ km}$ :  $\Omega_{\text{pix}} = 3.60000000 \times 10^{-11} \text{ sr}$  ( $\text{IFOV} = 6 \mu\text{rad}$ ).

**References.** [Holst 2008], [Wolfe & Zissis §1].

### 3.3.12 The effective pupil — the cold stop as the aperture stop

**Equation.** In a cooled instrument the cold stop *is* the system aperture stop, built slightly smaller than the geometric primary so that alignment and thermal tolerances can never let the focal plane see past it to warm structure. That one fact has one modelling consequence: the pupil the system works with is the cold stop's, and everything derived from the pupil uses it —

$$D_{\text{eff}} = (1-u) D, \quad \epsilon_{\text{eff}} = \max(\epsilon, \epsilon_{\text{cs}}), \quad N_{\text{eff}} = \frac{f}{D_{\text{eff}}} = \frac{N}{1-u}, \quad A_{\text{collect}} = \frac{\pi}{4} D_{\text{eff}}^2 (1-\epsilon_{\text{eff}}^2),$$

with  $u = \text{optics.cold\_stop\_undersize\_frac}$  (the fractional reduction of the pupil *diameter*),  $\epsilon_{\text{cs}} = \text{optics.cold\_stop\_obscuration\_ratio}$  (what the cold shield itself blocks, governing only when it exceeds the telescope's own obscuration), and  $\text{optics.f\_number}$  always the primary's —  $N_{\text{eff}}$  is derived, never entered. The collecting area (signal), the working f/# (pixel étendue), the complex pupil (diffraction PSF and MTF — one pupil, both spatial paths), the sampling ratio  $Q$ , the defocus fold, and the near-field acceptance cone  $\Omega_{\text{cone}} = 2\pi(1 - \cos \theta)$ ,  $\theta = \arctan[1/(2N_{\text{eff}})]$  (Appendix A) all read the effective pupil, so signal and warm-optics background scale together as they physically do. There is no cold-stop *efficiency*: in-cone emission arrives through the imaging path and cannot be blocked, out-of-cone warm structure is blocked completely, and undersizing is not a way to buy a darker background for free.

**Assumptions & validity.** A cold stop that undersizes by more than half the diameter ( $u \geq 0.5$ ) is no longer describing tolerancing margin and is refused; at the defaults ( $u = 0$ ,  $\epsilon_{\text{cs}} = 0$ ) the effective pupil is the primary pupil bit-for-bit.

**Pitfalls.** Entering the working f/# as  $\text{optics.f\_number}$  (it is the primary's; the working value is a published output); scaling the signal by  $(1-u)^2$  but leaving the PSF,  $Q$  or  $\Omega_{\text{cone}}$  on the primary; a per-element near-field solid angle instead of the one étendue cone.

**Numeric anchor.**  $u = 0.05$ :  $A_{collect}$  falls to  $0.95^2 = 90.25\%$  of the primary's,  $N_{eff}$  rises by  $1/0.95 = 1.0526$ , and  $\Omega_{cone}$  at  $f/4$  falls from  $0.048520$  sr to  $0.043838$  sr.

### 3.3.13 Sub-pixel fill fraction and radiance mixing

**Equation.** When the target's solid angle is smaller than the pixel's:

$$ff = \frac{\Omega_t}{\Omega_{pix}} = \frac{A_t}{R^2 \Omega_{pix}} \quad (0 < ff \leq 1, \text{ clamped at } 1)$$

The pixel-averaged apparent radiance is the area-weighted mix (with path radiance filling the whole pixel uniformly):

$$L_{mixed}(\lambda) = ff \cdot L_{target}(\lambda) \cdot EE_{box} + (1 - ff) L_{bg}(\lambda) + L_{path}(\lambda)$$

**Symbols.**  $ff$  [—] fill fraction;  $L_{target}$ ,  $L_{bg}$  [ $\text{W/m}^2/\text{sr}/\mu\text{m}$ ] pure target and background contributions at the aperture (path radiance separated out);  $L_{path}$  [ $\text{W/m}^2/\text{sr}/\mu\text{m}$ ] path radiance;  $EE_{box}$  [—] applied to the compact target term only.

**Derivation.** The pixel sees  $\Omega_{pix}$ ; the target fills  $\Omega_t$  of it and background fills the rest — hence the  $ff / (1 - ff)$  split. The **regime-consistency identity** ties the sub-pixel and point-source routes together: the target's contribution to aperture irradiance via the point-source route is  $L_t A_t \tau / R^2$ , and via the sub-pixel route it is  $L_t \cdot ff \cdot \Omega_{pix} \cdot \tau$ . Since  $ff \cdot \Omega_{pix} = A_t / R^2 = \Omega_t$  by construction, the two routes agree identically — that identity is what the fill-fraction anchor test pins. Path radiance is generated along the line of sight, not at the target, so it is added once, unweighted by  $ff$ , and never multiplied by  $EE_{box}$ .

**Assumptions & validity.** Areal mixing assumes the target sits fully within one IFOV footprint (no straddling-pixel split modeling). Clamps to  $ff = 1$  when the target overfills the pixel (the regime then belongs to extended-scene handling). Fill fraction from geometry requires positive area, range, pitches, and focal length; otherwise RADIANT falls back to the explicit `fill_fraction` parameter rather than guessing.

#### Pitfalls.

- Area vs side-length confusion:  $ff = A_t / \text{GSD}^2$ , not  $l_t / \text{GSD}$ .
- Forgetting the  $(1 - ff)$  background complement — the unfilled pixel area still sees background radiance.
- Applying  $EE_{box}$  to the background term — background is extended;  $EE$  applies to the compact target only, and RADIANT guards this explicitly.
- Splitting path radiance by  $ff$  — it fills the pixel uniformly.
- Deriving  $ff$  from nadir  $\text{GSD}^2$  off-nadir — the footprint grows with incidence.

**Numeric anchor.**  $A_t = 1 \text{ m}^2$ ,  $\text{GSD} = 3 \text{ m}$  (i.e.  $\Omega_{pix}$  matched to a  $3 \text{ m}$  footprint):  $ff = 0.111111111$  (dimensionless).

**References.** [Holst 2008], RADIANT Source/Target System doc §3.

### 3.3.14 Photon conversion — spectral power to photoelectron rate

**Equation.** Photon energy  $E_{\text{ph}} = hc/\lambda_m$  [J]. The photoelectron rate from spectral power  $\Phi(\lambda)$  [W/μm] at the detector is

$$\dot{n}_e = \int \text{QE}(\lambda) \Phi(\lambda) \frac{\lambda_m}{hc} d\lambda \quad [\text{e}^-/\text{s}], \quad \lambda_m = \lambda \cdot 10^{-6}$$

integrated over  $\lambda$  in μm; total signal  $N_e = \dot{n}_e \cdot t_{\text{int}}$  [e-]. RADIANT packages the same physics as an aperture-referred spectral responsivity

$$R(\lambda) = A_{\text{ap}} \Omega_{\text{pix}} \tau_{\text{opt}}(\lambda) \text{QE}(\lambda) \frac{\lambda_m}{hc} \quad [\text{e}^-/\text{s per W/m}^2/\text{sr}/\mu\text{m}]$$

with  $R_{\text{band}} = \int R(\lambda) d\lambda$  used for backward propagation ( $L_{\text{aperture}} = N_e/(R_{\text{band}} t_{\text{int}})$ ).

**Symbols.**  $\Phi(\lambda)$  [W/μm];  $\text{QE}(\lambda)$  [e-/photon, dimensionless];  $\lambda_m/(hc)$  [photons/J] converts watts to photons/s;  $t_{\text{int}}$  [s];  $A_{\text{ap}}$  [m<sup>2</sup>];  $\Omega_{\text{pix}}$  [sr];  $\tau_{\text{opt}}$  [—].

**Derivation.** A watt of monochromatic light at wavelength  $\lambda$  carries  $1/E_{\text{ph}} = \lambda_m/hc$  photons per second. Weight by QE per photon and integrate across the band. The  $10^{-6}$  lives in  $\lambda_m$ :  $\lambda$  in μm must be converted to meters so  $\lambda_m/(hc)$  has units 1/J.

**QE temperature dependence.** The base QE — the scalar `detector.qe_value` or the `qe_table_path` curve — is scaled by a linear factor in the detector temperature,

$$\text{QE}(\lambda, T) = \text{QE}_{\text{base}}(\lambda) [1 + c (T_{\text{det}} - T_{\text{ref}})],$$

$c = \text{detector.qe\_temperature\_coeff\_per\_K}$  (default 0: temperature-independent),  $T_{\text{ref}} = \text{detector.qe\_temperature\_ref\_K}$ . The product is clamped to  $[0, 1]$  with a warning when the factor pushes it out of range — never silently. Physics note: a linear coefficient is the small-excursion form of the cutoff-wavelength shift of a narrow-gap material; a 20 K excursion at  $c = -0.002 \text{ K}^{-1}$  is a 4 % change, and beyond a few tens of kelvin the curve, not the coefficient, is the right input.

**Assumptions & validity.** One photoelectron per detected photon, weighted by QE (no avalanche gain — gain is a separate downstream stage).  $\lambda/hc$  must stay *inside* the integral: photon energy varies across the band, and pulling out a band-center value biases wide bands (for 3–5 μm the edge-to-edge photon-energy ratio is 5/3). QE convention (per incident vs per absorbed photon) must be consistent, with fill factor and window transmission counted once — inside QE or as separate factors, never both. Shot noise follows as  $\sqrt{N_e}$  only if  $N_e$  is a true photon count: any premature gain scaling corrupts the Poisson statistics.

#### Pitfalls.

- The  $10^{-6}$ : using  $\lambda$  in μm directly in  $\lambda/(hc)$  — rate high by  $10^6$ .
- Energy-weighted vs photon-weighted band averages confused: QE tables are per-photon; responsivity in A/W is per-energy ( $R_{A/W} = \text{QE} \cdot q\lambda_m/hc$ ) — mixing them double-counts  $\lambda$ .
- Applying  $t_{\text{int}}$  inside a “rate” function so downstream multiplies by it again.
- The electron charge  $q = 1.602 \times 10^{-19} \text{ C}$  appearing where it does not belong — electron *counts* are dimensionless; charge enters only when converting to amperes.

**Numeric anchor.** Monochromatic  $\Phi = 10^{-12}$  W at  $\lambda = 4 \mu\text{m}$  with  $\text{QE} = 0.7$ :  $\dot{n}_e = 1.40955264 \times 10^7$  e-/s (check:  $E_{\text{ph}}(4 \mu\text{m}) = 4.96611464 \times 10^{-20}$  J, and  $0.7 \times 10^{-12}$  W /  $4.96611464 \times 10^{-20}$  J reproduces the rate).

**References.** [Holst 2008], [Janesick 2001], [CODATA 2018].

### 3.4 How the foundations feed the chain

The ten stages consume the foundations above in a fixed order. Each stage is a pure function of the chain state and the parameters; all inter-stage data flows through one immutable state object, which no stage mutates.

| # | Stage                | Radiometric role                                                                                                             | Foundations used                                                                                            | Detailed in  |
|---|----------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------|
| 0 | Geometry             | Slant range $R$ , incidence, solar geometry                                                                                  | — (feeds $R$ , $\theta_{\text{sun}}$ to everything)                                                         | Ch. 2        |
| 1 | Source               | Build $L_{\text{target}}(\lambda)$ , $L_{\text{bg}}(\lambda)$ ; tentative regime                                             | Planck, graybody, reflected solar, BRDF, brightness/radiance temperature converters, point-source intensity | this chapter |
| 2 | Atmosphere           | $L_{\text{ap}} = L_{\text{atm}} + L_{\text{path}}$                                                                           | band integration on the $\tau(\lambda)$ grid                                                                | Ch. 4        |
| 3 | Optics               | Throughput $\tau_{\text{opt}}$ , $A_{\text{ap}}$ , $\Omega_{\text{pix}}$ ; warm-optics self-emission; <b>final regime</b>    | Kirchhoff emissivity, Planck (self-emission), pixel solid angle                                             | Ch. 5        |
| 4 | Platform             | Smear/jitter degradation; $\text{EE}_{\text{box}}$ from the fully degraded PSF                                               | —                                                                                                           | Ch. 5        |
| 5 | Spectral Integration | Spectral $\rightarrow$ scalar, exactly once; $\text{EE}_{\text{box}}$ applied exactly once                                   | photon conversion, point-source/sub-pixel regime radiometry, fill-fraction mixing                           | this chapter |
| 6 | Detector             | Noise budget (16 terms)                                                                                                      | $\sqrt{N_e}$ shot statistics                                                                                | Ch. 6        |
| 7 | Readout              | TDI, binning, coadd, gain, ADC scaling                                                                                       | —                                                                                                           | Ch. 6        |
| 8 | Calibration          | Post-NUC residuals, drift, bias budget — calibration noise terms and accuracy inputs; the stage adds terms and never a frame | Planck at the calibration points, band responsivity                                                         | Ch. 7        |
| 9 | Performance          | SNR, NEDT, NIIRS, system MTF                                                                                                 | $\partial B / \partial T$ (NEDT kernel), band responsivity (backward propagation)                           | Ch. 8        |

Three chain-glue equations worth stating here because they are pure radiometry:

**At-aperture radiance** (Atmosphere stage):

$$L_{\text{ap}}(\lambda) = L_{\text{target}}(\lambda) \tau_{\text{atm}}(\lambda) + L_{\text{path}}(\lambda)$$

where  $\tau_{\text{atm}}$  follows Beer–Lambert along the slant path and  $L_{\text{path}}$  is atmospheric self-emission and scatter into the line of sight.

**Extended-scene signal electrons** (Spectral Integration stage — the only spectral-to-scalar collapse in the chain):

$$N_e = \int_{\lambda_1}^{\lambda_2} L_{\text{ap}}(\lambda) \tau_{\text{opt}}(\lambda) A_{\text{ap}} \Omega_{\text{pix}} \text{QE}(\lambda) \frac{\lambda_m}{hc} t_{\text{int}} d\lambda$$

with  $EE_{\text{box}}$  multiplying the target term only in point-source and sub-pixel regimes, never in extended scenes and never on the background term.

**SNR** (Performance stage):  $SNR = N_e / \sigma_{\text{total}}$  with  $\sigma_{\text{total}}$  the RSS of the noise budget; contrast SNR uses  $N_{e,\text{target}} - N_{e,\text{bg}}$  in the numerator. NEDT divides the noise-equivalent radiance by the band-integrated  $\partial B / \partial T$  — which is why the NEDT-kernel entry above carries per-K units end to end.

### 3.4.1 Dimensional Trace

| Stage                   | Input units                                     | Output units                                                                                      | Conversion                                                           |
|-------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Geometry                | orbit/attitude params                           | m, rad                                                                                            | trigonometry                                                         |
| Source                  | K, dimensionless $\epsilon/\rho$                | W/m <sup>2</sup> /sr/μm                                                                           | Planck + $\epsilon$ ; BRDF · E <sub>sun</sub> · cos θ                |
| Atmosphere              | W/m <sup>2</sup> /sr/μm                         | W/m <sup>2</sup> /sr/μm                                                                           | × $\tau_{\text{atm}} + L_{\text{path}}$                              |
| Optics                  | W/m <sup>2</sup> /sr/μm                         | W/m <sup>2</sup> /sr/μm (+ A <sub>ap</sub> [m <sup>2</sup> ], Ω <sub>pix</sub> [sr]<br>published) | × $\tau_{\text{opt}}$                                                |
| Platform                | — (radiometric<br>pass-through)                 | EE <sub>box</sub> [—]                                                                             | PSF ensquared energy                                                 |
| Spectral<br>Integration | W/m <sup>2</sup> /sr/μm, m <sup>2</sup> , sr, s | e-                                                                                                | × A · Ω · QE · $\lambda_m / hc$ · t <sub>int</sub> , $\int d\lambda$ |
| Detector                | e-                                              | e- RMS (noise terms)                                                                              | noise model                                                          |
| Readout                 | e-                                              | e- (scaled), DN                                                                                   | × N <sub>TDI</sub> · M <sub>bin</sub> · N <sub>coadd</sub> ; ÷ gain  |
| Performance             | e-                                              | dimensionless (SNR), K (NEDT), NIIRS                                                              | S/σ; $\sigma_L / \int (\partial B / \partial T) d\lambda$            |

Every row must check; integrating a per-μm density over μm is the only “conversion-free” collapse, and the two 10<sup>−6</sup> factors in the chain (Planck Jacobian, photon-energy  $\lambda_m$ ) each appear exactly once.

## 3.5 Assumptions and Limitations

- Lambertian target emission; BRDF structure available only through the Lambertian and normalized-Phong models (no microfacet, no measured BRDF tables)
- Sun as a collimated directional source; blackbody-5778 solar spectral shape (few-percent deviations from the real spectrum; tabulated ASTM E490 model planned)
- No molecular spectroscopy or fluorescence
- No atmospheric refraction or scintillation
- No ghost images, BSDF scatter, or chromatic aberration
- No optical crosstalk between pixels
- No temporal variability in scene
- The project’s scope-decision register records the full list of deferred effects

# Atmosphere Models

The physics of RADIANT’s two atmosphere model families — the closed-form **simple parametric** model and the **library-backed** interpolated / tabulated / MODTRAN-import models — organized by model rather than by change history. What each model computes, from what first principles, and which measurement or test holds each claim in place.

This document is the *physics*. The measured accuracy of every model against the MODTRAN run set is recorded in the MODTRAN-parity validation record; the architecture contract (AtmosphericQuantities, backend dispatch, guard structure) is specified in the atmosphere architecture document; which model an operator should reach for is the User’s Guide’s model-selection guidance. Atmospheric **turbulence** is a spatial effect and is documented with the rest of the MTF cascade in the spatial chapter’s *Atmospheric turbulence*; it is not repeated here.

**Symbols.**  $\lambda$  wavelength [ $\mu\text{m}$ ];  $\zeta$  zenith angle at a path segment’s **lower** endpoint [rad];  $\theta_s$  solar zenith [rad];  $\Theta$  scattering angle [rad];  $h$  altitude above mean sea level [m];  $H_i$  species scale height [m];  $\tau$  transmittance [–]; OD optical depth [–];  $m$  air mass [–];  $\sigma$  volume extinction coefficient [ $\text{km}^{-1}$ ];  $w$  precipitable water [cm];  $\omega_0$  single-scattering albedo [–];  $B(\lambda, T)$  Planck spectral radiance [ $\text{W}/\text{m}^2/\text{sr}/\mu\text{m}$ ];  $L$  radiance [ $\text{W}/\text{m}^2/\text{sr}/\mu\text{m}$ ];  $E$  irradiance [ $\text{W}/\text{m}^2/\mu\text{m}$ ];  $R_E = 6.371 \times 10^6$  m.

## 4.1 Two families, one contract

Every model in RADIANT delivers the same product bundle — a transmittance, an upwelling path radiance, a downwelling emission term, and (for the composed topologies) the segment products a path is assembled from. What differs is where the numbers come from:

| Family                 | Members                                         | Physics origin                                                        | Geometry response                                                                       |
|------------------------|-------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Closed-form parametric | simple                                          | Beer-Lambert on four species, calibrated band-by-band against MODTRAN | Recomputed for every geometry                                                           |
| Library-backed         | interpolated, tabulated, modtran (tape7 import) | Real MODTRAN 6 radiative transfer, stored and served                  | interpolated interpolates over declared axes; the other two are geometry-agnostic files |
| Identity               | exo                                             | $\tau \equiv 1, L \equiv 0$ — the cosmic vacuum                       | None (exact)                                                                            |

The simple model is the only backend that can serve an arbitrary path topology (down-looking, up-looking, level, grazing, twilight), because the segment evaluators are built on its species model. The library families serve the topologies their runs actually measured, and refuse the rest rather than approximating them.

## 4.2 The simple parametric model

### 4.2.1 Beer-Lambert structure

Transmittance along a path is the exponential of a slant optical depth built from four species, each carried on its own exponential density profile:

$$\tau_{atm}(\lambda) = \exp[-OD_{tot}(\lambda)], \quad OD_{tot}(\lambda) = \sum_i m_i OD_{i,vert}(\lambda)$$

with  $i \in \{\text{mol}, \text{aer}, \text{h}_2\text{O}, \text{gas}\}$  and  $m_i$  the species air mass of *Air mass* below. The **vertical** column each species presents between two altitudes is the analytic exponential integral

$$\text{col}_i(h_{lo}, h_{hi}) = \int_{h_{lo}}^{h_{hi}} e^{-h/H_i} dh = H_i \left( e^{-h_{lo}/H_i} - e^{-h_{hi}/H_i} \right) \quad [\text{km}]$$

which is endpoint-symmetric, so one segment read in either direction presents one column — the structural reason transmittance is single-valued (*Path topologies*).

Scale heights:  $H_{\text{mol}} = 8000$  m,  $H_{\text{aer}} = 1200$  m,  $H_{\text{h}_2\text{O}} = 2000$  m. The well-mixed-gas floor rides the molecular profile by construction (*The well-mixed-gas absorption floor*).

### 4.2.2 Molecular (Rayleigh) scattering

The published Rayleigh constant is a **total vertical optical depth**, dimensionless — not a per-km coefficient:

$$\tau_{R,vert}(\lambda) = 0.0088 \lambda_{\mu\text{m}}^{-4.09}$$

At  $\lambda = 0.55 \mu\text{m}$  this evaluates to 0.101484, against the published whole-atmosphere value 0.0973–0.10 ([Hansen & Travis 1974], [Bucholtz 1995]). The sea-level volume extinction the slant integral needs is **derived** from it through the exponential profile's own identity  $\tau_{vert} = \sigma_0 H$ :

$$\sigma_{\text{mol}}(\lambda) = \frac{\tau_{R,vert}(\lambda)}{H_{\text{mol}}} = 0.012686 \text{ km}^{-1} \text{ at } 0.55 \mu\text{m} \quad (\text{published } \approx 0.0116 \text{ km}^{-1})$$

and is scaled along the path by  $e^{-h/H_{\text{mol}}}$ .

Deriving  $\sigma_0$  rather than storing it independently is what prevents the classic error of consuming the published optical depth *as* the  $\text{km}^{-1}$  coefficient, which inflates every VIS/NIR molecular optical depth by exactly the column depth,  $\approx 8\times$ .

### 4.2.3 Aerosol (Mie) extinction

Aerosol extinction at the reference wavelength is fit from the Koschmieder visibility relation, then carried spectrally by an Ångström power law:

$$\sigma_{\text{aer}}(0.55 \mu\text{m}) = \frac{3.912}{V_{\text{km}}} [\text{km}^{-1}], \quad \sigma_{\text{aer}}(\lambda) = \sigma_{\text{aer}}(0.55) \left( \frac{\lambda}{0.55} \right)^{-\alpha}$$

The constant  $3.912 = -\ln(0.02)$  is the 2 % contrast threshold defining meteorological visibility. Three canonical aerosol regimes:

| Regime   | Ångström $\alpha$ | Aerosol SSA $\omega_{\text{aer}}$ |
|----------|-------------------|-----------------------------------|
| rural    | 1.3               | 0.95                              |
| urban    | 1.5               | 0.85                              |
| maritime | 0.7               | 0.99                              |

The power law is a **scattering** model: good in VIS/SWIR, weak but usable through the MWIR, and wrong in the LWIR where real IR aerosol extinction is absorption-dominated and roughly flat. It is therefore clamped at the MWIR–LWIR boundary  $\lambda_{\text{clamp}} = 5.0 \mu\text{m}$  — beyond it the extinction is frozen at its  $5 \mu\text{m}$  value rather than decaying unphysically toward zero, and the model warns once per run when the clamp engages.

**Background aerosol — the term visibility does not reach.** The Koschmieder relation describes the *boundary layer*, whose 1.2 km scale height empties by  $\sim 5$  km. Above it the atmosphere still carries free-tropospheric and stratospheric aerosol, at an amount a surface visibility reading says nothing about. RADIANT carries that separately, as a per-region vertical optical depth `aer_bg_od` on the molecular scale height:

$$\text{OD}_{\text{bg}}(\lambda) = s \cdot \text{aer\_bg\_od}(\lambda) \cdot \frac{\text{col}_{\text{mol}}}{H_{\text{mol}}}$$

with  $s = \text{atmosphere.background\_aerosol\_scale}$  (1.0 = the MODTRAN rural climatology the model is calibrated against; 0 = none). It is calibrated only where the anchors resolve it, the two regions below  $0.70 \mu\text{m}$ , and is zero elsewhere. It **scatters** — it enters  $\omega_0$  and the phase function with the aerosol’s single-scattering albedo, not as a pure absorber — which is the whole reason it is not left on the gas floor.

#### 4.2.4 Water vapor — the curve of growth

Water vapor is **not** linear in Beer’s law over a real band, because the strong lines saturate long before the windows between them do. The calibrated model is a 17-region curve of growth:

$$\text{OD}_{\text{H}_2\text{O}}(\lambda) = k(\lambda) w_{\text{eff}}^{b(\lambda)}$$

where  $w_{\text{eff}}$  [cm] is the path water amount — total precipitable water times the traversed fraction of the  $H_{\text{H}_2\text{O}} = 2$  km exponential column. The exponent is fit region by region against the real MODTRAN 6 water ladder (runs D4 / A1 / D5,  $\text{H}_2\text{O} \times 0.5 / \times 1 / \times 2$ ): **sub-linear**  $b \approx 0.2$ – $0.8$  in the saturated absorption bands, **super-linear**  $b \approx 1.3$ – $1.75$  in the LWIR where the e-type continuum dominates and the absorber’s own pressure broadening compounds with amount.

This replaced a five-Lorentzian line fit whose far wings made the MWIR water response  $\approx 5\times$  too steep.

Standard-profile water columns ([McClatchey 1972], carried into MODTRAN MODELS 1–6) supply the default when the operator selects a climate profile and leaves the water column at its schema default:

| Profile       | PWV [cm] | Profile          | PWV [cm] |
|---------------|----------|------------------|----------|
| tropical      | 4.11     | midlat_winter    | 0.85     |
| midlat_summer | 2.92     | subarctic_summer | 2.08     |
| us_standard   | 1.40     | subarctic_winter | 0.42     |

#### 4.2.5 The well-mixed-gas absorption floor

Each of the same 17 regions carries a **water-independent** vertical optical depth — the CO<sub>2</sub> 4.3 μm and 15 μm bands, N<sub>2</sub>O, the O<sub>3</sub> 9.6 μm band, and O<sub>2</sub> / CH<sub>4</sub> overtones — in excess of what Rayleigh and aerosol already supply:

$$\text{OD}_{\text{region}}(w) = \text{floor\_od} + k w_{\text{eff}}^b$$

The floor rides the molecular scale height — it is defined as a fraction of the molecular column — and it enters the single-scattering albedo denominator as a **pure absorber** (*Single-scatter solar path radiance*). Without it a model attributes the MWIR CO<sub>2</sub> floor to water and evaluates  $\omega_0 \approx 1$  for space columns.

**What the floor is, and what it is not (CU-337).** floor\_od is *gas*. The fit that produces it measures the band opacity beyond Rayleigh and the boundary-layer aerosol, and below 0.70 μm that quantity is mostly not gas: the 0.45–0.70 μm window’s chemistry (the ozone Chappuis band, with a narrow O<sub>2</sub> B contribution) supplies about 0.020 optical depths, against the 0.1375 the fit resolves. The remainder is the background aerosol of §2.3. The two visible rows are therefore split at their chemistry — 0.0030 and 0.0200 — with the rest carried as aer\_bg\_od; the **totals are unchanged**, so no transmittance moves. The rows above 0.70 μm sit close enough to their own chemistry that the anchors do not resolve a background, and they carry their whole floor as gas.

The calibrated table, exactly as shipped (\_CALIBRATED\_GAS\_REGIONS):

| Region [μm]      | floor_od      | <i>k</i> | <i>b</i> | Region [μm]      | floor_od      | <i>k</i> | <i>b</i> |
|------------------|---------------|----------|----------|------------------|---------------|----------|----------|
| <b>0.30–0.45</b> | <b>0.0030</b> | 0.0000   | 1.000    | <b>3.10–3.50</b> | <b>0.1370</b> | 0.5824   | 0.457    |
| <b>0.45–0.70</b> | <b>0.0200</b> | 0.0025   | 0.874    | <b>3.50–5.00</b> | <b>0.4494</b> | 0.0944   | 0.808    |
| <b>0.70–1.30</b> | <b>0.0402</b> | 0.1245   | 0.434    | 5.00–7.50        | 1.3543        | 1.7850   | 0.530    |
| 1.30–1.50        | 0.0000        | 1.0933   | 0.327    | 7.50–8.00        | 0.9424        | 0.9210   | 0.673    |
| <b>1.50–1.75</b> | <b>0.0217</b> | 0.0282   | 0.645    | 8.00–9.40        | 0.1494        | 0.0992   | 1.204    |
| 1.75–2.05        | 0.0000        | 1.1186   | 0.216    | 9.40–9.90        | 0.8877        | 0.0409   | 1.701    |
| <b>2.05–2.40</b> | <b>0.0747</b> | 0.0320   | 0.843    | 9.90–10.00       | 0.3013        | 0.0379   | 1.805    |
| <b>2.40–3.10</b> | <b>0.7440</b> | 0.9666   | 0.560    | 10.00–12.00      | 0.0471        | 0.0602   | 1.750    |
|                  |               |          |          | 12.00–14.29      | 0.5956        | 0.1398   | 1.583    |

The two visible rows carry a companion aer\_bg\_od of 0.1232 and 0.1175 — the background aerosol split out of them, so their pre-CU-337 totals (0.1262 and 0.1375) are conserved.

Spectral shape *within* a region is flat: the model's contract is band-integrated fidelity, not line structure. Wavelengths outside 0.30–14.29  $\mu\text{m}$  clamp to the edge regions' calibration.

**The eight bold floor<sub>od</sub> values are a re-fit of the visible, near-infrared and short-wave rows.** floor<sub>add</sub> is the measured opacity *in excess of* Rayleigh + aerosol and is clamped at zero, so an over-large Rayleigh reference pins the visible rows to that clamp and leaves the model too transmissive. Only floor<sub>od</sub> moves in such a re-fit:  $k$  and  $b$  are solved in closed form from the MODTRAN ladder alone and never see the RADIANT model, so their bit-identity across a re-fit is the check that the calibration *reference* changed and the fit did not.

**The fit is evaluated on the ladder's own grid.** floor<sub>add</sub> is a *difference* of two band optical depths, and the two have to be weighted alike. MODTRAN's native grid is uniform in wavenumber ( $1\text{ cm}^{-1}$ , so  $\Delta\lambda \propto \lambda^2$ ); a uniform- $\lambda$  reference disagrees with it wherever the spectrum is steep, always toward an over-large floor — a few hundredths of an optical depth in the visible,  $\leq 0.0004$  beyond 1.3  $\mu\text{m}$ . The generator evaluates the reference on the ladder's grid, so the two weightings match.

The same convention removes a *coverage* mismatch in the first row. The delivered tape7 grid starts at 0.374953  $\mu\text{m}$ , so the measured 0.30–0.45  $\mu\text{m}$  optical depth is really the 0.375–0.45  $\mu\text{m}$  mean, while an unrestricted reference spans the whole region — including 0.30–0.375  $\mu\text{m}$ , where Rayleigh alone is enormous. Measured over the same interval the row reads 0.642 against the ladder's 0.768 and carries **0.1262**, within 0.014 OD of the 0.45–0.70  $\mu\text{m}$  floor: a continuous short-wavelength deficit rather than an artificial 0.16 OD step at the 0.45  $\mu\text{m}$  edge, which is physically what one expects, since whatever supplies that deficit does not know where a table boundary is. **Its residual limitation: the row is fitted from 0.375–0.45  $\mu\text{m}$  and applied across 0.30–0.45  $\mu\text{m}$ ,** because no anchor data exists below 0.374953  $\mu\text{m}$ .

At the fit's own anchor (us\_standard, rural 23 km, nadir full column) the model's 0.45–0.70  $\mu\text{m}$  band optical depth reads 0.4566 against MODTRAN's 0.4561 — **0.1 %**. RMS band-mean  $\tau$  parity across thirteen full-column anchors is 0.0111 on 0.45–0.70  $\mu\text{m}$  and 0.0292 on 0.40–0.90  $\mu\text{m}$ .

One residual is recorded rather than tuned away: 0.14 optical depths is far more than real 0.45–0.70  $\mu\text{m}$  gas chemistry supplies — the O<sub>3</sub> Chappuis band contributes ~0.03 — so part of the visible floor is standing in for an aerosol-model deficit. The band total is right; the attribution between gas and aerosol is not resolved by this fit, and the 0.30–0.45  $\mu\text{m}$  row landing beside it is further evidence of that.

**The 8.00–9.40 / 9.40–9.90 / 9.90–10.00  $\mu\text{m}$  partition is an ozone split.** One flat row across 8.00–10.00  $\mu\text{m}$  is a 2  $\mu\text{m}$  slab covering both the clean window and the O<sub>3</sub>  $\nu_2$  fundamental, so the model carries no identifiable ozone opacity at all: on a nadir full column  $\tau(9.60)$  and  $\tau(8.70)$  agree to six figures. Cutting the partition at the band edges the delivered ladder shows — water-free OD rises 0.24  $\rightarrow$  0.89 across 9.372–9.416  $\mu\text{m}$  and falls 0.52  $\rightarrow$  0.33 across 9.901–9.911  $\mu\text{m}$  — gives a band-core floor **5.9 $\times$  the adjacent window's**, which is the ozone. Two consequences follow directly from the table:

- The **ozone share of the in-feature floor is arithmetic**, not fitted:  $(0.8877 - 0.1494)/0.8877 = 0.832$ , the band-core floor above the continuum floor the neighbouring window measures.
- The floor still rides the **molecular** scale height, so the identified ozone is placed at ~8 km rather than ~25 km. Full columns barely notice; partial columns do. Placing it correctly is separate work that this split unblocks and does not itself perform.

The whole table is one vintage, regenerated from a single run of the generator, which reproduces its own output bit-for-bit on a re-run.

#### 4.2.6 The region-edge smoothstep blend

The table is piecewise-constant, but it is **not** read as a step function. Across each of the sixteen interior region edges the three coefficients  $c \in \{\text{floor\_od}, k, b\}$  are joined by a  $C^1$  smoothstep ramp of half-width  $h_w = 0.02 \mu\text{m}$  (full width  $0.04 \mu\text{m}$ ):

$$u(\lambda) = \text{clip}\left(\frac{1}{2} + \frac{\lambda - \lambda_{\text{edge}}}{2h_w}, 0, 1\right), \quad S(u) = u^2(3 - 2u), \quad c(\lambda) = c_{lo} + (c_{hi} - c_{lo}) S(u)$$

with  $S(0) = 0$ ,  $S(1) = 1$ ,  $S'(0) = S'(1) = 0$  — the ramp meets the flat calibrated regions with matching value *and* slope — and  $S(\frac{1}{2}) = \frac{1}{2}$ , so the edge itself carries the exact arithmetic mean of the two regions. At the  $0.70 \mu\text{m}$  edge, for instance,  $k$  evaluates to  $(0.0025 + 0.1245)/2 = 0.0635$  exactly, and `floor_od` to  $(0.1375 + 0.0402)/2 = 0.08885$  — real work on both sides of the edge, where an uncalibrated pair of zeros would have contributed nothing.

Outside the ramps nothing changes: a  $\lambda$  at or beyond  $h_w$  from every edge keeps the bit-identical calibrated coefficient. Every region is wider than  $2h_w$  — the narrowest is  $0.10 \mu\text{m}$  at  $9.90\text{--}10.00 \mu\text{m}$ , two and a half times the full ramp width — so no two ramps overlap, which is the invariant that stops a future refit from silently invalidating the blend.

Read literally, the step table made  $\tau(\lambda)$  jump at every edge and made a band-mean  $\tau$  that straddled an edge **sampling-grid-dependent**. Both effects, and the size of the adopted blend, are in the parity document §2.7.

$h_w$  is a documented module constant, not a schema parameter (the KOSCHMIEDER precedent): it is a numerical property of the calibration table, not a tuneable physical quantity.

#### 4.2.7 Air mass — plane-parallel inside $80^\circ$ , per-species spherical past it

The slant path length over the *absorbing* vertical extent of the segment is

$$L_{\text{slant}} = \frac{\Delta h_{\text{absorbing}}}{\cos \zeta}, \quad \Delta h_{\text{absorbing}} = \min(|h_{\text{sen}} - h_{\text{tgt}}|, h_{\text{atm,top}} - \min(h_{\text{sen}}, h_{\text{tgt}}))$$

so a ground site viewing a  $700 \text{ km}$  target traverses  $100 \text{ km}$  of air, not  $700 \text{ km}$ . The stored air mass is therefore exactly  $m = \sec \zeta$ , and a wholly exo path returns  $m = 1$  by the ExoAtmosphere convention.

$\sec \zeta$  is the honest plane-parallel primitive and it is used unchanged for  $\zeta \leq \text{SPHERICAL\_SWITCH\_RAD} = 80^\circ$ . Past  $80^\circ$  every column — the observer column, the full ground-to-sensor column, the solar column, and any `ColumnSegmentSpec` — hands over to the exact spherical slant integral, **per species**:

$$m_i = \frac{S_i(r_0; h_{lo} \rightarrow h_{hi}; H_i)}{\text{col}_i}, \quad r_0 = (R_E + h_{lo}) \sin \zeta_{lo}$$

where  $S_i$  is the density-weighted spherical column about the ray's perigee radius  $r_0$  (`grazing_column`, `grazing_slant_column_km`, a graded-grid quadrature anchored analytically against Chapman's grazing limit). The well-mixed-gas floor rides  $m_{\text{mol}}$ , because it is defined as a fraction of the molecular column.

Three properties matter, and all three are measured rather than asserted:

- **It has to be per species.** Water vapor’s 2 km profile hugs the tangent point far harder than the 8 km molecular one, so a single corrected scalar cannot serve both — at  $\zeta = 89.4^\circ$  the two diverge by  $2.27 \times$  in error.
- **The direction is toward more signal.** The spherical air mass is always the smaller one, so transmittance and SNR move **up** past  $80^\circ$  and never down.
- **It is a step, not a blend.** The size of that step, and why  $80^\circ$  rather than the  $89.5^\circ$  ceiling, are in the parity document §2.4.

The solar column’s old  $89.5^\circ$  clamp retires with the hand-over: the spherical route has no ceiling, so a twilight scene at  $\theta_s = 89.9^\circ$  now gets its own column. ZENITH\_CEILING\_RAD ( $89.5^\circ$ ) still bounds the *observer* zenith, because that is the column-air-mass validity ceiling for the plane-parallel form.

#### 4.2.8 One linearization convention — the slant column

Two calibrated terms — the water curve of growth and the gas floor — are *column* optical depths, the integral of no local coefficient. Wherever the model needs a **local** extinction (the single-scatter weights of *Single-scatter solar path radiance*, the level arm of *Path topologies*), they must be linearized: divided by a reference column to produce an equivalent per-km coefficient.

Because the curve of growth is sub-linear, that choice is not neutral. Linearizing against the vertical column instead of the slant one scales the effective water weight by  $m_{\text{h2o}}^{b-1}$ , and  $\omega_0$  with it, wherever water absorbs.

**All three evaluators now linearize against the slant column** — the amount actually traversed, which is what the linearization is *of*. `column_segment_optical_depth` publishes `slant_column_mol_km / _aer_km / _h2o_km` provenance under the same key names the near-horizon branch already used, so the convention is inspectable.

Note the scope precisely: this is the convention for the **linearized local weights**. The optical depth itself is still built as *vertical column*  $\times$  *species air mass* (*Air mass*); the two are consistent because  $m_i$  is defined as the ratio of the slant to the vertical column.

At  $\zeta = 0$  the air mass is exactly 1, so slant  $\equiv$  vertical and every vertical anchor is bit-identical across this change.

#### 4.2.9 Single-scatter solar path radiance

The scattered term on any column is the classic single-scatter source weighted by the column’s own emissivity-equivalent  $1 - \tau$ :

$$L_{\text{path,scat}}(\lambda) = \frac{E_{\text{sun}}(\lambda)}{4\pi} \cos \theta_s \omega_0(\lambda) P(\Theta, \lambda) [1 - \tau_{\text{atm}}(\lambda)]$$

with  $E_{\text{sun}}$  the top-of-atmosphere solar spectral irradiance and  $4\pi$  the full-sphere phase-function normalization.

**Single-scattering albedo.** Extinction-weighted, with the pure absorbers in the denominator only:

$$\omega_0(\lambda) = \frac{\sigma_{\text{mol}} + \omega_{\text{aer}} \sigma_{\text{aer}}}{\sigma_{\text{mol}} + \sigma_{\text{aer}} + \sigma_{\text{h2o}} + \sigma_{\text{gas}}}$$

**Phase function.** Combined by *scattering* cross-section — a photon absorbed by water does not contribute scattered radiance at all — and normalized so an isotropic scatterer gives  $P \equiv 1$ :

$$P(\Theta, \lambda) = \frac{\sigma_{\text{mol}} P_R(\Theta) + \omega_{\text{aer}} \sigma_{\text{aer}} P_{HG}(\Theta)}{\sigma_{\text{mol}} + \omega_{\text{aer}} \sigma_{\text{aer}}},$$

$$P_R(\Theta) = \frac{3}{4} (1 + \cos^2 \Theta), \quad P_{HG}(\Theta) = \frac{1 - g^2}{(1 + g^2 - 2g \cos \Theta)^{3/2}}, \quad g = 0.7$$

**Where the species proportions are evaluated — the lower endpoint.**  $\omega_0$  and  $P(\Theta)$  depend on the *relative* proportions of the four species, and those change with altitude. All evaluators take them at the segment's **lower endpoint**: the densest air in the path, the end the `L_toward_lower` product emerges from, and the choice the grazing and level evaluators always made.

The retired alternative — the segment's arithmetic-mean altitude — put the weights, for any column taller than  $\approx 40$  km, above the altitude where the aerosol and water coefficients underflow to zero in double precision. The consequence was not a small bias but a silent model substitution:  $\omega_0$  evaluated to exactly 1 (no absorption at all) and the Henyey-Greenstein forward peak collapsed onto the isotropic-Rayleigh 1.5, so a tall column scattered as if the atmosphere held no aerosol whatever `visibility_km` said. The adoption measurement is in the parity document §2.3.

This term is **provisional below 3  $\mu\text{m}$** . Single scattering under-predicts the daytime VIS/NIR sky, where multiple scattering dominates; a `UserWarning` fires when the evaluation grid extends below `SCATTERED_SKY_PROVISIONAL_MAX_UM = 3.0  $\mu\text{m}$`  and a solar geometry with the sun above the local horizon is supplied. A pure-thermal MWIR/LWIR call warns about nothing, and neither does a night scene on a VIS grid.

#### 4.2.10 Thermal path radiance — Kirchhoff emission at a height-resolved temperature

The thermal term on any column is a one-slab Kirchhoff graybody whose emissivity is derived from the column's own transmittance by Kirchhoff's law, never taken as an independent input:

$$L_{\text{path,therm}}(\lambda) = [1 - \tau_{\text{atm}}(\lambda)] B(\lambda, T_{\text{eff}}(\lambda))$$

It applies whether or not the sun is up — a night down-looking scene has `scattered`  $\equiv 0$  and `thermal`  $> 0$  — and one implementation serves both directions of travel.

**What  $T_{\text{eff}}$  is.** It is the single temperature that makes this one-slab form reproduce the **layered formal solution** of the segment's own non-isothermal air. Slice the segment into  $N$  sub-layers; layer  $i$  has slant optical depth  $\delta_i(\lambda)$  and temperature  $T_i$ . Ordering the layers **from the end the radiation escapes**, the emergent radiance is the discrete formal solution of the non-scattering LTE transfer equation:

$$L(\lambda) = \sum_i B(\lambda, T_i) (1 - e^{-\delta_i(\lambda)}) e^{-c_i(\lambda)}, \quad c_i(\lambda) = \sum_{j < i} \delta_j(\lambda)$$

The weights telescope exactly:

$$\sum_i (1 - e^{-\delta_i}) e^{-c_i} = 1 - e^{-\sum_i \delta_i} = 1 - \tau(\lambda)$$

so  $L = (1 - \tau) \langle B \rangle$  with  $\langle B \rangle$  a convex combination of the layer Planck functions, and

$$T_{\text{eff}}(\lambda) = B^{-1}(\langle B \rangle(\lambda))$$

is guaranteed to lie between the coldest and warmest layer in the segment.

Three properties are structural, not fitted:

- **The total optical depth is untouched.** Only the altitude the emission is weighted at changes, so every  $\tau$  in the model is bit-identical across this construction.
- **Isothermal is exact.** Every  $T_i$  equal returns that temperature exactly, for every  $\tau$ , every direction and every layer count. A level arm is isothermal, so the level topologies keep a single exact graybody temperature.
- **Direction is geometry, not a fork.** escape names which endpoint the radiance leaves from. One model serves both directions: the down-looking evaluate term uses escape="upper" and the up-looking segment product escape="lower". A direction-blind optical-depth-weighted mean temperature was measured against the same anchors and rejected — it degrades the MWIR everywhere, which is the measurement that forces escape into the model.

**Where the opacity sits in altitude.** The curve of growth fixes each species' *total* column optical depth and says nothing about its vertical distribution, which is what the weighting needs. That distribution is taken from first principles, with no fitted coefficient:

- **Scattering species** (Rayleigh, aerosol) — extinction is proportional to number density, so the weighting profile is the species' own density scale height, unchanged.
- **Pressure-broadened absorbers** (the well-mixed-gas floor, water vapor) — a Lorentz line's absorption coefficient goes as number density *times* the collisional half-width, and the half-width goes as total pressure, so  $\alpha \propto \rho_a p_{\text{air}}$  and the emission weighting rides the harmonic combination

$$H_{\text{emit}} = \left( \frac{1}{H_a} + \frac{1}{H_{\text{air}}} \right)^{-1}$$

which takes the well-mixed floor from 8 km to **4 km** and water from 2 km to **1.6 km**.

- **Layered absorbers** (ozone) — a species produced photochemically in the mid stratosphere has no scale height at all, so it is placed on a Gaussian layer instead:

$$\rho_{\text{O}_3}(z) \propto \exp\left(-\frac{1}{2} \left(\frac{z - z_0}{w}\right)^2\right), \quad z_0 = 25 \text{ km}, \quad w = 5 \text{ km},$$

the US Standard Atmosphere ozone profile's peak and half-width.

**The ozone split.** The calibrated `floor_od` is one number per spectral region covering CO<sub>2</sub>, N<sub>2</sub>O, CH<sub>4</sub> and O<sub>3</sub>, and only the first three are well mixed. Inside the 9.40–9.90 μm O<sub>3</sub> ν<sub>2</sub> region the floor is therefore partitioned before it is placed:

$$\text{share}_{\text{O}_3}(\lambda) = 1 - \frac{\text{floor}_{\text{continuum}}(\lambda)}{\text{floor}(\lambda)} = \frac{0.8877 - 0.1494}{0.8877} = 0.8317 \quad \text{inside the band,}$$

where `floorcontinuum` is the same region table evaluated with the band's row carrying its clean-window neighbor's floor. That share of the gas floor rides the ozone layer; the remainder keeps the 4 km pressure-broadened placement. Three properties follow:

- **the share is arithmetic, not fitted** — it is read off the two committed table rows (the well-mixed-gas floor's 8.00–9.40 and 9.40–9.90 μm regions), so a re-fit moves it automatically and there is no coefficient in the emission model to go stale. An absorption *band* stands on top of the continuum its neighbors carry, so the excess — not the total — is ozone;
- **placement is continuous in λ** — both floors pass through the same region-edge smoothstep, so the share inherits its  $C^1$  ramp at 9.40 and 9.90 μm rather than needing a second ramp implementation that could drift from the first;
- **τ is untouched** — the two parts sum to the floor the model already had. This is a redistribution in altitude and nothing else, as every placement rule here is.

Outside the band the share is exactly zero, no layer is constructed, and  $T_{\text{eff}}$  is bit-identical to the four-species form. The 9.90–10.00 μm long-wave tail (floor 0.3013, 3.3× its continuum) is a **documented exception**: part of it is ozone too, but the split is scoped to the band core the parity is measured in, so a narrow-band product centered on 9.95 μm still places its ozone at 4 km.

The sub-layer count is a convergence-tested quadrature parameter, not a tuning knob: `EMISSION_LAYERS_PER_SPECIES` = 32, whose discretization error against a 512-per-species reference is  $\max |\Delta T_{\text{eff}}| = 0.016$  K over the whole anchor set — two orders of magnitude below the model's own  $\approx 4$  K accuracy against MODTRAN. It is therefore fixed, not exposed as a user parameter.

**Vacuum limit.** A segment with no opacity has no emission to weight; the function falls back to the temperature of its densest air so the value stays finite. The radiance it multiplies is exactly zero there, so it is unobservable — it exists only so nothing returns NaN.

#### 4.2.11 The hemispheric downwelling terms

Two products describe what falls on a *surface* — hemispheric irradiances, not directional radiances, and consumed by the reflected-diffuse terms of the target and ground-background arms.

**Thermal.** A target-anchored graybody on the **vertical** target  $\rightarrow h_{\text{atm,top}}$  column — the sky the target actually sees, so the sensor's altitude and viewing zenith deliberately do not enter a hemispheric flux at the target:

$$E_{\text{sky,thermal}}(\lambda) = \left[1 - \tau_{\text{sky,vert}}(\lambda)^D\right] \pi B(\lambda, T(h_{\text{tgt}} + z_{\text{em}})), \quad L_{\text{atm,down}}(\lambda) = \frac{E_{\text{sky,thermal}}(\lambda)}{\pi}$$

$T(\cdot)$  is the fixed-lapse ICAO standard-atmosphere lookup (6.5 K/km, floored at the 216.65 K tropopause above 11 km). The emission-height offset  $z_{em} = 200$  m is fitted to the real up-looking MODTRAN H-runs (downwelling is dominated by near-surface air).

**The flux-diffusivity exponent  $D$  is not fitted.** It is a geometric identity,

$$D = \sec 48.2^\circ = 1.50030 \dots,$$

the secant of the diffusivity angle the entire downwelling reference set was run at: every reference deck is an up-looking MODTRAN run at  $48.2^\circ$ , and the quantity this model is scored against is  $\pi L(48.2^\circ)$  — the hemispheric-flux proxy, itself validated to  $\sim 15\%$  against a true hemispheric-flux table. Scoring against a  $48.2^\circ$  pencil and then weighting the emissivity by any other factor mixes two different hemispheric approximations; taking the exponent from the reference geometry removes a free parameter instead of re-tuning one. (The textbook Elsasser diffusivity factor 1.66 is  $\sec 53.13^\circ$  — a different angle and a different approximation, not what this reference set uses.)

Because  $z_{em}$  is fitted *through this one closed form*, a directional path-radiance product cannot inherit it. That is why the height-resolved thermal path-radiance model and this graybody coexist deliberately: they are different products, not two versions of one.

**The obvious refinement was measured and declined.** Computing the downwelling directly — evaluating the sky column’s emergent radiance at  $48.2^\circ$  from the height-resolved machinery, and retiring both fitted constants — was scored against a nine-rung ladder of measured hemispheric-proxy downwelling from 0 to 80 km, as  $\text{RMS} |\ln(\text{model}/\pi L_{\text{MODTRAN}})|$  over nine rungs  $\times$  two bands. The shipped pairing of  $\sec 48.2^\circ$  with the near-surface emission height scores **1.9233**; swapping in the layered temperature scores 1.9385, and restricted to the four tropospheric rungs — the only ones where the two temperatures differ, the standard profile being isothermal above the tropopause — it is worse still, 0.3087 against 0.4771.

The band split behind that is worth knowing, because it says what would have to change first. Isolating the temperature shows the layered form is a  $3.8\times$  improvement in the LWIR — the near-surface proxy runs 5–22 K too warm on a semi-transparent window that genuinely samples the whole column — and a  $3.7\times$  degradation in the MWIR, running 3.5–10.6 K too cold. With the region-flat 3–5  $\mu\text{m}$  gas floor supplying too little opacity, the layered weighting reaches too high into cold air, and the fitted warm bias was compensating for a  $\tau$ -shape error rather than a temperature error. The layered temperature therefore waits on a finer MWIR gas model, not on the emission-placement machinery.

Adopting  $\sec 48.2^\circ$  raises the downwelling effective emissivity everywhere — by  $\sec 48.2^\circ / 1.1 = 1.364\times$  in the optically thin limit, and asymptotically not at all where the sky column already saturates — so every reflected-sky term rises.

**Scattered.** On the same vertical slab:

$$E_{\text{sky,scattered}}(\lambda) = E_{\text{TOA}}(\lambda) \cos \theta_s \omega_{0,\text{eff}}(\lambda, \text{aerosol}) [1 - \tau_{\text{down,vert}}(\lambda)]$$

$\omega_{0,\text{eff}}$  is **not** the internal column  $\omega_0$  of *Single-scatter solar path radiance*. It is a MODTRAN-derived effective single-scattering albedo: band-median values per aerosol regime over VIS 0.4–0.7 / NIR 0.7–1.4 / SWIR 1.4–2.5  $\mu\text{m}$ , edge-extended outside, obtained by inverting *this closed form* against the real ground-level diffuse-

flux tables — so it absorbs MODTRAN’s multiple-scatter contribution as an effective parameter of this formula. Examples: rural 0.791 / 0.698 / 0.187; urban 0.423 / 0.430 / 0.263.

The two  $\omega_0$  definitions coexist on purpose: the flux-fit table is not a valid substitute inside the phase-function-weighted  $L_{\text{path}}$  integral, and the internal column  $\omega_0$  over-predicted diffuse sky irradiance by  $\approx 1.3\times$  (VIS rural) to  $\approx 5\times$  (SWIR urban).

#### 4.2.12 Path topologies — segments, arms, and whole paths

Every non-trivial path is a **composition of path segments**: one piece of path between two points, evaluated once, read from either end. Two spec types, because there are two topologies and they are not variations of one form:

| Spec              | Fields                                                            | Topology                                                         | Air mass        |
|-------------------|-------------------------------------------------------------------|------------------------------------------------------------------|-----------------|
| ColumnSegmentSpec | $h_{\text{low}}, h_{\text{high}}$ [m], $\zeta_{\text{low}}$ [rad] | <b>endpoint-minimum</b> — the path’s lowest point is an endpoint | <i>Air mass</i> |
| LevelArmSpec      | altitude [m], length [m]                                          | <b>interior-tangent</b> — the lowest point is in the middle      | none at all     |

One evaluated product, `SegmentQuantities`, carries **one**  $\tau$  and **two** directional radiances  $L_{\text{toward upper}}$ ,  $L_{\text{toward lower}}$ . Transmittance is reciprocal, so a segment has one  $\tau$  no matter which way it is read; path radiance is not, because the emitting and scattering layers are weighted by the transmittance of the material *between* them and the receiver, and that weighting reverses with direction.

**The lower-endpoint convention.** Every column segment’s zenith is keyed to its **lower** endpoint. Two structural reasons: it is the one endpoint the two travel directions share, so a single scalar describes the segment rather than the reading of it; and it is the angle MODTRAN’s Card 3 wants when  $H_1 \leq H_2$ , so no convention translation sits between RADIANT and its truth source.

**The level arm.** A level path cannot be served by the column machinery at all, and the reason is structural rather than accuracy: a column’s optical depth is  $\int e^{-h/H} dh$  between two equal altitudes, which is **exactly zero**, and its air mass is  $\sec(\pi/2)$ , which is undefined. The arm is instead

$$\tau(\lambda) = \exp[-\alpha(\lambda, h) L], \quad L_{\text{path}}(\lambda) = [1 - \tau(\lambda)] B(\lambda, T_{\text{eff}}) + \text{single-scatter source}$$

with  $\alpha$  the **local** extinction at the arm’s altitude [ $\text{km}^{-1}$ ] and  $L$  the **true spherical chord** between the endpoints [km] — not a flat-Earth range. No new calibration is introduced: the water and gas terms are linearized exactly as *One linearization convention* prescribes, against a reference column independent of the arm’s own length, which is what makes  $\alpha$  a property of altitude alone and  $\tau$  a pure exponential in  $L$ .

The arm’s one approximation is constant density. On a spherical Earth the chord dips below its endpoints by the tangent-height depression

$$\Delta h \approx \frac{L^2}{8R_E}, \quad \text{mean sag over the chord} \approx \frac{2}{3}\Delta h$$

so the real path samples slightly *denser* air and the model **under-states** optical depth. The horizon guard bounds the error:  $\Delta h < 100$  m computes clean, up to 2 km warns, beyond 2 km raises. On the 2 km water

scale height that is a 3.4 % water-density error at the clean edge, 6.8 % on the longest grid arm ( $L = 100$  km), and a factor  $\approx 1.9$  at the raise threshold — which is why 2 km raises rather than warns.

$\tau(2L) = \tau(L)^2$  holds exactly for this arm, and it is precisely where a correlated- $k$  band model disagrees: strong lines saturate first and flux keeps leaking through the windows between them. That divergence is measured against the horizontal MODTRAN grid in the parity document §2.6.

**The whole level path.** The sky behind a level target is *not* an arm composed with a continuation joined at the target plane. A level ray is tangent at the chord **midpoint**, so the sensor sits on its descending half, and the traversed column is, per species,

$$S_i = 2S(r_p; h_p \rightarrow h_{\text{arm}}) + S(r_p; h_{\text{arm}} \rightarrow h_{\text{top}}), \quad r_p = \sqrt{r_{\text{arm}}^2 - (L/2)^2}$$

evaluated once, in one place. Rooting a single ascending arc at the sensor — the up-looking branch’s shape — would drop the arm entirely, recovering only 83.0 % of the true traversed molecular column for a 100 km arm at 3 km and 75.1 % for a 150 km arm at 10 km. A zero-length arm reduces the whole-path evaluator **exactly** to the grazing evaluator at  $\zeta = \pi/2$ , so the level and ascending sky topologies join without a step. The 8 km-at-sea-level case is degenerate — its perigee sits 1.3 m *below* the ellipsoid — so the integration floor is clamped at MSL and the model warns.

**Sky radiance along the LOS.** The radiance a receiver at  $h_{\text{start}}$  sees looking up along a ray of zenith  $\zeta$  with nothing behind the atmosphere but cold space:

$$L_{\text{sky}}(h_{\text{start}}, \zeta) = \text{SegmentQuantities}(\text{ColumnSegmentSpec}(h_{\text{start}}, h_{\text{atm, top}}, \zeta)) \cdot L_{\text{toward lower}} + \tau_{\text{seg}} L_{\text{beyond}}$$

with  $L_{\text{beyond}} \equiv 0$  — there is nothing behind the atmosphere but cold space.

The 2.7 K cosmic background contributes  $< 10^{-9}$  W/m<sup>2</sup>/sr/μm anywhere in the 0.3–14 μm working range, so the composition is a no-op and the module is a thin, well-named wrapper rather than a second physics implementation. **The ray starts at the sensor, not the target:** a background behind a target cannot depend on where along the ray the target sits, and splitting one column at the target plane swaps part of a warm ground-anchored graybody for a cold target-anchored one — the segment model is not additive across a join.

**Composition rules.** For two adjoining segments  $A$  (nearer the source) and  $B$ :

$$\tau(A \cup B) = \tau_A \tau_B, \quad L(A \cup B) = L_A \tau_B + L_B$$

which, with the vacuum identities  $\tau_V \equiv 1$ ,  $L_V \equiv 0$ , collapse an exo-altitude target’s path to the published fields with no arithmetic performed at all.

#### 4.2.13 Twilight — per-altitude illumination and the solar tangent transit

Whether a point is lit is decided per altitude rather than by a global  $\theta_s < \pi/2$  bound, so solar zenith spans the closed  $[0, \pi]$ :

$$\text{sunlit}(h, \theta_s) \iff \theta_s \leq \frac{\pi}{2} \quad \text{or} \quad (R_E + h) \sin \theta_s \geq R_E \iff h \geq R_E (\sec \delta - 1), \quad \delta = \theta_s - \frac{\pi}{2}$$

so a 60 km booster is sunlit at 5° solar depression while the ground beneath it (shadow height  $\approx 24$  km) is not. The assumption is a **sharp terminator** — opaque sphere, point Sun, no refraction; the  $\approx 200$  m penumbral blur and the  $\approx 0.5^\circ$  of unmodeled refractive lift are documented, not smoothed.

For a **sunlit** target with  $\theta_s > \pi/2$  the direct beam is a tangent transit, not a descending column, so the solar transmittance is the two-arm decomposition

$$\tau_{\text{sun}} = \tau(\text{tangent} \rightarrow \text{target}) \cdot \tau(\text{tangent} \rightarrow \text{TOA})$$

about the tangent radius  $r_0 = (R_E + h_{tgt}) \sin \theta_s$ . For a **shadowed** target  $\tau_{\text{sun}}$  is exactly 0 and the scattered-sky solar component is already identically zero; the thermal sky is untouched.

**This branch is provisional.** The twilight transit carries the largest optical depths anywhere in RADIANT (30–70 air masses), where both the exponential-in-column transmittance and the unmodeled refraction are at their worst. The twilight pair Q7/Q8 *was* delivered — with Card-3 ANGLE hand-set to 93° / 96° and LENN = 1, and the hand edit verified against the matrix by the Card-3 echo sweep — but both rows are dev\_only: no library family and no radiometric parity test consumes them, so the transit’s **transmittance remains unanchored** (parity document §1 and §3). Treat it as an order-of-magnitude bound.

Note also that RADIANT models the target as a horizontal Lambertian facet, so the direct solar term is multiplied by  $\cos \theta_s$  clamped at zero: for any  $\theta_s > \pi/2$  the direct term vanishes regardless of  $\tau_{\text{sun}}$ , because the beam arrives from below the facet.  $\tau_{\text{sun}}$  is still published correctly, because it is an inspectable physical quantity that a non-horizontal target model would consume.

## 4.3 The library-backed models

### 4.3.1 What a library family is

A family is a set of pre-computed MODTRAN states at discrete geometry points, plus a declaration of which geometry fields are **interpolation axes**. Each stored node carries a wavelength grid, a transmittance, a path radiance, a downwelling emission array, and the full five-field run geometry it was rendered at.

Shipped families, with their axes and direction:

| Family                       | Direction | Axes                                                  | Coverage                                             |
|------------------------------|-----------|-------------------------------------------------------|------------------------------------------------------|
| us_standard_zenith_fan       | down      | path_zenith_rad                                       | ground target, sensor 100 km, $\zeta$ 0–60°          |
| midlat_summer_ladders        | down      | sensor_altitude_m, target_altitude_m                  | targets 0–29 km; sensor 35 km / 100 km / GEO; nadir  |
| midlat_summer_sensor_ladder  | down      | sensor_altitude_m                                     | ground target; sensor 3–100 km + GEO; nadir          |
| midlat_summer_boost_offnadir | down      | sensor_altitude_m, target_altitude_m, path_zenith_rad | targets 0–100 km; sensor 100 km / GEO; $\zeta$ 0–60° |

| Family                                | Direction | Axes                                | Coverage                                                                 |
|---------------------------------------|-----------|-------------------------------------|--------------------------------------------------------------------------|
| midlat_summer_upwelling_offnadir      | down      | sensor_altitude_m,path_zenith_rad   | ground target; sensor 10/100 km / GEO; $\zeta$ 0–60°                     |
| midlat_summer_boost_ladder            | down      | sensor_altitude_m,target_altitude_m | targets 0–100 km, 12 rungs; explicit-dir only                            |
| midlat_summer_uplooking_ladder        | up        | target_altitude_m                   | ground sensor, targets 0–20 km, vertical                                 |
| midlat_summer_uplooking_zenith_fan    | up        | target_altitude_m,path_zenith_rad   | ground sensor, targets 0–20 km, $\zeta$ 0–60°                            |
| midlat_summer_uplooking_sensor_ladder | up        | sensor_altitude_m                   | observer 0–100 km, full column, fixed 48.2°                              |
| midlat_summer_sst_column_fan          | up        | path_zenith_rad                     | ground sensor, full column, $\zeta$ 0–78.5° (sec 1–5); explicit-dir only |
| midlat_summer_sst_column_fan_site900m | up        | path_zenith_rad                     | 900 m site, full column, $\zeta$ 0–78.5° (sec 1–5); explicit-dir only    |

Direction is a first-class property. An up-looking run measures one travel direction, so an up-looking family stores its radiance under a **different NPZ key** (`path_radiance_toward_lower`), carries a `los_direction` marker, and is served through a different query entry point; the two entry points refuse each other's families rather than reading the wrong product.

The two SST fans differ only in their rendered lower endpoint (0 m and 900 m) and are **siblings, not one family with a sensor axis**: two rungs cannot be interpolated between, the 900 m block has no 48.2° rung to make a rectangular grid with, and the 0 m fan had to stay byte-identical. A site at any other elevation is served by simple.

#### 4.3.2 Geometry interpolation in $\log\tau$

Transmittance is interpolated in **optical-depth** space, not in  $\tau$ :

$$\ln \tau_{\text{query}} = \text{interp}(\text{axes}; \ln \tau_{\text{nodes}}), \quad \tau_{\text{query}} = \exp(\ln \tau_{\text{query}})$$

Beer-Lambert gives  $\tau = e^{-\text{OD}}$  with OD linear in path length and absorber amount, so linear interpolation of  $\ln \tau$  preserves that linearity and linear interpolation of  $\tau$  does not.

Path radiance and downwelling emission interpolate **linearly**. They are additive quantities with no Beer-Lambert exponential in path length, so log space would buy nothing and would misbehave at zeros.

**No extrapolation.** A query outside the convex hull of the nodes always raises. The one query served from outside the node span is not an exception but an application of the rule — see *Vacuum-equivalence identities*.

#### 4.3.3 Wavelength resampling in $\log\tau$ , and why the two must commute

A chain runs on its own wavelength grid, which is normally not the stored one, so the geometry-interpolated spectra must be resampled. That resample runs in the **same**  $\log\tau$  space:

$$\tau(\lambda_{\text{query}}) = \exp[\text{interp}_{\lambda}(\ln \tau_{\text{stored}})]$$

Both operations are then linear in  $\ln \tau$ , so they **commute** and the answer no longer depends on their order. Resampling linearly in  $\tau$  instead returns the arithmetic mean of the bracketing samples where the physics gives the geometric mean — and by AM–GM the arithmetic mean is strictly larger, so the old convention was biased one way, upward.

The key equation the tests use as their truth anchor: for  $\tau(\lambda) = e^{-a\lambda}$  the optical depth is *exactly* linear in  $\lambda$ , so a log-linear resample is exact at every query wavelength and a cell midpoint must return  $\sqrt{\tau_i \tau_{i+1}}$ .

**The convention is universal across all three file-backed backends.** `tabulated` and `modtran` each perform only one resample — no operation order to get wrong — but a different *convention* meant the same stored MODTRAN column returned different numbers depending on which backend served it. All three now carry every  $\tau$ -like array (`transmittance`,  $\tau_{up}$ ,  $\tau_{sun}$ ,  $\tau_{full,up}$ ) through one shared implementation. Scope, in both directions:

- **log- $\tau$** : transmittance only, in every backend.
- **linear**:  $L_{path}$ ,  $L_{atm,down}$ , and every irradiance.

A query on the file’s own grid short-circuits the log round trip and is **bit-identical** to the stored array.

#### 4.3.4 Zenith axes interpolate in $\sec \zeta$

A zenith-angle axis is mapped  $\zeta \mapsto \sec \zeta$  before interpolation. Combined with the log- $\tau$  geometry interpolation above, this is Beer-Lambert-**exact** between nodes: the path length along a zenith axis is  $\propto \sec \zeta$ , so  $\ln \tau$  linear in  $\sec \zeta$  reproduces Beer at every query angle. Linear-in-angle carried a measured in-band bias at fan midpoints (parity document §2.10).

The transform is internal — user-facing coordinates, bounds and errors stay in radians. Angles  $\geq \text{\_MAX\_ZENITH\_RAD}$  ( $\approx 88.8^\circ$ ) are **refused**, because  $\sec \zeta$  diverges at the horizon.

Batch-2 gave the up-looking direction the same axis, sampled as a *uniform sec ladder* rather than a uniform angle ladder:  $\sec \zeta = 1.0/1.4999/2.0$  for the target fan and  $1/1.5/2/3/4/5$  for the full-column SST fan. Three sec rungs is the minimum that can **test** linearity in  $\sec$  rather than assume it. The  $85^\circ/88^\circ/89.5^\circ$  probes (M6–M8) were run but are deliberately **not shipped as nodes**: at and past the  $88.8^\circ$  ceiling the mapping is unvalidated, and a shipped node there would sit inside a hull the interpolator may traverse. They serve as physics anchors instead (parity document §2.4).

#### 4.3.5 TAU\_FLOOR — a lower clamp only

Stored transmittance is floored at  $\text{TAU\_FLOOR} = 10^{-30}$  (equivalently  $\text{OD} \approx 69$ ) before the logarithm, so  $\ln \tau$  is finite everywhere and an opaque band resamples to that floor rather than to  $-\infty$  or NaN. One definition, imported by every consumer.

The floor is deliberately **not** matched by a cap at 1.0. A  $\tau > 1$  array is invalid data, and capping it would convert a mis-scaled tape7 into a plausible-looking column — exactly the kind of silent repair RADIANT forbids. Instead it survives the resample and fails loudly downstream. Negative  $\tau$  raises inside the resample itself.

### 4.3.6 Vacuum-equivalence identities

Two identities let a family serve a query outside its node span *without* extrapolating, because the intervening path is provably vacuum. Both are gated on the shared constant `_VACUUM_EQUIVALENT_ALTITUDE_M = 100 km =  $h_{\text{atm,top}}$` .

**Sensor axis.** MODTRAN’s atmosphere ends at  $h_{\text{atm,top}}$ , so a sensor above a node at or above the column top sees an *exactly identical* column — the added path has zero extinction and zero emission. This is why the ladders duplicate their 100 km TOA state at a 40,000 km node: an orbital sensor then falls inside the hull, and the duplication is a physical identity, not an approximation. The same identity exempts such a query from the non-axis geometry-mismatch warning.

**Target axis.** The mirror. A family whose measured target ceiling reaches  $h_{\text{atm,top}}$  integrated the *entire* column, so its top-of-column run **is** the observer leg for any exo-altitude target; the query is clamped to the ceiling node and the clamp is recorded in the segment provenance under `exo_target_vacuum_clamp`.

The guard asks the **family**, never a hard-coded name: `uplooking_target_ceiling_m` reads the highest target altitude the family’s own runs measure — from the `target_altitude_m` axis hull when it carries one, from the recorded fixed value when it does not. A **partial-column** family (ceiling below  $h_{\text{atm,top}}$ ) is **refused**, with an actionable error naming its measured ceiling and the families that do serve the scene: composing its top rung with the vacuum identity would join a measured leg to an invented one. The clamp is gated at  $h_{\text{atm,top}}$  and nowhere else — a 50 km target through a 20 km ladder is 40 km of real atmosphere and still fails the hull check, unchanged.

### 4.3.7 The hybrid up-looking composition

An up-looking run family is **one leg of data**, so an up-looking chain run on the interpolated backend is a declared hybrid: the family serves the leg it measured, and a `SimpleAtmosphere` companion — built pre-chain from the same `atmosphere.*` parameters a `model = "simple"` run would use — serves the rest.

| Leg                                                | Served by                                   | Why                                                                                                                                                                  |
|----------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| observer (sensor → target)                         | the up-looking run family                   | this <i>is</i> the rendered column, and it dominates a ground-to-air scene                                                                                           |
| illumination (solar column + sky above the target) | the <code>SimpleAtmosphere</code> companion | no rung of a sensor→target ladder is the column <i>above</i> the target, and the down-looking proxy query an up-looking family would need is refused by construction |
| sky at aperture (sensor → $h_{\text{atm,top}}$ )   | the <code>SimpleAtmosphere</code> companion | reading a partial ladder’s top rung as “the sky” would be extrapolation past the hull                                                                                |

Two independently-calibrated models in one answer is a real modeling compromise, so it is never silent: a `UserWarning` is raised, an `INFO` record is logged, and `stage_outputs["atmosphere"]["topology_provenance"]["backend_split"]` names which leg came from which model. Where the two models must agree —  $\tau_{\text{sun}}$ ,  $E_{\text{TOA}}$ , and both  $E_{\text{sky}}$  terms, all served by the companion alone — they are bit-identical.

**The compromise is accepted only while it stays declared.** The warning, the `INFO` record and the `backend_split` marker are part of what makes it acceptable and must never be softened into silence. The measured divergence between the two legs is in the parity document §2.11.

The one re-audit condition: the split exists because an up-looking family is one leg of data. A family that is self-contained — carrying its own solar column and its own sensor  $\rightarrow h_{\text{atm,top}}$  sky — or a scene whose target is a blackbody, where the illumination terms vanish, makes the companion unnecessary *for that scene*, and the split should be dropped there rather than declared.

A **level** path on an up-looking family is refused, not approximated: a level arm has zero vertical extent and a local zenith of  $\pi/2$  everywhere, so no rung of a column ladder is that path and no interpolation between rungs produces it.

#### 4.4 Optical turbulence — from $C_n^2(h)$ to $r_0$

The spatial chapter consumes one number: the Fried parameter  $r_0$  at the band-centre wavelength of the scene’s spectral grid. This section is where that number comes from.

**Resolution policy** (`atmosphere.cn2_profile`). `direct` (the default) uses `atmosphere.r0_m` exactly as entered — 0 means turbulence off — rescaled to the band centre when `atmosphere.r0_reference_wavelength_um` says what wavelength it was quoted at:

$$r_0(\lambda_b) = r_0(\lambda_{ref}) \left( \frac{\lambda_b}{\lambda_{ref}} \right)^{6/5}.$$

`hufnagel_valley` and `tabulated` derive  $r_0$  from a profile by the path integral below. An explicit `r0_m` entered beside a profile wins, and the profile becomes a cross-check: the two must agree within 1 % or the resolver raises. A profile beside an explicit `r0_m = 0` is a contradiction (“compute the turbulence” and “there is none”) and raises.

**Path integral.** With  $k = 2\pi/\lambda$  and  $\zeta$  the LOS zenith angle at the segment’s *lower* endpoint,

$$r_0 = \left[ 0.423 k^2 \sec \zeta \int_{h_{low}}^{h_{high}} C_n^2(h) W(h) dh \right]^{-3/5},$$

integrated over the part of the line of sight inside the atmosphere — the endpoint altitudes clipped to  $[0, h_{\text{atm,top}}]$  — as a plane-parallel slab,  $\sec \zeta$  turning vertical thickness into path length. The weighting  $W$  is the wave type (`atmosphere.turbulence_wave_type`):  $W = 1$  for `plane` (the default: the source is effectively at infinity, the imaging case and the convention every published  $r_0$  uses) and  $W = u^{5/3}$  for `spherical`,  $u$  the fraction of the way from the *target* to the aperture — the finite-range point-source coherence radius, in which turbulence near the sensor dominates. A level arm is not a column ( $\sec \zeta$  diverges): it integrates along its length  $L$  at its altitude,  $\int C_n^2 W ds = C_n^2(h) L \times \{1, 3/8\}$  for plane and spherical, since  $\int_0^1 u^{5/3} du = 3/8$ . A space sensor has no column above it: the integral is empty,  $r_0$  saturates at a large sentinel flagged *negligible*, and the turbulence term is omitted rather than multiplied in as unity.

**Hufnagel-Valley profile** (`hufnagel_valley`):

$$C_n^2(h) = 0.00594 \left( \frac{w}{27} \right)^2 \left( 10^{-5} h \right)^{10} e^{-h/1000}$$

$$\bullet 2.7 \times 10^{-16} e^{-h/1500} + A e^{-(h - h_{\text{site}})/100}$$

with  $h$  in metres above mean sea level and  $C_n^2$  in  $\text{m}^{-2/3}$ : a jet-stream term scaled by the RMS upper-atmosphere wind  $w$  (`atmosphere.cn2_hv_wind_rms_m_s`; the  $h^{10}e^{-h/1000}$  shape peaks at exactly 10 km), a fixed middle-atmosphere term, and a 100 m-scale-height surface layer scaled by the ground strength  $A$  (`atmosphere.cn2_hv_ground_strength`) and referenced to the site elevation `geometry.site_elevation_m`. The defaults are HV-5/7 —  $w = 21 \text{ m/s}$ ,  $A = 1.7 \times 10^{-14} \text{ m}^{-2/3}$  — named for the  $r_0 = 5 \text{ cm}$  and  $\theta_0 = 7 \text{ } \mu\text{rad}$  it produces at  $0.5 \text{ } \mu\text{m}$  on a vertical path.  $A$  is a site-quality parameter: solve the path integral for the  $A$  that reproduces a site’s measured seeing.

**Tabulated profile** (`tabulated`, `atmosphere.cn2_tabulated_file`): a two-column altitude /  $C_n^2$  table interpolated *linearly in log  $C_n^2$*  against altitude — exponential inside each interval, the shape every analytic profile has — with a linear fallback across a zero-valued node. Outside the table the profile is zero; the integral warns with how much of the path was uncovered rather than extrapolating a power law it was not given.

**Units.**  $C_n^2 [\text{m}^{-2/3}] \times dh [\text{m}] = \text{m}^{1/3}$ ;  $\times k^2 [\text{m}^{-2}] = \text{m}^{-5/3}$ ; to the  $-3/5$  power gives metres.

**Assumptions & validity.** Kolmogorov statistics; plane-parallel slab (the profile is evaluated against altitude, so the integral is exact only away from the horizon — the same zenith ceiling the column models carry); HV is a climatological fit, not a measurement, and its surface term describes daytime sea-level turbulence unless  $A$  is re-derived for the site.

**Pitfalls.** Applying an  $r_0$  quoted at  $0.5 \text{ } \mu\text{m}$  unscaled in the infrared (at  $4 \text{ } \mu\text{m}$  it is  $12\times$  larger, so the MTF is badly pessimistic); reversing the spherical weighting (it peaks at the aperture, not the source); carrying the HV-5/7 ground strength to a mountain site now that the surface term is site-referenced, which double-counts the elevation; reading  $\zeta$  at the wrong endpoint of an up-looking path.

**Numeric anchors.** HV-5/7, vertical, plane wave,  $0.5 \text{ } \mu\text{m}$ :  $r_0 = 5 \text{ cm}$  (the preset’s defining value);  $\int_0^1 u^{5/3} du = 3/8$  exactly;  $r_0(4 \text{ } \mu\text{m})/r_0(0.5 \text{ } \mu\text{m}) = 8^{6/5} = 12.1257$ ; the Paranal median of  $0.80 \text{ arcsec}$  at  $2635 \text{ m}$  is reproduced by  $A = 2.70 \times 10^{-15} \text{ m}^{-2/3}$ .

## 4.5 The no-atmosphere sub-cases

A scene can declare that no atmosphere lies between sensor and target: the target descriptor’s `no_atmosphere` location with one of three sub-cases — **space** (both endpoints above the atmosphere: satellite-to-satellite, space surveillance), **ground\_test** (an outdoor test range short enough to neglect) and **lab\_test** (a chamber). The atmospheric products are then the identity family’s —  $\tau \equiv 1$ ,  $L_{\text{path}} \equiv 0$ ,  $L_{\text{down}} \equiv 0$ , the exo backend — which is a statement that the intervening medium has unity transmittance and no emission, not an absence of physics: the cosmic background still arrives as a  $2.725 \text{ K}$  blackbody from the source stage.

**Preconditions, refused rather than defaulted.** The space sub-case requires a sensor altitude on the line of sight and a path that clears the Earth’s limb (there is no `earthlimb` model). The two test sub-cases require an explicit spectral background — the chamber or range wall — because there is no sky to supply one: the source stage builds a grey-body wall,  $L_{bg}(\lambda) = \varepsilon_{bg} B(\lambda, T_{bg})$ , from `source.background.temperature` and `source.background.emissivity`, warning when the temperature was left at its default rather than baking an ambient value in silently; a measured  $L_{bg}(\lambda)$  can be injected in its place.

**Illumination assertion.** `source.lab_test_mode = "dark"` declares a chamber with no external illumination — no lamp, no sun, thermal self-emission only, the dark-calibration sub-mode — and is validated: a user-set target reflectance contradicts it and is rejected, since a reflectance-driven target has nothing to reflect in a dark chamber. `"lit"` records an externally illuminated bench for readability (unvalidated until a lamp surface exists); the empty default asserts nothing.

**Pitfalls.** Treating the vacuum case as “skip the atmosphere stage” (the stage still runs and publishes unity and zero, so every downstream consumer sees one contract); a ground-test range long enough for real extinction declared as no-atmosphere; a lab scene with a reflectance target and no illumination assertion, which computes a reflected term from the schema’s default solar geometry.

---

## 4.6 What the models do not represent

Recorded here because a physics document that omits its own boundaries is misleading. Each item is tracked in the repository, where the measured consequences are tabulated in full.

- **Line structure inside a calibrated region.** The simple model’s spectral shape is flat within each of the 17 regions; the 0.04  $\mu\text{m}$  edge ramps remove the discontinuity, not the underlying flatness. It is the dominant residual of the thermal path radiance.
- **Multiple scattering.** The single-scatter source under-predicts the daytime VIS/NIR sky; the sub-3  $\mu\text{m}$  provisional warning is what says so to an operator.
- **Refraction.** The geometry is unrefracted. It is the dominant geometric error inside the horizon guard’s warn band, and the decks that would calibrate it have not been run.
- **Stratospheric structure.** The fixed-lapse ICAO profile is isothermal above the tropopause, so real stratospheric warming is not represented. Measured on the downwelling ladder: MODTRAN’s recovered emission temperature reaches 268 K at the 50 km stratopause and falls to 178 K at 80 km, while the profile answers 222.65 K at every one of those altitudes. This is why the five stratospheric rungs of the downwelling ladder are insensitive to *any* emission-placement choice — every candidate form returns the same temperature there — and why they carry no information about the hemispheric downwelling terms’ fitted constants.
- **Grazing-arc opacity distribution.** A grazing arc’s air lies along the arc, not along the vertical between its endpoints; the emission weighting is approximate there, though the *total* optical depth is exact. Measured on a ground-rooted column at 85/88/89.5°, the two placements differ by  $\leq 1.2\%$  in band-mean thermal radiance — under the 3–6 % model/MODTRAN residual there, so those runs cannot settle it. The effect is only measurable on an arc rooted at an elevated tangent point *below* the tropopause (modeled separation 7.9 %/9.8 % at 5 km, 16.1 %/13.1 % at 8 km, MWIR/LWIR, and exactly 0 at 15 km for the isothermal reason above); the reference runs for that geometry are authored and not yet executed.
- **O<sub>3</sub> emission altitude.** The well-mixed-gas floor lumps CO<sub>2</sub>/N<sub>2</sub>O/CH<sub>4</sub> with O<sub>3</sub>, which peaks near 25 km, so 9.6  $\mu\text{m}$  emission is placed too low. The  $\tau$  side is settled: the 8–10  $\mu\text{m}$  region is split at the band edges (*The well-mixed-gas absorption floor*), so the ozone share of the in-feature floor is arithmetic — 0.832 — rather than a free parameter. The *placement* is not: the identified opacity still rides the molecular scale height, so the model puts ozone near 8 km instead of 25 km. Measured on fourteen matched pairs with the split table in place, the 9.4–9.9  $\mu\text{m}$  feature reads RMS  $|\ln \text{ratio}|$  0.3581 (it was 0.1519 when the flat slab under-supplied the in-band opacity), and moving the  $\tau$ -determined share onto

a 25 km layer recovers it to 0.1456 — within 7 % of the sweep’s unconstrained best, 0.1365 at share 0.70. The remaining degeneracy in the layer’s center and width is unchanged in kind (0.133–0.191 across 20–30 km centers and 3–8 km widths) because the profile is isothermal wherever the layer sits.

- **Polarization, 3D/heterogeneous atmospheres, time dependence, adjacency, aurora/airglow, cloud microphysics.** Out of scope for this version of the model.
- 

## 4.7 References

This chapter cites, by the keys of the manual’s single References list: [Hansen & Travis 1974] and [Bucholtz 1995] for the Rayleigh whole-atmosphere optical depth, [McClatchey 1972] for the standard-profile water columns, [Henyey & Greenstein 1941] for the aerosol phase function, [Elsasser 1942] for the diffusivity-factor context of  $D$ , and [Chapman 1931] for the grazing-limit anchor of the spherical slant column.

# Spatial Model

PSF construction, the MTF budget, ensquared energy, sampling, and the dual-path architecture that keeps them consistent — as implemented in RADIANT. Numeric anchors are literature values re-derived independently of the implementation, then checked against it.

**Symbols used throughout:**  $\lambda$  wavelength [ $\mu\text{m}$ ] ( $\lambda_m = \lambda \cdot 10^{-6}$  m where SI is needed);  $D$  aperture diameter [m];  $f$  focal length [m];  $F_{\#} = f/D$ ;  $\epsilon$  central-obscuration ratio;  $p$  pixel pitch [m];  $\nu$  spatial frequency at the focal plane [cy/m] (cy/mm in worked numbers);  $\nu_c = 1/(\lambda F_{\#})$  optical cutoff;  $\nu_{Nyq} = 1/(2p)$  Nyquist.

## 5.1 The two spatial paths

RADIANT maintains **two parallel spatial representations, both rooted in the same complex pupil**:

- the **PSF path** — every spatial degradation enters as a convolution kernel on one EffectivePSF; EE\_box, RER, FWHM, Strehl, LSF and ERF are computed only from it;
- the **MTF product path** — the frequency-domain budget  $\text{MTF}_{sys}(\nu) = \prod_i \text{MTF}_i(\nu)$  consumed by MTF-at-Nyquist, folded MTF, and GIQE/NIIRS.

The two are the same information: for incoherent imaging the OTF is simultaneously the Fourier transform of the PSF and the normalized autocorrelation of the complex pupil  $P = A e^{i2\pi W/\lambda}$ ,

$$\text{OTF}(\nu) = \frac{\mathcal{F}\{\text{PSF}\}(\nu)}{\mathcal{F}\{\text{PSF}\}(0)} = \frac{(P \star P)(\lambda f \nu)}{(P \star P)(0)},$$

so the FFT of the fully convolved EffectivePSF must agree with the MTF product. RADIANT checks this on every chain run that computes the spatial path, to an absolute tolerance of  $2 \times 10^{-2}$  ( $\approx 2\times$  the measured full-chain discretization floor).

**Why a separate  $\text{MTF}_{diffraction} \times \text{MTF}_{aberration}$  factoring is forbidden:** aberrations are a phase modification *inside* the pupil, entering the autocorrelation as a phase *difference across the shear* under the integral — they cannot be factored into (clear-aperture MTF)  $\times$  (aberration-only multiplier). True aberrated OTFs can go negative (defocus contrast reversal); no positive scalar multiplier can reproduce that. Only genuinely independent *image-plane convolutions* (detector aperture, diffusion, IPC, jitter, smear) and statistically independent ensemble averages (turbulence) multiply. The one deliberate MTF-only term is TDI mis-registration (*TDI mis-registration — the one MTF-only term*, below) — a readout-timing effect with no instantaneous kernel, excluded from the consistency comparison.

## 5.2 Diffraction-limited PSF

**Equation.** Unaberrated circular pupil (Airy):

$$I(r) = I_0 \left[ \frac{2J_1(v)}{v} \right]^2, \quad v = \frac{\pi r}{\lambda F_{\#}},$$

first zero at  $v_1 = 3.8317060$ , i.e.  $r_1 = 1.2196699 \lambda F_{\#}$ . Encircled energy  $EE(v) = 1 - J_0^2(v) - J_1^2(v)$  (unobscured only): 0.837785 at the first dark ring. With central obscuration  $\epsilon$ :

$$I(v) = \frac{I_0}{(1 - \epsilon^2)^2} \left[ \frac{2J_1(v)}{v} - \epsilon^2 \frac{2J_1(\epsilon v)}{\epsilon v} \right]^2$$

— core narrows, energy moves into the rings.

**Implementation.** RADIANT does not evaluate the Bessel form; it FFT-propagates the sampled complex pupil (aperture mask incl. obscuration and spider vanes, plus wavefront phase) and takes  $|\cdot|^2$  — the Airy pattern is the *test oracle*, not the algorithm. Polychromatic PSFs are photon-flux-weighted sums of monochromatic PSFs, one per wavelength on a sub-sampled grid whose size is a parameter.

**Assumptions & validity.** Scalar Fraunhofer diffraction;  $F_{\#} \gtrsim 2$  for the scalar approximation; grid effects controlled by the pupil-sampling and PSF-oversampling parameters, whose defaults are 128 pupil samples across and 8× oversampling.

**Pitfalls.** 1.22 vs the exact 1.2196699 (0.03 % — matters for tight baselines); amplitude vs irradiance PSF; forgetting the  $(1 - \epsilon^2)^{-2}$  energy renormalization; applying the unobscured EE formula to an annular pupil.

**Numeric anchor.**  $r_1 = 19.5147 \mu\text{m}$  at  $\lambda = 4 \mu\text{m}$ ,  $F_{\#} = 4$ .

## 5.3 Optical MTF from the pupil autocorrelation

**Equation.** Clear circular pupil, incoherent:

$$\text{MTF}(\tilde{\nu}) = \frac{2}{\pi} \left[ \arccos \tilde{\nu} - \tilde{\nu} \sqrt{1 - \tilde{\nu}^2} \right], \quad \tilde{\nu} = \nu / \nu_c \leq 1, \quad \nu_c = \frac{1}{\lambda F_{\#}}.$$

RADIANT computes the general case (obscuration, spiders, aberrations) by direct autocorrelation of the complex pupil — a single `MTF_optics` term, never a factored product (*The two spatial paths*, above).

**Assumptions & validity.** Incoherent illumination (the coherent cutoff is  $\nu_c/2$  — a classic factor-2 trap); monochromatic per- $\lambda$ , spectrally weighted for broadband.

**Pitfalls.**  $\lambda$ -unit slips in  $\nu_c$  ( $\lambda$  must be in mm for cy/mm); NaN instead of 0 above cutoff; treating the annular-pupil MTF as clear-MTF × scalar (the true ratio crosses 1:  $\approx 0.78$  at  $0.3\nu_c$  but  $\approx 1.10$  at  $0.8\nu_c$  for  $\epsilon = 0.3$ ).

**Numeric anchor.**  $\text{MTF}(0.5\nu_c) = \frac{2}{\pi} \left( \frac{\pi}{3} - \frac{\sqrt{3}}{4} \right) = 0.391002$ ;  $\nu_c = 62.5 \text{ cy/mm}$  at  $\lambda = 4 \mu\text{m}$ ,  $F_{\#} = 4$ .

**Halo terms outside the pupil.** Two optical degradations are not pupil phenomena and enter as kernel / MTF pairs beside the pupil term, each an exact Fourier pair on the two paths. **Surface scatter.** Micro-roughness scatters a fraction of the specular beam into a wide halo; in the smooth-surface (Rayleigh–Rice) limit at normal incidence the total integrated scatter of the train’s effective RMS roughness  $\sigma_s$  (`optics.surface_roughness_nm`, 0 = none) is

$$\text{TIS} = 1 - \exp\left[-\left(\frac{4\pi\sigma_s}{\lambda}\right)^2\right] \approx \left(\frac{4\pi\sigma_s}{\lambda}\right)^2 \quad (\sigma_s \ll \lambda),$$

evaluated at the band centre, and the scattered energy lands in an isotropic Gaussian halo of focal-plane width  $\sigma_h$  (`optics.scatter_halo_sigma_um`):  $k(r) = (1 - \text{TIS})\delta(r) + \text{TIS}G(r; \sigma_h)$ ,  $\text{MTF}(\nu) = (1 - \text{TIS}) + \text{TIS}e^{-2\pi^2\sigma_h^2\nu^2}$  — a contrast floor of  $1 - \text{TIS}$  at every frequency the halo cannot resolve. A warning fires above  $\text{TIS} = 0.3$ , where the smooth-surface limit degrades; the halo width is a model input to tune to a measured halo, not a derived quantity. **Veiling glare.** The veiling-glare fraction of the noise chapter’s stray-light model is, by default, a radiometric pedestal only; with `optics.stray.veiling_glare_mtf` on, the same fraction is re-imaged as a Gaussian halo of width `optics.stray.halo_sigma_um` with the identical kernel / MTF form — the low-frequency contrast loss a pedestal cannot express. Both halos are truncated at the PSF grid edge, so the halo widths must fit the grid for the pairs to stay exact.

**Numeric anchors.**  $\sigma_s = 20$  nm:  $\text{TIS} = 0.2233$  at  $0.5\ \mu\text{m}$  but  $0.00394$  at  $4\ \mu\text{m}$  — the  $\lambda^{-2}$  scaling is why the same polish is a VIS problem and an MWIR non-event.

## 5.4 Wavefront error, Zernike modes, and Strehl

**Equations.** Wavefront OPD enters the pupil phase as  $e^{i2\pi W/\lambda}$ . Noll-indexed Zernike defocus  $Z_4 = \sqrt{3}(2\rho^2 - 1)$ : with  $W = a_4 Z_4$ , the RMS OPD is exactly  $a_4$  and the peak-to-valley is  $2\sqrt{3}\sigma$ . Maréchal Strehl approximation from RMS OPD  $\sigma_{OPD}$ :

$$S \approx \exp\left[-\left(\frac{2\pi\sigma_{OPD}}{\lambda}\right)^2\right].$$

RADIANT reports two Strehls: the **PSF-derived** `strehl` — degraded-PSF peak over the diffraction-limited `reference_psf` peak, with the *same* detector kernels applied to both so detector effects cancel — and the analytic `strehl_marechal` diagnostic.

**Defocus as a pupil term.** A detector-plane displacement  $\delta$  from best focus (`optics.defocus_um`, sign irrelevant) is a quadratic wavefront: the marginal ray at half-angle  $u$ ,  $\tan u = 1/(2N_{eff})$ , acquires OPD  $\delta(1 - \cos u) \approx \delta/(8N_{eff}^2)$  at the pupil edge, so  $W(\rho) = \delta\rho^2/(8N_{eff}^2)$ , peak-to-valley  $\delta/(8N_{eff}^2)$  — the depth-of-focus statement  $\lambda/4 \text{ P-V} \Leftrightarrow \delta = 2\lambda N_{eff}^2$ . Fitted to Noll  $Z_4$  (P-V =  $2\sqrt{3}a_4$ ) the RMS coefficient is

$$a_4 = \frac{\delta}{16\sqrt{3}\lambda N_{eff}^2} \text{ waves.}$$

RADIANT folds  $\delta$  into the *same* complex pupil as the wavefront error — added to an existing  $Z_4$ , or carried beside a scalar-RMS screen — once, before either spatial path, so PSF and MTF derive defocus from one pupil and the consistency invariant holds by construction; there is no separate defocus kernel.

**Assumptions & validity.** Maréchal reliable for  $\sigma \lesssim \lambda/10$ ; Noll normalization (coefficient = RMS) — the Wyant convention differs by  $\sqrt{3}$ -type factors. Zernike modes are orthonormal on the unobscured unit disk; annular pupils strictly need annular polynomials.

**Pitfalls.** Waves vs radians in the Maréchal exponent (factor  $(2\pi)^2$ ); P-V vs RMS (the  $2\sqrt{3}$  is defocus-specific); using Maréchal to reconstruct a PSF or MTF (it is a scalar diagnostic only).

**Numeric anchors.**  $S(\sigma = \lambda/14) = 0.817569$  (the classic  $\approx 0.8$  diffraction-limited threshold); quarter-wave P-V defocus  $\rightarrow S = 0.814$ , reached physically at  $\delta = 2\lambda N^2 = 128 \mu\text{m}$  for  $\lambda = 4 \mu\text{m}$ ,  $f/4$ .

## 5.5 Detector-plane kernels: pixel aperture, diffusion, IPC, electronics

**Pixel aperture.** A photosite of linear width  $w$  integrates the image — a rect convolution:

$$\text{MTF}_{det}(\nu) = \left| \frac{\sin(\pi w \nu)}{\pi w \nu} \right|,$$

first zero at  $\nu = 1/w$ . With areal fill factor FF,  $w = p\sqrt{\text{FF}}$  — the same width the PSF path's pixel kernel uses, sampled by area overlap, so the two paths agree.

**Charge diffusion.** Gaussian carrier spread of RMS  $\sigma_d$ :  $\text{MTF}_{diff}(\nu) = \exp(-2\pi^2\sigma_d^2\nu^2)$ .

**Inter-pixel capacitance.** Pitch-spaced coupling kernel with fraction  $\alpha$ :  $\text{MTF}_{IPC}(\nu) = (1 - 4\alpha) + 2\alpha \cos(2\pi\nu p)$  per axis (nearest-neighbor form).

**Electronics.** Finite readout-amplifier bandwidth low-passes the video waveform; at the pixel clock rate the temporal response maps onto the focal plane as a one-dimensional blur along the readout (cross-scan,  $x$ ) axis with equivalent Gaussian sigma  $\sigma_e$  (`readout.electronics_sigma_um`, 0 = ideal):  $\text{MTF}_{elec}(\nu_x) = \exp(-2\pi^2\sigma_e^2\nu_x^2)$ , unity along  $y$ . It is a readout-side term but a genuine spatial one — a Gaussian-in- $x$  kernel on the PSF path and this analytic factor on the MTF path — so unlike TDI mis-registration it takes part in the dual-path consistency check. **Numeric anchor.**  $\sigma_e = 5 \mu\text{m}$  at 25 cy/mm (Nyquist for a 20  $\mu\text{m}$  pitch):  $\text{MTF}_{elec} = 0.7346$ .

**Pitfalls.** The sinc convention: NumPy's `np.sinc(x)` already includes  $\pi$  — `np.sinc(w·v)` is correct, `np.sinc(π·w·v)` double-counts  $\pi$  and moves the first zero to  $1/(\pi w)$ . Pitch vs aperture width when  $\text{FF} < 1$ . IPC and diffusion both exist in kernel form for the PSF path and analytic form for the MTF path — adding one side only breaks the agreement between them.

**Numeric anchor.** 100 % fill at Nyquist:  $\sin(\pi/2)/(\pi/2) = 2/\pi = 0.636620$ .

## 5.6 Platform kernels: jitter and smear

**Jitter** (random LOS motion, many cycles per integration): Gaussian blur of RMS  $\sigma$  at the focal plane ( $\sigma = \sigma_j f$  for the RMS angular jitter  $\sigma_j$  of the notation table):

$$\text{MTF}_{jit}(\nu) = \exp(-2\pi^2\sigma^2\nu^2).$$

**Smear** (deterministic image motion  $d = v_{image} t_{int}$  during integration): rect blur,

$$\text{MTF}_{smear}(\nu) = \left| \frac{\sin(\pi d \nu)}{\pi d \nu} \right|,$$

applied along the motion direction only.  $v_{image}$  derives from the **ground-track** velocity ( $v_g = v R/(R+h)$ , see the geometry chapter's *GSD — nadir and off-nadir*) times the magnification  $f/R_s$  — using orbital  $v$  instead of  $v_g$  inflates smear by  $h/R$  (+7.8 % at 500 km).

**Assumptions & validity.** Jitter Gaussian form requires jitter frequency  $\gg 1/t_{int}$ ; comparable-period motion is neither pure jitter nor pure smear. A  $d/\sqrt{12}$  “equivalent Gaussian” for smear is a second-moment match, not the correct MTF.

**Pitfalls.** The jitter exponent is  $2\pi^2\sigma^2\nu^2$  exactly (equivalently  $e^{-(2\pi\sigma\nu)^2/2}$ );  $\sigma$  vs FWHM ( $\times 2.3548$ ); a jitter MTF with a zero crossing indicates a wrongly applied sinc model.

**Numeric anchors.**  $\sigma = 0.25 p$  at Nyquist:  $e^{-\pi^2/32} = 0.734603$ ;  $d = 0.5 p$  at Nyquist:  $\sin(\pi/4)/(\pi/4) = 0.900316$ .

## 5.7 Atmospheric turbulence

**Equations.** Fried parameter  $r_0 \propto \lambda^{6/5}$  (so  $r_0(\lambda_2) = r_0(\lambda_1)(\lambda_2/\lambda_1)^{6/5}$ ); long-exposure Kolmogorov MTF in angular frequency  $f_a$  [cy/rad]:

$$\text{MTF}_{LE}(f_a) = \exp \left[ -3.44 \left( \frac{\lambda f_a}{r_0} \right)^{5/3} \right].$$

The 3.44 is half the structure-function constant 6.88. Turbulence is the one contributor legitimately multiplied into the budget without pupil-level treatment: in the long-exposure ensemble average the atmosphere is statistically independent of the pupil. The PSF path receives the matching kernel.

**Assumptions & validity.** Kolmogorov spectrum (infinite outer scale), weak fluctuations, full tilt averaging (long exposure). Short-exposure (tilt-removed) imaging needs Fried's corrected form — not the shipped default.

**Pitfalls.** Quoting  $r_0$  at  $0.5 \mu\text{m}$  and using it unscaled in the IR (at  $4 \mu\text{m}$ ,  $r_0$  is  $12\times$  larger); 3.44 vs 6.88; gating on platform type instead of the resolved  $r_0 > 0$  (RADIANT gates on the Fried parameter the atmosphere chapter's *Optical turbulence* section resolves — entered directly or integrated from a  $C_n^2$  profile — and the

MTF term is written by **PerformanceStage**, not **AtmosphereStage** — the atmosphere stage publishes `r0_m` only).

**Numeric anchors.**  $\text{MTF}_{LE} = 0.338398$  at  $\lambda f_a / r_0 = 0.5$ ;  $r_0 = 0.10 \text{ m} @ 0.5 \mu\text{m} \rightarrow 1.21257 \text{ m} @ 4 \mu\text{m}$ .

## 5.8 TDI mis-registration — the one MTF-only term

**Equation.** Cross-track drift angle  $\theta$  over  $N$  TDI stages displaces the  $N$  summed samples by  $d = p \tan \theta$  each; averaging  $N$  displaced copies gives the Dirichlet kernel, which for sub-pixel per-stage drift reduces to a sinc in the **total** drift  $Np\theta$ :

$$\text{MTF}_{TDI}(\nu) \approx \left| \frac{\sin(\pi \nu N p \theta)}{\pi \nu N p \theta} \right|.$$

This is a readout-timing/registration effect on the time-aggregated image — there is no instantaneous spatial kernel, so it enters **only** the MTF product and is excluded from the Rule-4 consistency comparison.

**Pitfalls.** Tolerancing  $\theta$  without  $N$  (degradation scales with total drift, so the  $\theta$  tolerance tightens as  $N$  grows); the discrete Dirichlet form exceeds the sinc form by up to  $\sim 0.6\%$  at small  $N$ .

**Numeric anchor.** Total drift  $0.5 p$  at Nyquist,  $N = 16$ : 0.9007 (Dirichlet) vs 0.900316 (sinc).

## 5.9 Ensquared energy (EE\_box)

**Equation.** Fraction of the fully degraded PSF's energy inside the centered  $n \times n$  pixel box:

$$EE_{n \times n} = \int_{-np/2}^{np/2} \int_{-np/2}^{np/2} \text{PSF}(x, y) dx dy.$$

Computed in **PlatformStage** from the **fully degraded** **EffectivePSF** (jitter, smear, turbulence included) and applied exactly once, in **SpectralIntegrationStage**, to point-source and sub-pixel target signals only — never to the background term, never in the extended regime.

**Discretization.** Each PSF cell is weighted by the fraction of its area inside the box (cell-area overlap) — second-order accurate; the earlier full-weight edge-cell scheme carried an  $O(dx)$  bias ( $+24\%$  at  $Q=2$  at default sampling) that overstated point-source SNR.

**Pixel sampling phase (straddle).** The box integral above is the box *centred on the image point*. `detector.pixel_phase_mode` selects where the geometric image point sits on the pixel grid: `average` (the default) is the expectation over a source placed uniformly across one pitch; `centered` puts it on a pixel centre; `worst_case` on a four-pixel corner,  $\delta = (\frac{1}{2}, \frac{1}{2})p$ ; specified at `(detector.pixel_phase_x, detector.pixel_phase_y)` in pitches, each in  $[-\frac{1}{2}, \frac{1}{2}]$ . With the photosite rect already convolved into the PSF, the energy a full-pitch box at phase  $\delta$  collects is that pixel-convolved PSF read at  $\delta$ ,

$$EE(\delta) = \iint \text{PSF}(\mathbf{x}) \text{rect}_p(\mathbf{x} - \delta) d\mathbf{x} = (\text{PSF} \circledast \text{rect}_p)(\delta),$$

and its average over one pitch is the pitch-wide box integral of the pixel-convolved PSF (a rect convolved with a rect is a triangle) — which is exactly what the default computes, so the phase average is the value the chain has always applied. The platform stage publishes  $EE_{box}$  at the selected phase,  $EE_{box,centered}$  as the reference, and the straddle factor  $EE_{box}/EE_{box,centered}$  beside them. Offsets are measured from the PSF grid centre (the chief-ray image point), never from the degraded PSF's centroid; the fill factor is applied downstream, so every value is normalized to the pitch box.

**Pitfalls.** Ensquared  $\neq$  encircled (a square of side  $p$  is not a circle of diameter  $p$  — quoting the 83.8 % first-ring figure for a pixel box is a category error); Airy ring tails decay as  $1/u^2$ , so truncated normalization biases EE; referencing a specified phase to the PSF centroid instead of the chief-ray point (a smeared PSF moves its centroid, not the pixel grid); applying the straddle factor to the background term, which never sees  $EE_{box}$ .

**Numeric anchor.** Unaberrated Airy at critical sampling ( $Q = 2$ ):  $EE_{1 \times 1} = 0.177327$  — only ~18 % of a point source's energy lands in the center pixel.

## 5.10 Sampling: Nyquist, Q, and folded MTF

**Equations.**  $\nu_{Nyq} = 1/(2p)$ ;

$$Q = \frac{\lambda F_{\#}}{p}, \quad \frac{\nu_c}{\nu_{Nyq}} = \frac{2}{Q}.$$

- $Q = 2$ : **critically sampled** — the optics pass nothing above Nyquist; no aliasing.
- $Q < 2$ : undersampled (typical EO design point, chosen for SNR/GSD); the band  $[\nu_{Nyq}, \nu_c]$  folds.
- $Q > 2$ : oversampled.

The folded MTF adds the aliased response back onto the baseband. Sampling on a pitch  $p$  replicates the pre-sampling spectrum at integer multiples of the **sampling** frequency  $\nu_s = 1/p = 2\nu_{Nyq}$ , so

$$\text{MTF}_{fold}(\nu) = \sum_{k=-N}^{+N} \text{MTF}_{opt}(|\nu + k \nu_s|).$$

At  $\nu = \nu_{Nyq}$  the  $k = -1$  replica lands back on  $\nu_{Nyq}$ , so the folded value there is twice the pre-sampling MTF (alias fraction  $\rightarrow 1/2$ ) whenever the higher orders are negligible — and it is *zero* for optics that cut off below Nyquist, which is the oversampled sanity check. The alias fraction  $(\text{MTF}_{fold} - \text{MTF}_{opt})/\text{MTF}_{fold}$  is evaluated only where  $\text{MTF}_{fold} > 10^{-9}$  of its DC value; below that floor both terms are round-off, and the reported fraction is exactly zero — an oversampled design has no aliased energy. GIQE consumes RER from the PSF path and the MTF budget, as the Performance Metrics chapter describes.

**Pitfalls.** “ $Q = 1$  critical” is a different (sampling-frequency) convention — RADIANT's statement is  $\nu_c/\nu_{Nyq} = 2/Q$ , critical at  $Q = 2$ ; sampling frequency  $1/p$  vs Nyquist  $1/(2p)$  — replicating the spectrum at  $\nu_{Nyq}$  instead of  $\nu_s$  puts the  $k = -1$  copy on DC and adds  $\text{MTF}(0) = 1$  to every system; ground-projecting Nyquist twice.

**Numeric anchor.**  $\lambda = 0.55 \mu\text{m}$ ,  $F_{\#} = 4$ ,  $p = 10 \mu\text{m}$ :  $Q = 0.22$  —  $\nu_c/\nu_{Nyq} = 9.1$ , heavily aliased, the classic high-resolution VNIR design point.

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## 5.11 RER and the edge response

**Equation.** The edge response function is the cumulative integral of the line spread function (itself the 1-D projection of the PSF); the relative edge response per axis is

$$\text{RER}_x = \text{ERF}_x(+p/2) - \text{ERF}_x(-p/2),$$

and the reported scalar is the **geometric mean**  $\sqrt{\text{RER}_x \text{RER}_y}$  (GIQE-5 usage). All of LSF/ERF/RER derive from that one degraded PSF.

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## 5.12 Scope and deferred effects

No ghost images, no measured-BSDF scatter beyond the total-integrated-scatter and halo model, no chromatic aberration model, no short-exposure turbulence. The project's scope-decision register records the full list.

# Noise Model

Complete noise taxonomy, equations, scaling rules, and acquisition timing as implemented in RADIANT. All noise in electrons RMS, the manual’s canonical noise unit. Numeric anchors were re-derived from the literature independently of the implementation, then checked against it.

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## 6.1 Overview — RSS composition

RADIANT models 16 independent noise sources, each in e- RMS. With statistical independence, variances add:

$$\sigma_{\text{total}} = \sqrt{\sum_{i=1}^{16} \sigma_i^2}$$

**The load-bearing assumption is independence** — each mechanism arises in a physically distinct process, so cross-covariances vanish. It is deliberately violated by fixed-pattern terms across frames of the same pixel (they are a static spatial pattern: they belong in a single-frame spatial budget but do not average down over frames — see TDI/co-add scaling below). All terms are input-referred to electrons at the sense node before combining; mixing DN-domain and e-domain terms in one RSS is the classic error.

Terms are classed **temporal** (frame-independent, average down as  $\sqrt{N}$ ) or **spatial** (fixed-pattern).

**Numeric anchor.**  $N_{\text{target}} = 10000$ ,  $N_{bg} = 5000$ ,  $N_{\text{dark}} = 1000$  e-,  $\sigma_{\text{read}} = 50$  e-, PRNU 0.1 % ( $\sigma_{PRNU} = 10$  e-):  $\sigma_{\text{tot}} = \sqrt{16000 + 2500 + 100} = 136.381817$  e- RMS.

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## 6.2 Noise Taxonomy

### 6.2.1 Photon shot noise (4 terms)

Poisson statistics of photon arrival: for  $N_e$  photoelectrons,  $\sigma = \sqrt{N_e}$ . QE < 1 preserves Poisson statistics (binomial thinning), so the square root is taken of **electrons**, never photons or DN ( $\sigma_{DN} = \sqrt{S_{DN}/g}$ , not  $\sqrt{S_{DN}}$ ).

- Signal shot:  $\sigma = \sqrt{S}$  (target electrons)
- Background shot:  $\sigma = \sqrt{S_{bg}}$
- Nearfield shot:  $\sqrt{S_{nf}}$  (warm-optics self-emission)
- Straylight shot:  $\sqrt{S_{stray}}$

**Pitfalls.** Adding  $\sigma$ 's instead of variances; dark-frame subtraction removes the dark *mean*, never its shot noise (single-shot subtraction doubles the dark shot variance).

### 6.2.2 Detector-material noise (4 terms)

**Dark shot:**  $\sigma = \sqrt{J_{\text{dark}} t_{\text{int}}}$  with  $J_{\text{dark}}$  in e-/s. Temperature scaling is Arrhenius:

$$J(T) = J_{\text{ref}} \exp \left[ \frac{E_a}{k_B} \left( \frac{1}{T_{\text{ref}}} - \frac{1}{T} \right) \right]$$

(`detector.dark_activation_energy_eV`, `dark_reference_temperature_K`). Physics note: diffusion-limited material has  $E_a \approx E_g$ , generation–recombination-limited  $E_a \approx E_g/2$  — conflating them mispredicts the cooling benefit by squaring the true ratio (audit anchor:  $E_g = 0.1$  eV diffusion, 77→84 K: dark current  $\times 4.56$ ).

**G-R noise:**  $\sigma_{gr} = \sqrt{2g_{gr}J_{\text{dark}}t_{\text{int}}}$  ( $g_{gr} \approx 0$  PV,  $\sim 1$  PC — the same generation-AND-recombination doubling that costs photoconductors  $\sqrt{2}$  in D\*).

**Johnson noise** (from  $R_0A$ ) and **1/f noise** (flicker coefficient integrated over a frequency band) follow the standard material forms.

### 6.2.3 ROIC noise (3 terms)

**Read noise:** fixed floor `read_noise_e_rms`; includes CDS benefit when `read_noise_is_post_cds = True` (default).

**kTC reset noise:**  $\sigma_{kTC} = \sqrt{k_B T C_{\text{node}}}/q$  — suppressed to zero when CDS is enabled (`readout.cds_enabled = 1`, default).

**ADC quantization:**

$$\sigma_{ADC} = \frac{g}{\sqrt{12}}, \quad g = \frac{N_{\text{fullwell}}}{2^n} \text{ [e-/DN]}.$$

Valid when input noise  $\gtrsim 1$  LSB (dithered); the error is uniform on  $\pm \text{LSB}/2$  and its variance is  $\text{LSB}^2/12$ .

**Pitfalls.** Dividing by 12 instead of  $\sqrt{12}$  (3.46 $\times$  understatement); counting quantization twice when a vendor's measured “read noise in e-” already includes the ADC; leaving  $\sigma_{\text{ADC}}$  in DN when the RSS is in e-.

**Numeric anchor.** 14-bit, 100 ke- full well:  $g = 6.10352$  e-/DN,  $\sigma_{ADC} = 1.76193$  e- RMS.

### 6.2.4 Spatial (fixed-pattern) noise (3 terms)

**PRNU** — multiplicative gain dispersion, **linear in signal** (not  $\sqrt{N}$ ):  $\sigma_{PRNU} = k S$  with  $k$  the residual non-uniformity after NUC. Shot/PRNU crossover at  $N = 1/k^2$  (0.1 %  $\rightarrow 10^6$  e-): above it, more integration cannot improve single-frame spatial SNR — only better flat-fielding can.

**DSNU** — dark-signal non-uniformity, a fixed spatial floor (`dsnu_e_rms`).

**Clutter** — scene-induced spatial variance,  $\sigma_c \cdot S_{bg}$  (`detector.clutter_sigma`).

**Pitfalls.** Applying  $\sqrt{\cdot}$  to PRNU; claiming temporal co-adding reduces FPN (perfectly frame-correlated for a staring pixel); applying  $k$  to signal+dark (dark dispersion is DSNU’s job).

### 6.2.5 Other (2 terms)

**Persistence** — residual charge from the prior exposure, on an exponential-decay model — and **ROIC glow shot**,  $\sqrt{R_{glow} t_{int}}$ .

### 6.2.6 Stray light — pedestal and halo

Stray light adds electrons, and their shot noise, to every pixel uniformly; it is never part of the signal numerator. The in-band irradiance at the focal plane comes from one of three input modes (`optics.stray.input_mode`): **veiling glare**,  $E_{stray}(\lambda) = g E_{in-FOV}(\lambda)$  with  $g = \text{optics.stray.veiling\_glare\_fraction}$ ; **absolute irradiance**, a total in-band  $E_{stray}$  (`optics.stray.absolute_irradiance_Wm2`) spread as a flat spectral density over the band; or a **spectral file** injected before the chain runs. A point-source transmittance (PST) mode is not offered: it needs a scene radiance distribution the model does not carry. `optics.stray.includes_thermal` says a measured stray-light figure already contains the warm-optics scatter, and suppresses the near-field term so it is not counted twice. The pedestal is radiometric only; the *spatial* consequence of veiling glare — a low-frequency contrast loss — is the optional halo of the spatial chapter (*Halo terms outside the pupil*).

## 6.3 Acquisition scaling

### 6.3.1 TDI and co-adding

Summing  $N$  stages/frames: signal  $\times N$ ; uncorrelated temporal noise  $\times \sqrt{N}$ ; SNR  $\times \sqrt{N}$ . The architecture determines the read-noise and FPN behavior:

| Noise category                                           | Charge-domain (analog) TDI | Digital co-add                     |
|----------------------------------------------------------|----------------------------|------------------------------------|
| Signal                                                   | $\times N$                 | $\times N$                         |
| Shot-like temporal                                       | $\times \sqrt{N}$          | $\times \sqrt{N}$                  |
| Read / kTC / quantization                                | $\times 1$ (one read)      | $\times \sqrt{N}$ (read per frame) |
| PRNU/DSNU across <b>distinct</b> pixels (cross-scan TDI) | effective $k/\sqrt{N}$     | —                                  |
| FPN, <b>same</b> pixels (staring co-add)                 | —                          | $\times N$ (coherent; no SNR gain) |
| Clutter (scene-correlated)                               | $\times N$                 | $\times N$                         |

**Well-fill cap:** charge-domain TDI accumulates into one well —  $N(S_1 + N_{d,1}) \leq N_{fullwell}$  bounds the useful  $N$  (audit anchor: 10 ke- per stage against a 100 ke- well caps  $N$  at 10, SNR gain at  $\sqrt{10}$ ).

### 6.3.2 Binning

On-chip binning merges charge before read (read noise once per binned super-pixel); off-chip binning averages after read (read noise per contributing pixel,  $\times \sqrt{n}$  in the sum). Signal scales with the binned area in both.

### 6.3.3 Frame timing

Frame period is stored independently of integration time:

$$f_{frame} = \frac{1}{T_{frame}}, \quad \text{duty} = \frac{t_{int}}{T_{frame}} \leq 1.$$

Unset `readout.frame_period_s` (0.0) defaults to  $t_{int}$  (duty 1.0, continuous readout);  $\text{duty} > 1$  is rejected with an actionable error. Outputs `frame_period_s/frame_rate_hz/duty_cycle/frame_period_` defaulted in `stage_outputs["readout"]`.

## 6.4 Digital-pixel counting readout

`readout.architecture = "digital_counting"` replaces the analog charge well with an in-pixel comparator and an  $N$ -bit counter: every time the integrated charge crosses the packet  $Q_{pkt}$  (`readout.count_packet_e`, required under counting) the counter increments and  $Q_{pkt}$  is subtracted from the well — the charge-subtraction reset of a digital-pixel readout (DROIC / DFPA). Everything upstream of the readout is unchanged; the counting branch replaces four things in the budget: the saturation bound, the quantization term, the per-frame kTC term, and the DN conversion.

**Effective well and saturation.**

$$Q_{eff} = 2^N Q_{pkt}, \quad Q_{dead} = f_{max} t_{int} Q_{pkt}, \quad Q_{sat} = \min(Q_{eff}, Q_{dead}),$$

with  $N = \text{readout.counter_bits}$  and  $f_{max} = \text{readout.max_count_rate_hz}$ , the comparator's dead-time flux ceiling (0 = no ceiling; rollover alone governs). Counter rollover is treated as saturation — the word clips, it does not wrap. An explicit analog full well under counting is rejected as over-specification: the effective well is  $2^N Q_{pkt}$ .

**Count conversion.**  $n = \lfloor Q_{int}/Q_{pkt} \rfloor$  counts and an analog residue  $Q_{res} = Q_{int} \bmod Q_{pkt} \in [0, Q_{pkt})$ .

**Quantization noise.** With the residue discarded (`readout.residue_readout = 0`) the quantizer step is the packet itself; with the residue digitized by the existing ADC model scoped to a full scale of one packet (the default), the step is the residue LSB:

$$\sigma_q = \frac{Q_{pkt}}{\sqrt{12}} \quad (\text{bare counter}), \quad \sigma_q = \frac{Q_{pkt}/2^M}{\sqrt{12}} \quad (\text{residue ADC, } M = \text{adc_bits}).$$

Both are the uniform-quantizer result and are flux-ensemble statements: valid when the signal spans several steps. Below  $Q_{int} \lesssim Q_{pkt}$  without residue readout the residue is the signal and the uniform assumption degrades.

**Packet reset noise.** Each charge subtraction is a reset, so the kTC term accumulates over the counts of the frame instead of entering once:  $\sigma_{reset} = \sqrt{n} \sigma_{kTC}$  (budget term `packet_reset`).

**Up/down differential counting** (`readout.counting_mode = "up_down"`). The counter becomes a signed modulo accumulator that increments during the scene (up) phase and decrements during a reference (down) phase, ending at

$$\Delta Q = Q_{up} - Q_{down}, \quad |\Delta Q| \leq 2^{N-1} Q_{pkt},$$

the in-pixel background subtraction for a dim target on a bright common background. Wrap during the up phase is not a failure — the down phase unwinds it — so the capacity constraint moves from rollover to the signed differential (clip mechanism `differential_overflow`). The down phase integrates for  $t_{down}$  (`readout.reference_integration_s`; 0 = equal to  $t_{int}$ , the balanced case) against a reference flux that is either the chain's own background term (sub-pixel and point-source regimes, where target and background are separate terms) or a user rate (`readout.reference_rate_e_per_s`), plus the dark, glow, near-field and stray electrons scaled by  $t_{down}/t_{up}$ . The means cancel; the noise does not: the reference phase pays its own shot noise,  $\sigma_{ref} = \sqrt{Q_{down}}$  — up to a  $\sqrt{2}$  penalty on the background noise relative to a noiseless reference. Packet reset noise accumulates over  $n_{up} + n_{down}$ , and the dead-time ceiling is checked on both phases. The signed count uses floor semantics,  $n = \lfloor \Delta Q / Q_{pkt} \rfloor$  with a non-negative residue — a two's-complement word plus the same residue convention as up-counting.

**Assumptions & validity.** Ideal comparator (no threshold noise beyond the kTC reset); equal packets; flux constant within a phase; the uniform-quantizer formula holds only when the signal spans several steps.

**Pitfalls.** Reading the counter word alone as the signal at low flux with the residue discarded; subtracting the reference *mean* without adding its shot noise — the flattering model that up/down counting forbids here; quoting  $2^N Q_{pkt}$  as the up/down capacity (it is  $2^{N-1} Q_{pkt}$ ); an analog `full_well_capacity_e` set beside the counting parameters.

**Numeric anchors.**  $N = 16$ ,  $Q_{pkt} = 2000$  e-:  $Q_{eff} = 131.07$  Me-, up/down capacity 65.54 Me-;  $\sigma_q = 577.35$  e- for the bare counter and 0.5638 e- with a 10-bit residue ADC ( $2000/1024/\sqrt{12}$ ).  $N = 14$ ,  $Q_{pkt} = 2000$  e-:  $Q_{eff} = 32.77$  Me- (the up-counting bound of scenario 2.9).

## 6.5 Noise regime selection

`detector.noise_regime` controls the budget composition: "imaging" (temporal terms only, default) vs "detection" (adds PRNU, DSNU, clutter). The regime distinction exists because single-frame detection against a scene sees the spatial pattern as noise, while a calibrated imaging chain removes the static component.

## 6.6 When each term dominates

| Regime                    | Dominant noise  | Typical scenario                                                                                              |
|---------------------------|-----------------|---------------------------------------------------------------------------------------------------------------|
| BLIP (background-limited) | Background shot | IR, long $t_{int}$ , low read noise — the Performance Metrics chapter gives the BLIP criterion and $f_{BLIP}$ |

| Regime               | Dominant noise | Typical scenario                |
|----------------------|----------------|---------------------------------|
| Read-noise limited   | Read           | Short $t_{int}$ , low flux      |
| Dark-current limited | Dark shot      | Long $t_{int}$ , warm detector  |
| FPN-limited          | PRNU/clutter   | $S > 1/k^2$ , no scan averaging |

A system can exit BLIP *from above* as flux grows:  $\sigma_{PRNU} = kN_{bg}$  outpaces  $\sqrt{N_{bg}}$  past the crossover.

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## 6.7 Parameter cross-reference

Volume III's Parameter Reference lists every detector and readout parameter with its type, default, and bounds.

# Calibration Error Model

What a radiometric calibration leaves behind. A non-uniformity correction removes the pixel-to-pixel gain and offset dispersion exactly at the signal levels where it was derived; everywhere else a residual survives, and that residual — not the temporal noise floor — frequently sets the achieved NEDT of a cooled infrared system. This chapter states the residual model, the drift terms, the accuracy (bias) budget, and how each enters the noise composition of Chapter 6.

Two facts drive the whole design. **Correlated errors do not average down:** TDI stages, co-added frames, and binned pixels reduce temporal noise as  $\sqrt{N}$ , but a calibration residual is the same error in every frame, so it is exempt. **Precision is not accuracy:** a cal-source temperature uncertainty moves the entire array's radiometric scale and must never be root-sum-squared into  $\sigma_{tot}$ , where it would masquerade as noise that integration could beat down.

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## 7.1 Overview — a terms-only stage after readout

The calibration stage sits between readout and the metric layer, and it transforms nothing: it collapses no spectrum, moves no frame, and — this matters — contributes no MTF term. Residual fixed-pattern noise is spatial *noise*, not a spatial *degradation*: it has no point spread function kernel, so neither spatial path of Chapter 5 gains a contributor and the dual-path consistency check is untouched.

What the stage contributes is three kinds of term:

- **Residual noise terms** [e- RMS], all classed spatial and all calibration-correlated: post-NUC residual, cal-source non-uniformity, gain drift, offset drift, and — on the internal-shutter cal path only — narcissus fixed-pattern noise.
- **Bias terms** [dimensionless fractions of radiance], the accuracy budget: cal-source temperature and emissivity, spectral calibration, internal-cal offset, and absolute gain.
- **Published totals:** the calibration RSS  $\sigma_{cal}$ , the post-calibration total  $\sigma_{tot}$ , the bias total, and the calibration-limited NEDT.

**The ordering is the physics.** Readout applies the  $\sqrt{N}$  scaling first; calibration appends afterwards. No term needs to be tagged “do not average me” for the exemption to hold — the chain position enforces it. Equivalently: the residual's size *relative to the signal* is invariant in  $N$ , because both the residual and the full-scale reference it is computed against live in the same summed domain.

**The type separation is the guarantee.** A noise term and a bias term are different types. SNR and NEDT read noise terms only; the radiometric-accuracy metric reads bias terms only. Nothing in the model can quietly RSS one into the other.

**Noise-regime interplay.** The imaging noise regime excludes the raw PRNU, DSNU, and clutter terms on the grounds that they are “calibrated out”. The calibration residuals are precisely what survives that calibration, so they enter the total in *both* the imaging and the detection regimes. With an active scheme the detector’s PRNU and DSNU are re-read as **pre-correction** dispersions feeding the residual model rather than as noise terms of their own, so nothing is counted twice.

## 7.2 Cal points — where the correction is anchored

A correction scheme is defined by the signal levels at which it is exact. RADIANT supports one-, two-, and three-point schemes, and two ways of declaring the points.

### 7.2.1 Temperature-declared points

The physical form: the cal source is a blackbody at a stated temperature. Because the stage runs after the single spectral collapse, only scalars are available, so the mapping from temperature to signal is a band photon-radiance ratio anchored at the scene:

$$S(T_{cal}) = S_{scene} \frac{B_q(T_{cal})}{B_q(T_{scene})}, \quad B_q(T) = \int_{\lambda_1}^{\lambda_2} \frac{B(\lambda, T)}{E_{ph}(\lambda)} d\lambda$$

with  $B_q$  in photon/s/m<sup>2</sup>/sr and  $E_{ph} = hc/\lambda_m$  [J]. Photon weighting, not energy weighting, is what the band integral uses, because the collected signal counts photons. Cal-point *fluxes* are never independent inputs in this mode — declaring both a cal temperature and its flux would over-specify the source.

**Assumption.** The spectral response  $QE(\lambda)\tau(\lambda)$  varies slowly enough across the band that a flat-weighted photon-radiance ratio stands in for the response-weighted signal ratio. Only the *ratio* enters, and numerator and denominator are anchored at nearby temperatures, so the first-order response slope largely cancels; the approximation degrades for a strongly sloped response curve.

**Numeric anchor.** MWIR band 3.0–5.0  $\mu\text{m}$ , scene at 300.0 K:  $S(280 \text{ K}) = 0.46475 S_{scene}$  and  $S(320 \text{ K}) = 1.96525 S_{scene}$ . For a scene signal of  $5.000 \times 10^4$  e- this places the cal points at  $S_1 = 2.3238 \times 10^4$  e- and  $S_2 = 9.8262 \times 10^4$  e-.

### 7.2.2 Flux-declared points

The laboratory form: an integrating sphere or flat-field source produces a known level relative to the scene, with no thermal anchor at all. The cal points are then declared directly as fractions of the scene signal,

$$S_i = f_i S_{scene}, \quad f_i \text{ dimensionless,}$$

with  $f_1 < f_2 < f_3$  required. In this mode every temperature-anchored input — the cal temperatures, the source temperature uncertainty, the source non-uniformity in K, the band-center uncertainty, and the source emissivity uncertainty — is **rejected** as over-specification rather than silently ignored. A set parameter that does nothing is exactly the failure class the model exists to prevent. Drift, the direct gain-uncertainty bias, and the internal-shutter path carry no temperature anchor and behave identically in both modes.

### 7.2.3 When the thermal anchor does not describe the scene

The temperature mapping anchors at the declared scene temperature. On a reflective scene that temperature describes none of the collected signal: a visible/near-infrared band looking at a 300 K surface collects Kirchhoff-reflected sunlight, while a 300 K blackbody puts essentially nothing into that band. The diagnostic quantity is the in-band share of the scene-temperature blackbody's photon exitance,

$$\Phi_{band}(T) = \frac{\int_{\lambda_1}^{\lambda_2} B_q(\lambda, T) d\lambda}{\int_{0.05 \mu m}^{1000 \mu m} B_q(\lambda, T) d\lambda} \quad [\text{dimensionless}]$$

**Numeric anchor.**  $\Phi_{band}(300 \text{ K})$  is  $4.30 \times 10^{-22}$  for a 0.50–0.85  $\mu m$  band,  $3.20 \times 10^{-3}$  for a 3.0–5.0  $\mu m$  band; a 77 K laboratory scene in an 8–12  $\mu m$  band gives  $1.99 \times 10^{-5}$ . The discrimination spans nearly twenty orders of magnitude, so the  $10^{-9}$  floor the model warns below is a diagnostic bound, not tuned physics.

Below that floor — or for a target declared purely reflective — the cal points are deterministic stand-ins: the residual *structure* (the shape of the parabola, the  $\sqrt{N}$  exemption, the plateau between cal points) survives, but the absolute level rides the stand-in. The run proceeds under a loud advisory rather than silently, and the correct remedy is to declare the cal points as flux fractions.

### 7.2.4 The full-scale reference

The nonlinearity coefficient of *Post-NUC residual* below is referenced to full scale, so the reference  $S_{ref}$  must live in the same summed domain as the signal it is compared against: the counting detector's effective well when that branch ran, otherwise the per-pixel full-well capacity scaled by the accumulation gain the signal actually received through TDI, binning, and co-adding. One domain for numerator and denominator is what makes the residual's signal-relative size invariant in  $N$ .

## 7.3 Post-NUC residual — the correctability parabola

### 7.3.1 Two-point correction

Model each pixel's response as linear plus a small quadratic term,  $y = g(S + \beta S^2/S_{ref}) + o$ , with the curvature coefficient  $\beta$  dispersed across the array with  $1\sigma$  magnitude  $\sigma_\beta$ . A two-point correction solves for each pixel's gain and offset so that the corrected output is exact at  $S_1$  and  $S_2$ . Carrying the exact linear-correction algebra through in the small- $\beta$  limit leaves the per-pixel error

$$\Delta S = \beta \frac{(S - S_1)(S - S_2)}{S_{ref}},$$

so the array RMS residual is

$$\sigma_{NUC}(S) = \beta \frac{|(S - S_1)(S - S_2)|}{S_{ref}} \quad [\text{e- RMS}]$$

— a parabola that vanishes at both cal points, peaks between them, and grows quadratically outside them. The peak sits at the midpoint of the cal span and has the value  $\beta (S_2 - S_1)^2 / (4S_{ref})$ , which is the one number worth memorizing: the residual is quadratic in the *span*, so halving the separation between cal points quarters the worst-case residual inside it. Gain and offset dispersion (PRNU and DSNU) are removed exactly by the correction itself and re-enter only through the drift terms of *Drift between calibration events*.

**Symbols.**  $S$  scene signal [e-];  $S_1, S_2$  cal-point signals [e-];  $\beta$   $1\sigma$  nonlinearity fraction at full scale [dimensionless];  $S_{ref}$  full-scale reference [e-].

**Pitfalls.** Quoting the residual at a cal point (it is zero there by construction, which is why *Cal-source spatial non-uniformity* exists); referencing  $\beta$  to the signal instead of to full scale, which makes the residual cubic; taking  $S_{ref}$  in the per-pixel domain while  $S$  is in the summed TDI domain, which makes the residual appear to shrink with  $N_{TDI}$ .

**Numeric anchor.**  $\beta = 0.5\%$ ,  $S_1 = 2.000 \times 10^4$  e-,  $S_2 = 8.000 \times 10^4$  e-,  $S_{ref} = 1.000 \times 10^5$  e-: at the midpoint  $S = 5.000 \times 10^4$  e- the residual is 45.00 e- RMS, and it is 0.00 e- RMS at either cal point.

### 7.3.2 One-point correction

An offset-only correction leaves the gain dispersion intact, acting on the departure from the single cal point:

$$\sigma_{1pt}(S) = k |S - S_1| \quad [\text{e- RMS}]$$

with  $k$  the pre-correction PRNU fraction. The contrast with the two-point correction above is the point of running two points at all: the one-point residual is *linear* in the departure and carries the full PRNU coefficient, so it is typically an order of magnitude larger.

**Numeric anchor.**  $k = 0.5\%$ ,  $S_1 = 2.000 \times 10^4$  e-,  $S = 5.000 \times 10^4$  e-: 150.0 e- RMS — 3.3× the two-point residual at the same signal in the anchor above.

### 7.3.3 Three-point correction

With a third point between the extremes, the correction is piecewise: inside each bracketing segment it is exactly that pair's two-point solve, so the residual is that segment's own parabola, vanishing at all three cal points and peaking at a quarter of the corresponding two-point value when the mid point bisects the span. Outside the calibrated range the nearest segment extrapolates, exactly as the two-point model extrapolates past its own span. The model stops at three points by design; beyond that a real instrument uses a response curve, not more anchors.

**Numeric anchor.** Same  $\beta$  and  $S_{ref}$  as the two-point correction, with a mid point at  $5.000 \times 10^4$  e-: the residual at  $3.500 \times 10^4$  e- (the midpoint of the lower segment) is 11.25 e- RMS, a quarter of the 45.00 e- RMS two-point peak.

## 7.4 Cal-source spatial non-uniformity

A real calibration blackbody holds only  $\pm 0.01$ – $0.05$  K ( $1\sigma$ ) across its aperture. At cal time each pixel therefore views a slightly different source temperature and the correction imprints that pattern into its own coefficients.

With  $\delta T$  the pixel's local departure and  $D_j = dS/dT$  evaluated at cal point  $j$  [e-/K], the imprinted error at cal view  $j$  is  $\delta S_j = \delta T D_j$ , and propagating it through the correction solve gives

$$\sigma_{unif}(S) = \Delta T_{unif} D_1 \quad (\text{one-point}), \quad \sigma_{unif}(S) = \Delta T_{unif} \frac{|D_1(S_2 - S) + D_2(S - S_1)|}{S_2 - S_1} \quad (\text{two-point})$$

with the three-point form interpolating piecewise between the bracketing pair's constants.

**The combination is linear, not RSS.** The same plate is viewed at both cal points and a cavity gradient is temperature-independent in kelvin to first order, so the two imprints are fully correlated. Treating them as independent and root-sum-squaring the weights would under-predict the residual inside the cal span.

**Why this term matters out of proportion to its size.** Unlike the nonlinearity parabola, it does **not** vanish at the cal points:  $\sigma_{unif}(S_1) = \Delta T_{unif} D_1$  exactly. It is what retires the comfortable but false picture of a calibration that is perfect where it was taken. When it dominates, it sets a calibration-limited NEDT floor of approximately  $\Delta T_{unif}$  itself, because the term and the scene derivative it is divided by are both proportional to  $dS/dT$ .

**Numeric anchor.** MWIR 3.0–5.0  $\mu\text{m}$ , scene 300.0 K, scene signal  $5.000 \times 10^4$  e-, cal points 280.0 K and 320.0 K:  $D_1 = 950.3$  e-/K and  $D_2 = 3123.9$  e-/K. With  $\Delta T_{unif} = 0.020$  K the residual at the scene signal is 34.51 e- RMS. Against the scene's own derivative  $dS/dT = 1795$  e-/K that is 0.0192 K — within 4 % of  $\Delta T_{unif}$ , as the floor argument predicts.

## 7.5 Drift between calibration events

Between the cal event and the observation, the corrected gain and offset both walk. The v1 model is time-linear in both, with  $1\sigma$  rates:

$$\sigma_{gain}(S) = r_g t_{cal} S \quad [\text{e- RMS}], \quad \sigma_{offset} = r_o t_{cal} \quad [\text{e- RMS}]$$

with  $r_g$  in  $\text{s}^{-1}$  (entered as %/hour),  $r_o$  in e-/s (entered as e-/hour), and  $t_{cal}$  the elapsed time since calibration [s]. Gain drift is proportional to signal and therefore scales with scene brightness; offset drift is signal-independent and dominates at low flux. Both are spatial and calibration-correlated, so both are exempt from  $\sqrt{N}$  averaging.

The practical use of these two terms is cadence: they are what turns “how often must we calibrate” into a number, since they are the only residual terms that grow without bound between events.

**Numeric anchor.**  $r_g = 0.20\%$ /hour and  $r_o = 10.0$  e-/hour, 1800 s (30 min) since the cal event, scene signal  $5.000 \times 10^4$  e-:  $\sigma_{gain} = 50.00$  e- RMS and  $\sigma_{offset} = 5.000$  e- RMS.

**Deferred.** Drift driven by focal-plane temperature change is not modeled; the v1 rates are time-linear only.

## 7.6 The internal-shutter cal path

A calibration flag inside the optical train sees only itself and the elements *behind* it. The correction derived from that view absorbs the aft-element emission but never the emission of the elements in *front* of the flag, which returns in operation as an uncorrected offset. The split is the photon-weighted in-band fraction of the near-field focal-plane irradiance contributed by the fore elements:

$$f_{fore} = \frac{\int_{\lambda_1}^{\lambda_2} \sum_{i \in \text{fore}} E_i(\lambda) \lambda d\lambda}{\int_{\lambda_1}^{\lambda_2} \sum_{i \in \text{all}} E_i(\lambda) \lambda d\lambda} \quad [\text{dimensionless}]$$

with  $E_i(\lambda)$  the per-element near-field irradiance at the focal plane [W/m<sup>2</sup>/μm]. Each watt of in-band power carries  $\lambda/hc$  photons, so  $hc$  cancels in the ratio and only the factor  $\lambda$  survives as the weighting. The electron magnitude then rides the detector's own near-field conversion,  $S_{fore} = S_{nf} f_{fore}$  [e-]: no radiometry is re-derived here, which is what keeps this term consistent with the warm-optics treatment of Chapter 3 and the appendix.

That uncorrected offset produces two distinct effects. Its mean is an **offset bias**,  $S_{fore}/S$  as a fraction of the scene signal, which belongs in the accuracy budget of *The bias budget* below. The part of it that varies across the array — cold-stop reflections, vignetting of the warm fore-optics — is **narcissus fixed-pattern noise**, a spatial noise term equal to a declared fraction of  $S_{fore}$ .

**Assumption.** The split weights per-element irradiance by photon count rather than by the QE spectrum. It is exact for flat QE across the band and first-order otherwise, because the per-element spectra are same-family graybody curves and the QE weighting largely cancels in the ratio.

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## 7.7 Composition — how calibration enters the budget

The five residual terms are mutually independent mechanisms, so they combine in quadrature, and the calibration RSS combines the same way with the post-readout total:

$$\sigma_{cal} = \sqrt{\sigma_{NUC}^2 + \sigma_{unif}^2 + \sigma_{gain}^2 + \sigma_{offset}^2 + \sigma_{narc}^2}, \quad \sigma_{tot} = \sigma_{readout} \oplus \sigma_{cal}$$

Every term is appended in the electron domain, input-referred at the sense node, exactly as Chapter 6 requires of any noise term. All five are classed **spatial**, so a single-frame spatial budget contains them; none is temporal, so none averages down over frames. The downstream metrics prefer this post-calibration total wherever it is published: SNR and contrast SNR use  $\sigma_{tot}$  above, and the signal-to-clutter-plus-noise ratio adds  $\sigma_{cal}$  into its spatial RSS.

**Worked example (MWIR staring, two-point cal).** Band 3.0–5.0 μm, scene 300.0 K, scene signal  $5.000 \times 10^4$  e-, full-scale reference  $1.000 \times 10^5$  e-, cal points 280.0 K and 320.0 K (so  $S_1 = 2.3238 \times 10^4$  e-,  $S_2 = 9.8262 \times 10^4$  e-), nonlinearity  $\beta = 0.5\%$ , source uniformity  $\Delta T_{unif} = 0.020$  K, drift rates  $r_g = 0.20\%$ /hour and  $r_o = 10.0$  e-/hour at  $t_{cal} = 1800$  s, full-aperture cal path:

| Term                                             | Value               | Class                   |
|--------------------------------------------------|---------------------|-------------------------|
| Post-NUC residual                                | 64.58 e- RMS        | spatial, cal-correlated |
| Cal-source uniformity                            | 34.51 e- RMS        | spatial, cal-correlated |
| Gain drift                                       | 50.00 e- RMS        | spatial, cal-correlated |
| Offset drift                                     | 5.000 e- RMS        | spatial, cal-correlated |
| <b>Calibration RSS <math>\sigma_{cal}</math></b> | <b>88.81 e- RMS</b> | —                       |

Against a temporal (post-readout) total of 300.0 e- RMS this gives  $\sigma_{tot} = 312.9$  e- RMS: the SNR falls from 166.7 to 159.8, a 4.1 % loss that no amount of integration time recovers.

**Rule-4 non-interaction.** None of these terms has a PSF kernel or an MTF factor. Adding an “FPN MTF” would double-count a spatial *noise* as a spatial *degradation*.

## 7.8 Calibration-limited NEDT

The reason the stage exists. Dividing the calibration RSS by the scene’s own thermal derivative gives the temperature resolution the calibration alone permits:

$$\text{NEDT}_{cal} = \frac{\sigma_{cal}}{dS/dT} \quad [\text{K}]$$

with  $dS/dT$  [e-/K] the band-integrated derivative the chain already computes for NEDT. This is a *floor*, not a contribution to be traded: because  $\sigma_{cal}$  is exempt from  $\sqrt{N}$  averaging, lengthening the integration or adding TDI stages raises the signal and the residual together, leaving  $\text{NEDT}_{cal}$  where it was. A system whose temporal NEDT has been pushed below its calibration floor has bought nothing.

**Numeric anchor.** Continuing the *Composition* example,  $dS/dT = 1795$  e-/K at the 300.0 K scene:

| Quantity                               | Value    |
|----------------------------------------|----------|
| Temporal NEDT (readout total alone)    | 167.1 mK |
| Calibration-limited NEDT               | 49.5 mK  |
| Achieved NEDT (post-calibration total) | 174.3 mK |

The calibration floor costs only 7.2 mK on this 167.1 mK system. Halve the temporal noise and the *same* floor costs 13.5 mK on an 83.6 mK system; halve it again and it costs 23.0 mK on a 41.8 mK system, which now achieves 64.8 mK. The floor does not move, so every improvement bought with integration time, cooling, or TDI stages is worth progressively less until the calibration — not the detector — is the thing to fix. That crossover, not the absolute numbers, is the design content of this chapter.

## 7.9 The bias budget — accuracy, not precision

A calibration-source error does not disperse pixel to pixel; it moves the whole array's radiometric scale. Such an error is reported as a fractional radiance bias and never appears in  $\sigma_{tot}$ .

**Cal-source temperature.** A  $1\sigma$  uncertainty  $\Delta T_{src}$  in the cal source's own temperature propagates through the photon-weighted band Planck log-derivative at the cal temperature:

$$b_{src} = \frac{\int_{\lambda_1}^{\lambda_2} (\partial B_q / \partial T) d\lambda}{\int_{\lambda_1}^{\lambda_2} B_q d\lambda} \Delta T_{src} \quad [\text{dimensionless}]$$

**Cal-source emissivity.** A pure scale error,  $b_\varepsilon = \Delta\varepsilon / \varepsilon_{src}$ .

**Spectral calibration.** A band-center error  $\delta\lambda$  [ $\mu\text{m}$ ] rigidly shifts the band the radiometry is computed over. The calibration absorbs the resulting scale error *at its own source temperature*, because the gain is derived there, so what survives on a scene is the difference of the band-shift log-derivatives:

$$g(T) = \frac{B_q(\lambda_2, T) - B_q(\lambda_1, T)}{\int_{\lambda_1}^{\lambda_2} B_q(\lambda, T) d\lambda} [\mu\text{m}^{-1}], \quad b_{spec} = |g(T_{scene}) - g(T_{cal})| \delta\lambda$$

The expression follows from the Leibniz rule with both band edges moving together. It is exactly zero when the scene sits at the cal temperature and grows with the scene-versus-cal separation; short-wave (Wien-side) bands are the sensitive ones. Band-*width* drift is a distinct and typically smaller mechanism and is not modeled.

**Internal-cal offset.** The uncorrected fore-optics emission of *The internal-shutter cal path* as a fraction of the scene signal,  $S_{fore}/S$ .

**Absolute gain.** A directly declared fractional uncertainty in the radiometric gain.

**Composition.** Independent sources RSS *within* the budget, and the total converts to a temperature at the scene through the chain's own derivative:

$$b_{tot} = \sqrt{\sum_i b_i^2}, \quad \Delta T = \frac{b_{tot} S}{dS/dT} \quad [\text{K}]$$

A scene with no thermal derivative reports the fractional value with a named failure for the kelvin conversion rather than a silent NaN.

**Numeric anchors.**  $\Delta T_{src} = 0.10$  K on a 300.0 K cal source gives  $b_{src} = 3.590 \times 10^{-3}$  (0.359 %) in a 3.0–5.0  $\mu\text{m}$  band and  $1.615 \times 10^{-3}$  (0.161 %) in an 8.0–12.0  $\mu\text{m}$  band — the MWIR band's steeper Planck slope makes it 2.2 $\times$  as unforgiving of source-temperature error. An emissivity uncertainty of 0.002 on an  $\varepsilon_{src} = 0.99$  cavity gives  $b_\varepsilon = 2.020 \times 10^{-3}$ . A band-center uncertainty  $\delta\lambda = 0.005$   $\mu\text{m}$  with the scene at 300.0 K and the cal at 320.0 K in the 3.0–5.0  $\mu\text{m}$  band gives  $b_{spec} = 6.121 \times 10^{-4}$  (0.061 %), from  $g(300 \text{ K}) = 1.5459 \mu\text{m}^{-1}$  and  $g(320 \text{ K}) = 1.4235 \mu\text{m}^{-1}$ .

**Pitfalls.** RSS-ing a bias into the noise total, which makes an accuracy error look like something  $\sqrt{N}$  can fix; quoting the spectral-calibration bias at the cal temperature (it is zero there); reporting a bias in kelvin without stating the scene temperature it was converted at.

## 7.10 Assumptions, validity, and what is deferred

| Assumption                                  | Why it holds                                                                                                     | What breaks it                                                                                                                                                              |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quadratic response over the cal span        | The departure from linearity of a well-behaved photodiode is smooth and small over the calibrated range          | Approach to saturation, where the response is not polynomial and the residual is a lower bound                                                                              |
| Band photon-radiance ratio for cal points   | Only the ratio enters, anchored at nearby temperatures, so the response slope largely cancels                    | Strongly sloped $QE\tau$ across the band; a scene whose signal is not thermal at the declared temperature (see <i>When the thermal anchor does not describe the scene</i> ) |
| Fully correlated source-uniformity imprints | The same plate is viewed at every cal point and a cavity gradient is temperature-independent in K to first order | A source whose gradient changes shape with temperature                                                                                                                      |
| Time-linear drift                           | Adequate over a cal interval short compared with the thermal time constants driving it                           | Long intervals, or drift driven by focal-plane temperature excursions (not modeled)                                                                                         |
| Full PRNU correlation along a TDI column    | Takes the correlated limit, an upper bound on the residual floor                                                 | Long TDI columns with partially decorrelated pixel gains — the true residual is then lower than modeled                                                                     |
| Photon-weighted fore-optics split           | Exact for flat QE; the per-element spectra are same-family graybody curves                                       | Sharply structured QE across the band                                                                                                                                       |

**Deferred by ruling.** Focal-plane-temperature-driven drift; operations-level calibration (cal cadence trades, scene-based non-uniformity correction); partial along-column decorrelation of PRNU under long TDI. Each has a home in this stage when it arrives.

**Boundary conditions.** With the scheme off, the stage emits nothing and results are bit-identical to a chain without it — enabling the model is a deliberate act, never a default that quietly moves numbers. Cal points that coincide, are unordered, or are left unset with a scheme active are rejected with an actionable error rather than ill-conditioned arithmetic: as  $S_2 \rightarrow S_1$  the two-point solve degenerates, and the model refuses instead of dividing by a vanishing span.

## 7.11 Parameter cross-reference

Volume III's Parameter Reference lists the `calibration.*` namespace with types, defaults, bounds, and entry units. Every cal-related default is the model-off limit.

# Performance Metrics

The metric layer: how RADIANT turns the signal/noise/spatial chain state into SNR, NEDT, NIIRS, detection range, and their relatives. Numeric anchors were re-derived from the literature independently of the implementation. This is the one layer permitted to report a *named failure* instead of a number — a metric that cannot be computed for a scene says so, with a reason the caller can read; silently returning NaN remains forbidden anywhere in RADIANT.

**Symbols:**  $S$  target signal [e-];  $S_{bg}$  background signal [e-];  $\sigma_{tot}$  total noise [e- RMS] (see the Noise Model chapter);  $t_{int}$  [s];  $p$  pixel pitch [m]; GSD [m].

## 8.1 SNR family

**Equations.**

$$\text{SNR} = \frac{S}{\sigma_{tot}}, \quad \text{CSNR} = \frac{S - S_{bg}}{\sigma_{tot}}, \quad \text{SCNR} = \frac{S - S_{bg}}{\sigma_{tot} \oplus \sigma_{clutter}}$$

( $\oplus$  = RSS). **The denominator always contains the background and dark shot noise:** subtracting the background estimate in the numerator does not remove its Poisson fluctuations. In the thermal IR, where  $\Delta S \ll S_{bg}$ , using  $\sqrt{S - S_{bg}}$  in the denominator overstates contrast SNR by large factors (1.9× at the audit anchor).

**Numeric anchor.**  $S = 10000$ ,  $S_{bg} = 5000$ ,  $N_{dark} = 1000$  e-,  $\sigma_{read} = 50$  e-, PRNU 0.1 %:  $\sigma_{tot} = 136.38$  e-, SNR = 73.32, CSNR = 36.66.

## 8.2 NEDT

**Equation.**

$$\text{NEDT} = \frac{\sigma_{tot}}{dS/dT} \quad [\text{K}],$$

with the thermal responsivity band-integrated in the photon domain:

$$\frac{dS}{dT} = t_{int} G \int \frac{\lambda_m}{hc} QE(\lambda) \tau(\lambda) \frac{\partial L_\lambda}{\partial T} d\lambda \quad [\text{e}^-/\text{K}],$$

$\partial L_\lambda / \partial T$  the analytic Planck derivative (the Radiometric Chain chapter),  $G$  the étendue. RADIANT computes  $dS/dT$  in `SpectralIntegrationStage` (`ds_dt_e_per_K` stage output) and divides in the metric layer; a fallback `compute_nedt_from_snr` uses the SNR route when the derivative output is absent.

**Pitfalls.** Paraxial étendue  $A_d \pi / (4F_\#^2)$  vs exact  $A_d \pi / (4F_\#^2 + 1)$  — 6.25 % at  $f/2$ , 25 % at  $f/1$  (understates NEDT); finite-difference  $\Delta T = 1$  K instead of the analytic derivative; energy-domain radiance without the  $\lambda_m/hc$  photon factor; quoting mK NEDT that implies an impossible well fill (the audit's LWIR example: an  $f/2$ , 1 ms configuration implies 800× a 100 ke- well — always sanity-check  $S_{bg}$  against `total_well_e`).

**Numeric anchors.**  $dS/dT = 5000$  e-/K,  $\sigma = 60$  e-  $\rightarrow$  NEDT = 12.0 mK;  $\int_8^{12} (\partial L_\lambda / \partial T) d\lambda = 0.6297$  W/m<sup>2</sup>/sr/K at 300 K.

### 8.3 NEI, NEP, and D\*

**Equations.** Noise-equivalent irradiance at the aperture (photon domain):

$$\text{NEI} = \frac{\sigma_{tot}}{\eta_{sys} EE A_{ap} t_{int}} \quad [\text{ph/s/cm}^2 \text{ or } / \text{m}^2],$$

power form via  $hc/\lambda_m$ . Specific detectivity:

$$D^* = \frac{\sqrt{A_d \Delta f}}{\text{NEP}} \quad [\text{cm}\sqrt{\text{Hz}}/\text{W}], \quad \Delta f = \frac{1}{2 t_{int}},$$

with  $A_d$  in **cm<sup>2</sup>** (Jones convention). BLIP-limit PV detectivity  $(\lambda_m/hc)\sqrt{\eta/(2Q_b)}$ ; photoconductors are  $\sqrt{2}$  lower (generation *and* recombination noise).

**Symbols in this section.**  $\eta$  is the detector **quantum efficiency** [dimensionless] and  $\eta_{sys}$  the end-to-end **photon-to-electron efficiency** (optical throughput  $\times$  QE), following the detector literature these expressions come from. This is the one place in the manual where  $\eta$  is not the look angle of the geometry chapter (notation §9); the two never appear in one equation.  $Q_b$  is the **background photon flux at the detector** [photon/s/cm<sup>2</sup>] that sets the BLIP limit — the same in-band background the noise chapter integrates, expressed as a photon irradiance on  $A_d$  rather than as electrons per frame.

**Pitfalls.**  $\Delta f = 1/t_{int}$  vs  $1/(2t_{int})$  — a  $\sqrt{2}$  error of exactly the PV/PC magnitude, easily conflated;  $A_d$  in m<sup>2</sup> (inflates  $D^* \times 100$ ); omitting the ensquared-energy factor in NEI (optimistic by  $1/EE$ ); photon/energy domain mixing (the  $\lambda_m/hc$  lost or doubled — the NEP $\leftrightarrow$ electrons converter carries an absolute anchor for exactly this).

**Numeric anchors.**  $\text{NEP} = 10^{-14}$  W,  $A_d = (20 \text{ }\mu\text{m})^2$ ,  $\Delta f = 1000$  Hz  $\rightarrow D^* = 6.3246 \times 10^{12}$  cm $\sqrt{\text{Hz}}/\text{W}$ ; NEP anchor  $\sigma = 100$  e-,  $\eta = 0.7$ ,  $\lambda = 10 \text{ }\mu\text{m}$ ,  $t = 10$  ms  $\rightarrow 2.8378 \times 10^{-16}$  W.

### 8.4 NIIRS via GIQE-5 (and the IIRS placeholder)

**Equation.**

$$\text{NIIRS} = c_0 + c_1 \log_{10} \text{GSD}_{\text{in}} + c_2 \log_{10} \text{RER} + c_3 \log_{10} \text{SNR} + c_4 H + c_5 G$$

with the literature coefficients (9.57, −3.32, 3.32, 1.559, −0.334, −0.01) pinned exactly by test; GSD in **inches** (geometric mean of the two ground-sample directions the geometry chapter derives), RER the geometric mean from the PSF path,  $H$  edge overshoot,  $G$  noise gain.

**Fit-envelope gating.** GIQE-5 was fit over GSD 1.18–31.5 in, RER 0.2–0.95, SNR 2–130. Outside the envelope RADIANT refuses by default (result-typed failure with `failure_reason`), computes only when `performance.niirs.allow_extrapolated = true`, and flags `niirs_extrapolated = 1`.

**IIRS status.** The MWIR/LWIR interpretability metric currently **reuses GIQE-5 verbatim** — same formula, envelope, and coefficients. Treat IR “IIRS” outputs as GIQE-5-on-IR until a real IIRS lands.

**Numeric anchor.** GSD = 1 m, RER = 0.9, SNR = 50 → NIIRS = 6.4268 (pure-literal test value).

## 8.5 Johnson criteria and minimum resolvable

**Equations.** Cycles resolved across a target’s critical dimension  $d_c$  at range  $R_s$ :

$$N_{cyc} = \frac{d_c}{2 \text{ IFOV } R_s}$$

(the factor 2 converting pixels to cycles), compared against the Johnson thresholds (detect  $\approx 1$ , recognize  $\approx 4$ , identify  $\approx 8$  cycles, 50 % probability). `johnson_range_m` inverts for range at a given task. Minimum-resolvable temperature/contrast couple the MTF budget to NEDT/SNR thresholds.

**Pitfalls.** Cycles vs pixels (factor 2); using nadir IFOV footprint at slant geometry; 50%-probability thresholds quoted as certainties.

## 8.6 Detection range

**Equation.** For a point source with atmospheric extinction, the signal-vs-range equation mixes  $1/R_s^2$  geometric falloff with transmission along the path,  $S(R) = S_{ref}(R_{ref}/R)^2 \tau(R)/\tau(R_{ref})$ . The noise is **not** frozen at the reference range: the target’s own shot variance in electrons is the signal itself, so

$$\sigma^2(R) = S(R) + N_0^2, \quad N_0^2 = \sigma_{ref}^2 - S_{ref}$$

with  $N_0$  the target-free floor (background shot, dark, read, kTC, quantization). RADIANT solves

$$\frac{S(R)}{\sqrt{S(R) + N_0^2}} = \text{SNR}_{threshold}$$

by bisection (point-source regime only; threshold parameter default 5.0). The criterion inverts in closed form — the signal a threshold  $T$  demands is  $S^* = \frac{1}{2} \left( T^2 + \sqrt{T^4 + 4T^2 N_0^2} \right)$ , so in vacuum the answer is  $R = R_{ref} \sqrt{S_{ref}/S^*}$  with no root finding at all. Both forms agree at  $R_{ref}$  by construction, so the answer no longer depends on the range the chain was evaluated at. The superseded frozen-noise form solved  $S(R)/\sigma_{ref} = T$  instead, and gave 123.4 km referenced at 25 km against 182.5 km referenced at 100 km for one unchanged configuration — a range that depended on where you asked from.

**Pitfalls.** Applying extended-scene SNR to the point-source range equation; forgetting that EE\_box and jitter enter through the signal chain, not as post-hoc factors;  $\alpha$  from a band-average  $\tau$  over a very different path length; passing a signal and a total noise from different operating points, which leaves  $\sigma^2 - S$  negative and is refused.

---

## 8.7 Saturation, dynamic range, BLIP

**Equations.** Well margin  $20 \log_{10}(\text{FWC}/S_{tot})$  dB and ADC margin  $20 \log_{10}(DN_{max}/S_{DN})$  dB; dynamic range

$$\text{DR} = 20 \log_{10} \frac{\text{FWC}}{\sigma_{floor}} \text{ dB}$$

with  $\sigma_{floor}$  the **dark-scene** floor (read + dark shot + ADC), not the shot-inflated full-well noise. BLIP: background shot exceeds all other noise RSS'd;  $f_{BLIP} = \sigma_{bg,shot}/\sigma_{tot}$  ( $\rightarrow 1$  fully BLIP; threshold  $1/\sqrt{2}$ ); `blip_rate_e_per_s` reports the photon rate scale.

**Pitfalls.** 10 log vs 20 log (halves the dB); ADC full scale vs true full well (take the min); counting target shot as “background” in the BLIP test; quoting BLIP at one  $t_{int}$  as if general.

**Numeric anchors.** 100 ke-/50 e-  $\rightarrow$  66.02 dB; the audit’s marginal-BLIP case:  $\sigma_{bg} = 70.7$  vs other-RSS 60.0 e-,  $f_{BLIP} = 0.762$ .

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## 8.8 Radiometric inversions and sensitivity relatives

**NE $\Delta$ L / NE $\Delta$ p** — noise-equivalent radiance and reflectance differences, computed as radiance/SNR and reflectance/SNR. They are available as helpers but are **not part of the chain’s metric surface**, so a run does not report them. **Temperature retrieval** — band-radiance inversion with emissivity and temperature Jacobians; the forward band-integrated Planck radiance it inverts is anchored at 38.5 W/m<sup>2</sup>/sr for a 300 K scene over 8–12  $\mu\text{m}$ .

## 8.9 Metric selection

Metrics are computed in five groups — radiometric, sampling/geometry, spatial-MTF, saturation, interpretability — each of which can be switched off. All five are on by default, and switching one off stops the computation rather than hiding the result: a metric another enabled group depends on is still computed, because the selection closes over dependencies.

Deselecting the whole Spatial-MTF group is the one case that saves real time. When nothing else needs a spatial input, the PSF convolution stack and the MTF product are skipped entirely, and with them the consistency check that normally compares the two spatial paths on every run — there is no spatial computation left to check.

# Radiometric Model — Mixed Refractive and Reflective Optical Train

**This appendix is an implementation recipe, and it is written like one.** The chapters before it state each equation with its derivation, its validity limits and a pinned numeric anchor; this appendix instead gives the quantities in the order a program must compute them, in the ASCII notation the implementation uses, so that a reader working through the code can follow it line for line. There are no derivations or pitfalls sections here, and the equation blocks are deliberately not typeset as display mathematics — they are the recipe, not the physics. The physics they assemble is in the radiometric-chain and spatial chapters.

Two separate radiometric paths are modeled:

1. **Signal path** — transmission/reflection of source photons through the full optical train
2. **Thermal background path** — self-emission of each optical element propagated to the focal plane

Elements are typed as either REFRACTIVE or REFLECTIVE. The per-element equations differ fundamentally between the two types.

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## A.1 Element types and their radiometric character

### A.1.1 Refractive element (lens, window, filter, beamsplitter substrate)

- Signal passes **through** the element
- Has bulk absorption (Beer-Lambert), entry and exit surface coatings
- System quantity is **transmittance**  $T_{sys,i}(\lambda)$
- Thermal emission sourced from **bulk substrate + both surfaces**
- Cavity etalon effect between R1 and R2

### A.1.2 Reflective element (mirror, cold stop, baffle)

- Signal **reflects off** the element
  - No bulk propagation — surface interaction only
  - System quantity is **reflectance**  $Rho_{sys,i}(\lambda)$
  - Thermal emission sourced from **surface coating only**
  - No etalon — single surface interaction
  - Emissivity governed by Kirchhoff:  $eps_i(\lambda) = 1 - Rho_{sys,i}(\lambda) - A_{coat,i}(\lambda)$  where  $A_{coat,i}$  is the coating absorption (if known); if not:  $eps_i = 1 - Rho_{sys,i}$
-

## A.2 Per-element radiometric quantities

### A.2.1 Refractive element i

$$\begin{aligned} \text{beer}_i(\lambda) &= \exp(-\alpha_i(\lambda) \cdot d_i / \cos(\theta_{r,i})) \\ \text{denom}_i(\lambda) &= 1 - R1_i(\lambda) \cdot R2_i(\lambda) \cdot \text{beer}_i(\lambda)^2 \end{aligned}$$

$$T_{\text{sys},i}(\lambda) = \frac{T1_i(\lambda) \cdot \text{beer}_i(\lambda) \cdot T2_i(\lambda)}{1 - R1_i(\lambda) \cdot R2_i(\lambda) \cdot \text{beer}_i(\lambda)^2}$$

$$\text{eps}_{\text{eff},i}(\lambda) = \frac{T2_i(\lambda) \cdot n_i(\lambda)^2 \cdot (1 - \text{beer}_i(\lambda))}{1 - R1_i(\lambda) \cdot R2_i(\lambda) \cdot \text{beer}_i(\lambda)^2}$$

$$L_{\text{thermal},i}(\lambda) = \text{eps}_{\text{eff},i}(\lambda) \cdot B(\lambda, T_i)$$

The element **transfer factor** applied to upstream signals passing through it:

$$C_i(\lambda) = T_{\text{sys},i}(\lambda) \quad [\text{refractive} - \text{transmission}]$$

### A.2.2 Reflective element i

No bulk absorption. Single surface interaction. The coating has reflectance  $Rho_i(\lambda)$  and absorptance  $A_{\text{coat},i}(\lambda)$ . Transmittance is zero (opaque mirror).

$$Rho_{\text{sys},i}(\lambda) = Rho_i(\lambda) \quad [\text{mirror reflectance}]$$

$$\text{eps}_i(\lambda) = 1 - Rho_i(\lambda) - A_{\text{coat},i}(\lambda) \quad [\text{surface emissivity via Kirchhoff}]$$

If coating absorptance is not separately characterized, conservative form:

$$\text{eps}_i(\lambda) = 1 - Rho_i(\lambda) \quad [\text{upper bound on emissivity}]$$

Thermal radiance from the mirror surface:

$$L_{\text{thermal},i}(\lambda) = \text{eps}_i(\lambda) \cdot B(\lambda, T_i)$$

The element **transfer factor** applied to upstream signals reflecting off it:

$$C_i(\lambda) = Rho_{\text{sys},i}(\lambda) \quad [\text{reflective} - \text{reflectance}]$$

### A.2.3 Planck blackbody radiance (common to both types)

$$B(\lambda, T_i) = \frac{2 h c^2 / \lambda^5}{\exp(h c / \lambda k_B T_i) - 1}$$

Constants are the CODATA 2018 values of the Notation chapter's physical-constants table; they are defined once in the code and never re-entered here.

$B$  is a **per-wavelength** spectral radiance in  $\text{W}/\text{m}^2/\text{sr}/\mu\text{m}$ . Evaluate the expression above with  $\lambda$  in meters and it returns  $\text{W}/\text{m}^2/\text{sr}/\text{m}$ ; the per- $\mu\text{m}$  form used everywhere in this manual carries the Jacobian factor  $10^{-6}$  that converts the spectral density from per-meter to per-micrometer. Mixing the two is the single most common radiometry error.

### A.3 Signal path through the full optical train

#### A.3.1 Cascaded system transfer

Each element applies its transfer factor  $C_i(\lambda)$  — either  $T_{sys,i}$  or  $Rho_{sys,i}$  depending on type. The total signal transfer from source to focal plane is:

$$C_{total}(\lambda) = \prod C_i(\lambda) \quad \text{for } i = 1 \text{ to } N$$

Explicitly for a mixed train of  $N$  elements:

$$C_{total}(\lambda) = C_1(\lambda) \cdot C_2(\lambda) \cdot C_3(\lambda) \cdot \dots \cdot C_N(\lambda)$$

Where each  $C_i$  is either  $T_{sys,i}$  (refractive) or  $Rho_{sys,i}$  (reflective).

#### A.3.2 Signal irradiance at the focal plane

$$E_{signal,FP}(\lambda) = \pi \cdot L_{source}(\lambda) \cdot \sin^2(\theta_{FP}) \cdot C_{total}(\lambda)$$

Or equivalently using f-number:

$$E_{signal,FP}(\lambda) = \frac{\pi \cdot L_{source}(\lambda) \cdot C_{total}(\lambda)}{4 \cdot (f/\#)^2 \cdot (1 + M)^2}$$

#### A.3.3 Total in-band signal irradiance

$$E_{signal,total} = \int [\lambda_1 \text{ to } \lambda_2] E_{signal,FP}(\lambda) d\lambda$$

### A.4 Thermal background path from each element

Each element  $i$  emits thermal radiation that propagates through all downstream elements  $j > i$  before reaching the focal plane.

#### A.4.1 Downstream cumulative transfer from element $i$ to focal plane

The thermal emission from element  $i$  is transferred by every downstream element using that element's transfer factor  $C_j(\lambda)$ :

$$\tau_{i,j}(\lambda) = \prod C_j(\lambda) \quad \text{for } j = i+1 \text{ to } N$$

Note:  $C_j$  is  $T_{sys,j}$  if element  $j$  is refractive,  $Rho_{sys,j}$  if reflective. Downstream reflective elements both redirect **and** attenuate the upstream thermal flux.

For the last element:  $\tau_{i,N}(\lambda) = 1.0$

Explicitly for a 4-element train:

$$\begin{aligned}\tau_1(\lambda) &= C_2 \cdot C_3 \cdot C_4 \\ \tau_2(\lambda) &= C_3 \cdot C_4 \\ \tau_3(\lambda) &= C_4 \\ \tau_4(\lambda) &= 1.0\end{aligned}$$

#### A.4.2 The acceptance cone — one geometry for every element

The Lagrange invariant caps how much solid angle the focal plane can accept from an in-beam element. Whatever the internal layout, each element is seen through the reimaging optics and can fill at most the cone the working f-number sets:

$$\Omega_{\text{cone}} = 2 \cdot \pi \cdot (1 - \cos(\theta)), \quad \theta = \arctan(1 / (2 \cdot N_{\text{eff}}))$$

Where:

- $N_{\text{eff}}$  = effective (post-cold-stop) f-number,  $f / D_{\text{eff}}$
- $\theta$  = half-angle of the marginal ray

The exact form is used rather than the paraxial  $\pi / (4 \cdot N_{\text{eff}}^2)$ : the two agree to 0.5 % at  $f/6$  and 5 % at  $f/2$ , and the paraxial form over-states the cone, exceeding  $2\pi$  as  $N_{\text{eff}} \rightarrow 0$ , which no solid angle may do.

There is **no per-element geometry**. An element does not gain acceptance by sitting close to the focal plane: a private  $\Omega_i = \pi (D_i/2)^2 / z_i^2$  is not bounded by the invariant — a 0.3 m mirror 1.0 m from the focal plane would claim 0.0707 sr against an  $f/6$  cone of 0.0217 sr, 3.2× more than physics permits. Nor is there a cold-stop attenuation factor: in-cone emission arrives through the imaging path itself and cannot be blocked, while out-of-cone warm structure is taken to be blocked completely. What a cold stop does control is the size of the pupil, hence  $N_{\text{eff}}$ , hence  $\Omega_{\text{cone}}$ .

#### A.4.3 Focal plane irradiance from each element

Every element, last or upstream, emits as a graybody into that one cone, attenuated by everything downstream of it:

$$E_{\text{FP},i}(\lambda) = \Omega_{\text{cone}} \cdot L_{\text{thermal},i}(\lambda) \cdot \tau_i(\lambda)$$

Fully expanded:

**Refractive element i:**

$$E_{\text{FP},i}(\lambda) = \Omega_{\text{cone}} \cdot \epsilon_{\text{eff},i}(\lambda) \cdot B(\lambda, T_i) \cdot \prod_{j=i+1 \text{ to } N} C_j(\lambda)$$

**Reflective element i:**

$$E_{\text{FP},i}(\lambda) = \Omega_{\text{cone}} \cdot \epsilon_i(\lambda) \cdot B(\lambda, T_i) \cdot \prod_{j=i+1 \text{ to } N} C_j(\lambda)$$

The structure is identical — only the emissivity model differs between the two types. The last element is not a special case:  $\tau_N = 1$ , so it contributes  $\Omega_{\text{cone}} \cdot L_{\text{thermal},N}(\lambda)$ .

Dimensionally,  $\text{sr} \times [-] \times \text{W/m}^2/\text{sr}/\mu\text{m} \times [-] = \text{W/m}^2/\mu\text{m}$ , the spectral irradiance the focal plane sees.

#### A.4.4 Total thermal background irradiance at the focal plane

$$E_{\text{background},\text{FP}}(\lambda) = \sum_{i=1 \text{ to } N} E_{\text{FP},i}(\lambda)$$

Total in-band thermal background:

$$E_{\text{background},\text{total}} = \int_{[\lambda_1 \text{ to } \lambda_2]} E_{\text{background},\text{FP}}(\lambda) d\lambda$$


---

#### A.5 Combined focal plane irradiance

$$E_{\text{FP},\text{total}}(\lambda) = E_{\text{signal},\text{FP}}(\lambda) + E_{\text{background},\text{FP}}(\lambda)$$

Signal-to-background ratio:

$$\text{SBR}(\lambda) = E_{\text{signal},\text{FP}}(\lambda) / E_{\text{background},\text{FP}}(\lambda)$$


---

#### A.6 Kirchhoff self-consistency check

##### A.6.1 Refractive element i

System reflectance (accounting for cavity):

$$R_{\text{sys},i}(\lambda) = \frac{R_{1,i} + T_{1,i}^2 \cdot R_{2,i} \cdot \text{beer}_i^2}{1 - R_{1,i} \cdot R_{2,i} \cdot \text{beer}_i^2}$$

Check:

$$\begin{aligned} A_{\text{total},i}(\lambda) &= 1 - R_{\text{sys},i}(\lambda) - T_{\text{sys},i}(\lambda) \\ \text{eps}_{\text{eff},i}(\lambda) &\approx A_{\text{total},i}(\lambda) \end{aligned} \quad \leftarrow \text{must hold at each } \lambda$$

##### A.6.2 Reflective element i

$$\begin{aligned} A_{\text{total},i}(\lambda) &= 1 - \text{Rho}_{\text{sys},i}(\lambda) - A_{\text{coat},i}(\lambda) && [\text{if } A_{\text{coat}} \text{ known}] \\ &= 1 - \text{Rho}_{\text{sys},i}(\lambda) && [\text{conservative bound}] \\ \text{eps}_i(\lambda) &\approx A_{\text{total},i}(\lambda) && \leftarrow \text{must hold at each } \lambda \end{aligned}$$


---

#### A.7 Implementation recipe

Inputs per element i:

type\_i = REFRACTIVE or REFLECTIVE

If REFRACTIVE:

|                  |                             |
|------------------|-----------------------------|
| R1_i(λ), T1_i(λ) | entry surface coating       |
| R2_i(λ), T2_i(λ) | exit surface coating        |
| alpha_i(λ)       | bulk absorption coefficient |

$n_i(\lambda)$                       refractive index  
 $d_i$                               substrate thickness  
 $T_i$                               temperature  
 $\theta_{r,i}$                           refracted angle inside substrate

If REFLECTIVE:

$\rho_i(\lambda)$                       surface reflectance  
 $A_{coat,i}(\lambda)$               coating absorptance (optional; 0 if unknown)  
 $T_i$                               temperature

System geometry (all elements – one cone, not one per element):

$\theta_{FP}$                           focal plane convergence half-angle  
 $N_{eff}$                         effective (post-cold-stop) f-number  
 $\Omega_{cone}$                        $2 \cdot \pi \cdot (1 - \cos(\arctan(1 / (2 \cdot N_{eff}))))$  [sr]

Per element, compute transfer factor  $C_i$  and emissivity:

If REFRACTIVE:

$beer_i = \exp(-\alpha_i \cdot d_i / \cos(\theta_{r,i}))$   
 $denom_i = 1 - R1_i \cdot R2_i \cdot beer_i^2$   
 $C_i = T1_i \cdot beer_i \cdot T2_i / denom_i$  [transmittance]  
 $eps_i = T2_i \cdot n_i^2 \cdot (1 - beer_i) / denom_i$  [cavity emissivity]

If REFLECTIVE:

$C_i = \rho_i$  [reflectance]  
 $eps_i = 1 - \rho_i - A_{coat,i}$  [surface emissivity]

Both types:

$L_i(\lambda) = eps_i(\lambda) \cdot B(\lambda, T_i)$

Signal path:

$C_{total}(\lambda) = \prod C_i(\lambda)$  for  $i = 1$  to  $N$   
 $E_{signal}(\lambda) = \pi \cdot L_{source}(\lambda) \cdot \sin^2(\theta_{FP}) \cdot C_{total}(\lambda)$

Thermal background path (build tau as reverse cumulative product):

$\tau_N = 1.0$   
 $\tau_i = \prod C_j(\lambda)$  for  $j = i+1$  to  $N$  [reverse cumulative product]

For  $i = 1$  to  $N$ :

$E_{FP,i}(\lambda) = \Omega_{cone} \cdot L_i(\lambda) \cdot \tau_i(\lambda)$

$E_{background}(\lambda) = \sum_{i=1}^N E_{FP,i}(\lambda)$

Total:

$E_{FP,total}(\lambda) = E_{signal}(\lambda) + E_{background}(\lambda)$

$$\text{SBR}(\lambda) = E_{\text{signal}}(\lambda) / E_{\text{background}}(\lambda)$$

---

Validation:

Refractive:  $\text{eps\_eff}, i \approx 1 - R_{\text{sys}, i} - T_{\text{sys}, i}$  for each optic

Reflective:  $\text{eps}_i \approx 1 - \text{Rho}_i - A_{\text{coat}, i}$  for each mirror

---

## A.8 Key physical distinctions summary

| Property                 | Refractive element                  | Reflective element                            |
|--------------------------|-------------------------------------|-----------------------------------------------|
| Signal transfer factor   | $T_{\text{sys}, i}$ (transmittance) | $\text{Rho}_{\text{sys}, i}$ (reflectance)    |
| Bulk emission            | Yes — Beer-Lambert path             | No                                            |
| Surface emission         | Both surfaces (via $T_2, n^2$ )     | Single surface only                           |
| Cavity / etalon effect   | Yes — $R_1 \cdot R_2$ denominator   | No                                            |
| Emissivity model         | Full cavity $\text{eps\_eff}$       | Kirchhoff: $1 - \text{Rho} - A_{\text{coat}}$ |
| Downstream attenuation   | $T_{\text{sys}, j}$ per element     | $\text{Rho}_{\text{sys}, j}$ per element      |
| Sensitive to temperature | Bulk + surface                      | Surface only                                  |
| Dominant in cold systems | Windows, filters                    | Warm fold mirrors                             |

# References

Full citations for the short keys used across the theory chapters (e.g. [Holst 2008], [Fried 1966]). One list, alphabetical by key; chapters cite by key only, adding a section locator after the key where one helps — [Holst 2008 §3.2].

| Key                       | Full citation                                                                                                                                                                |
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